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Understanding and Designing Multi-Page Visually Rich Document Understanding Systems

Lewei Xu

Supervised by
Dr. Yihao Ding
Dr. Daochang Liu

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The University of Western Australia

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Abstract

Multi-page visually rich document understanding (MP-VRDU) concerns systems that interpret long documents whose relevant information may be distributed across pages, modalities, and structural relationships, often beyond the practical context and compute budgets of multimodal large language models. This thesis is presented as a compilation of a survey paper, an empirical attribution study, and a final system-level method that translates their findings into a working MP-VRDU system.

First, we systematically survey the field and introduce a taxonomy that organises MP-VRDU systems along four axes: architecture, multimodal representation, inference-time adaptation, and training strategy. Through an evidence-management perspective, we identify the key design choices and techniques that distinguish multi-page understanding from the single-page setting, and trace how successive architectures increasingly shift evidence management outside the MLLM backbone. We further consolidate MP-VRDU datasets and identify open challenges in efficiency, faithfulness, evidence pipelines, supervision integrity, and reproducible evaluation. Second, we conduct a controlled empirical attribution study of representation, selection, and reasoning failures, with complementary checks across document collections and model families. We show that complementary modalities reduce conversion sensitivity, while added document context can both repair and break answers despite complete evidence coverage. Reasoning and abstention instructions also interact differently across backbones, affecting the balance between supported answering and false refusal. These findings, supported by qualitative examples and accuracy–cost comparisons, provide practical guidance for designing MP-VRDU systems under varying task and resource constraints.

Finally, we translate these findings into a deliberately simple method for MP-VRDU that combines broad evidence acquisition, selective evidence presentation, and structured reasoning. The resulting system achieves performance comparable to state-of-the-art methods on principal MP-VRDU benchmarks without task-specific training and the sophisticated retrieval or orchestration techniques used by many competing systems, thereby providing a system-level validation of the preceding survey and empirical findings. Taken together, this thesis shows that effective MP-VRDU depends not only on stronger multimodal models, but on managing evidence according to the information, reasoning, and resource demands of the task, providing a principled basis for designing more capable and deployable long-document understanding systems.

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Contents

Abstract

List of Figures v

List of Tables vi

1 Introduction and Background 1
- 1.1 Research Aims . . . 2
- 1.2 Thesis by Compilation . . . 2
- 1.3 Background . . . 3
  - 1.3.1 A Brief History of Document Understanding . . . 3
  - 1.3.2 General-Domain MLLMs and MP-VRDU . . . 4
  - 1.3.3 Document Understanding in Practice . . . 4
  - 1.3.4 Single-Page VRDU Overview . . . 5

2 Survey of Architectures and Techniques 7
- 2.1 Introduction . . . 7
- 2.2 Methodology and Scope . . . 8
- 2.3 Multi-Page Architectural Overview . . . 9
  - 2.3.1 Page-to-Document Encoding Models . . . 9
  - 2.3.2 MLLM-Centric Adaptation Models . . . 10
  - 2.3.3 Retrieval-Augmented Pipelines . . . 10
  - 2.3.4 Adaptive-Trajectory Pipelines . . . 10
- 2.4 MP-VRDU Multimodal Representation . . . 11
  - 2.4.1 Text Modality . . . 11
  - 2.4.2 Visual Modality . . . 12
  - 2.4.3 Cross-Page Structure Modality . . . 12
- 2.5 Inference-Time Adaptation Strategies . . . 13
  - 2.5.1 Retrieval and Navigation . . . 13
  - 2.5.2 Reasoning Strategies . . . 14
  - 2.5.3 Agentic Methods . . . 15
- 2.6 Training Strategies . . . 16
  - 2.6.1 Trained Retrieval Components . . . 17
  - 2.6.2 Trained Reasoning over Assembled Evidence . . . 17
  - 2.6.3 Trained Agentic Methods . . . 18
  - 2.6.4 Cross-Cutting Training Techniques . . . 18
- 2.7 Datasets . . . 19

2.7.1 Pretraining . . . 19
2.7.2 Supervised Fine-Tuning . . . 19
2.7.3 Benchmarking . . . 20
2.8 Conclusion . . . 21

3 Empirical Attribution of Failure Modes **23**
3.1 Introduction . . . 23
3.2 Related Work . . . 24
3.2.1 Dataset and Diagnostic Studies . . . 24
3.2.2 Individual Component Studies . . . 24
3.2.3 System Design and Deployment . . . 25
3.3 Attribution Framework . . . 25
3.3.1 Representation . . . 25
3.3.2 Selection . . . 26
3.3.3 Reasoning . . . 26
3.4 Methodology . . . 26
3.4.1 Attribution by Construction . . . 27
3.4.2 Representation Experiments . . . 27
3.4.3 Selection Experiments . . . 29
3.4.4 Reasoning Experiments . . . 29
3.4.5 Additional Experiments . . . 30
3.4.6 Datasets . . . 32
3.4.7 Evaluation . . . 33
3.5 Empirical Findings . . . 34
3.5.1 Representation . . . 34
3.5.2 Selection . . . 35
3.5.3 Reasoning . . . 37
3.6 Practical Discussion . . . 38
3.6.1 Representation . . . 38
3.6.2 Selection . . . 39
3.6.3 Reasoning . . . 40
3.7 Conclusion . . . 41

4 Method and Conclusion **43**
4.1 A Simple Adaptive-Trajectory Method . . . 43
4.1.1 Method Overview . . . 43
4.1.2 Evaluation Setup . . . 44
4.1.3 Results . . . 45
4.2 Open Challenges and Future Work . . . 47
4.2.1 Evidence Pipeline Bottleneck . . . 47
4.2.2 Faithfulness and Abstention . . . 47
4.2.3 Efficiency and Deployability . . . 48
4.2.4 Supervision Scarcity and Contamination . . . 48
4.2.5 Evaluation Methodology and Scope . . . 48
4.3 Concluding Remarks . . . 48

References **50**

A Additional Survey Details **66**
A.1 Comparison with Prior Surveys . . . 66
A.2 Full Survey Methodology . . . 67
A.2.1 Search keywords. . . . 67
A.2.2 Inclusion and exclusion criteria. . . . 67
A.2.3 Knowledge extraction and verification. . . . 67
A.2.4 Dataset catalogue. . . . 67
A.3 Additional Resources . . . 69
A.3.1 Reported Performance by Architecture . . . 69
A.3.2 General Domain MLLM for MP-VRDU . . . 70
A.3.3 Model Components Details . . . 70

B Additional Attribution Details **73**
B.1 Extended Result Tables . . . 73
B.1.1 Representation . . . 73
B.1.2 Selection . . . 74
B.1.3 Reasoning . . . 76
B.2 Statistical Uncertainty . . . 77
B.3 Error Analysis . . . 78
B.3.1 Error Categories . . . 78
B.3.2 Error Analysis Results . . . 79

C Additional Method Details **81**
C.1 Implementation Details . . . 81
C.1.1 Role Prompts . . . 81
C.1.2 Evidence Assembly . . . 81
C.1.3 Configuration . . . 81

List of Figures

1.1 Representative pages from MMLongBench-Doc. . . . 3
1.2 Example of a multi-hop question over a long visually rich document. . . . 4
1.3 OCR-dependent and OCR-free framework architectures for MLLM-based VRDU. . . . 5
1.4 Multimodal feature representation and fusion mechanisms. . . . 6

2.1 Illustration of MP-VRDU architectures. . . . 9
2.2 ColPali late-interaction retrieval . . . 14
2.3 Evolution of reasoning topologies used in prompting schemes. . . . 15
2.4 Architecture of DocAgent. . . . 16
2.5 Adapter-based parameter-efficient tuning. . . . 18
2.6 MP-VRDU taxonomy. . . . 22

3.1 Failure loci and mechanisms . . . 25
3.2 Reasoner prompt template . . . 27
3.3 Judge rubric . . . 33
3.4 Conversion fidelity: text source and image resolution . . . 35
3.5 Evidence coverage: verdict transitions after removing gold pages . . . 36
3.6 Distractor exposure: accuracy and verdict transitions as pages are added . . . 36
3.7 Evidence integration: single- and multi-hop accuracy by model scale . . . 37
3.8 Response calibration: accuracy and refusal across prompt modes . . . 37
3.9 Qualitative examples of the six failure mechanisms . . . 38
3.10 Multi-pass reasoning strategies and their accuracy–cost trade-offs . . . 40
3.11 Additional worked examples for the six failure mechanisms . . . 42

4.1 Architecture of the proposed method. . . . 44
4.2 Trajectory of our simple method . . . 46

B.1 Page-level retrieval recall by depth . . . 76
B.2 Judge analysis rubric . . . 78
B.3 Error profile by prompt mode . . . 79
B.4 Error profile by evidence source . . . 80

C.1 Reader prompt structure. . . . 82
C.2 Synthesizer prompt structure. . . . 83
C.3 Arbiter prompt structure. . . . 84

List of Tables

2.1 Survey of MP-VRDU models grouped by architecture. . . . 8
2.2 Datasets referenced in the MP-VRDU survey. . . . 20

3.1 Modality ceiling experiments. . . . 28
3.2 Conversion fidelity experiments. . . . 28
3.3 Selection Experiments. . . . 29
3.4 Reasoning prompt modes for the `{instruction}` field. . . . 30
3.5 Selection: retriever exerpiments. . . . 31
3.6 Reasoner variations. . . . 31
3.7 MMLongBench-Doc composition and question labels. . . . 32
3.8 Abstention and accuracy by evidence source and representation. . . . 34
3.9 Performance and cost of each representation, retriever and reasoner. . . . 39

4.1 Performance of the simple adaptive-trajectory method against Qwen3-VL-8B systems. . . . 45
4.2 Inference cost of the pipeline per agent turn and question across both datasets. . . . 47

A.1 Comparison of this survey with prior VRDU surveys. . . . 66
A.2 The 46 surveyed MP-VRDU systems. . . . 68
A.3 Performance of MP-VRDU models grouped by architecture category. . . . 69
A.4 Per-model components of the surveyed MP-VRDU systems. . . . 70

B.1 Accuracy by text source, evidence source and scan status 73
B.2 Accuracy by image resolution and evidence source 74
B.3 Representation ladder by document domain and render resolution. . . . 74
B.4 LongDocURL accuracy by task and question type 75
B.5 Page-level retrieval across depth on both corpora. . . . 75
B.6 LongDocURL verdict transitions under gold-page removal 76
B.7 Accuracy for further reasoner configurations 77
B.8 Prompt-mode results by evidence source 77
B.9 Bootstrap confidence intervals for the main contrasts 78
B.10 The error taxonomy in judge priority order. . . . 79

C.1 Configuration of the simple adaptive-trajectory method. . . . 84

Chapter 1

Introduction and Background

Visually rich document understanding (VRDU) concerns systems that extract or infer information from documents by jointly interpreting their textual, visual, structural, and layout information [1, 2]. Documents such as contracts, financial reports, scientific papers, technical manuals, and regulatory filings are rarely plain text: they combine prose with tables, figures, diagrams, typography, spatial organisation, and other layout cues that contribute to meaning. VRDU therefore differs from conventional natural language processing because information may be encoded not only in the words themselves, but also in where they appear, how they are grouped, and how they relate to surrounding visual elements. Document-understanding tasks can involve extracting fields, identifying entities, comparing values, or interpreting visual elements, often requiring the ability to read small text, interpret tables and figures, resolve spatial relationships, and combine textual and visual evidence [3, 4]. In practice, these tasks are commonly unified under document question answering, where the information need is expressed as a natural-language question over the document [5].

Although VRDU was historically dominated by single-page research, the past several years have seen rapid growth in multi-page document understanding [6, 7]. This reflects the fact that real-world documents commonly extend across tens or hundreds of pages, creating a document-scale setting in which answer-relevant evidence may be sparse, distributed across distant pages, or expressed through several modalities and structural relationships. Simply providing every page to a model is often impractical: textual and visual tokens accumulate with document length, high-resolution page images are expensive to process, and sufficiently long documents can exceed a model's usable context budget. Even when the document fits, long-context models can struggle to use evidence buried within large inputs reliably [8]. The system must therefore decide not only how each page should be represented, but also which evidence should be retained or selected and how information from different pages should be combined. This is particularly important when a question depends on several pages, such as comparing values across tables, following references between sections, or combining evidence that is individually insufficient [9, 10]. MP-VRDU consequently introduces document-scale evidence-management and cross-page reasoning problems that do not arise to the same extent in the single-page setting, making it a distinct problem rather than a straightforward extension of VRDU [11].

These challenges motivate a systematic investigation of MP-VRDU, extending beyond individual architectures or model capabilities to examine the broader decisions that determine system effectiveness. In particular, understanding how existing approaches address document-scale evidence, why they fail, and how these findings can inform practical system design forms the basis of this thesis.

1.1 Research Aims

This thesis addresses three complementary research aims, progressing from a systematic understanding of existing MP-VRDU approaches, through empirical investigation of their limitations, to the development and evaluation of a practical system:

Aim 1: comprehensively survey techniques and architectures for multi-page document understanding, treating it as a problem distinct from the single-page setting. We set out to organise the current landscape of MP-VRDU through an evidence-management perspective, focusing on the design decisions and techniques that arise specifically from long multi-page documents. Prior surveys remain anchored to the single-page setting or organise the field by narrower distinctions such as OCR dependency [12, 13]; none treats multi-page understanding as a distinct problem or takes evidence management as its organising lens.

Aim 2: determine where and why MP-VRDU systems fail through controlled empirical attribution. We set out to isolate failures in representation, selection, and reasoning by intervening on one part of a common MP-VRDU pipeline while holding the others fixed. Existing diagnostic work typically studies individual components or aspects of system behaviour in isolation, and much of it is developed outside the document-understanding domain [14–16]. By testing how these findings generalise across datasets and model families, and examining performance–cost trade-offs across system components, we aim to derive practical guidance for MP-VRDU system design and deployment.

Aim 3: design a simple MP-VRDU system that translates the thesis's conceptual and empirical findings into a competitive working method. We set out to further validate the evidence-management perspective from Aim 1 and the empirical findings from Aim 2 by translating them directly into the design of an MP-VRDU system. The goal is to achieve performance comparable to or exceeding state-of-the-art systems using a deliberately simple design, demonstrating that these findings provide a practical basis for building effective document-understanding systems.

1.2 Thesis by Compilation

This thesis is structured around two papers addressing Aims 1 and 2 respectively, with Aim 3 developed as an additional contribution in Chapter 4. The remainder of this chapter provides the background for both studies. Chapter 2 surveys MP-VRDU architectures and techniques, while Chapter 3 presents the empirical attribution study of representation, selection, and reasoning failures. Finally, Chapter 4 introduces and evaluates the system developed for Aim 3, synthesises the thesis findings, and discusses open challenges and future work. The two constituent papers are as follows:

1. **Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding.** Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu. *Main Conference of EMNLP 2026.* Available at https://doi.org/10.13140/RG.2.2.23979.60965.

2. **Locating Failure in Multi-Page Visually Rich Document Understanding Systems: An Empirical Attribution.** Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu. *Under review at ACL ARR.* Available at https://arxiv.org/abs/2608.07943.

Additional explanatory material, figures, and tables have been added to Chapters 2 and 3 to improve readability and provide context beyond the conventions and space constraints of the original venues. This thesis also follows the conventional honours thesis structure, with Chapter 2 serving as the literature review and Chapter 3 providing a comprehensive related work section. The full experimental methodology is presented in Chapter 3, incorporating details previously relegated to the paper's appendix to meet the methodological reporting requirements of an honours thesis.

1.3 Background

1.3.1 A Brief History of Document Understanding

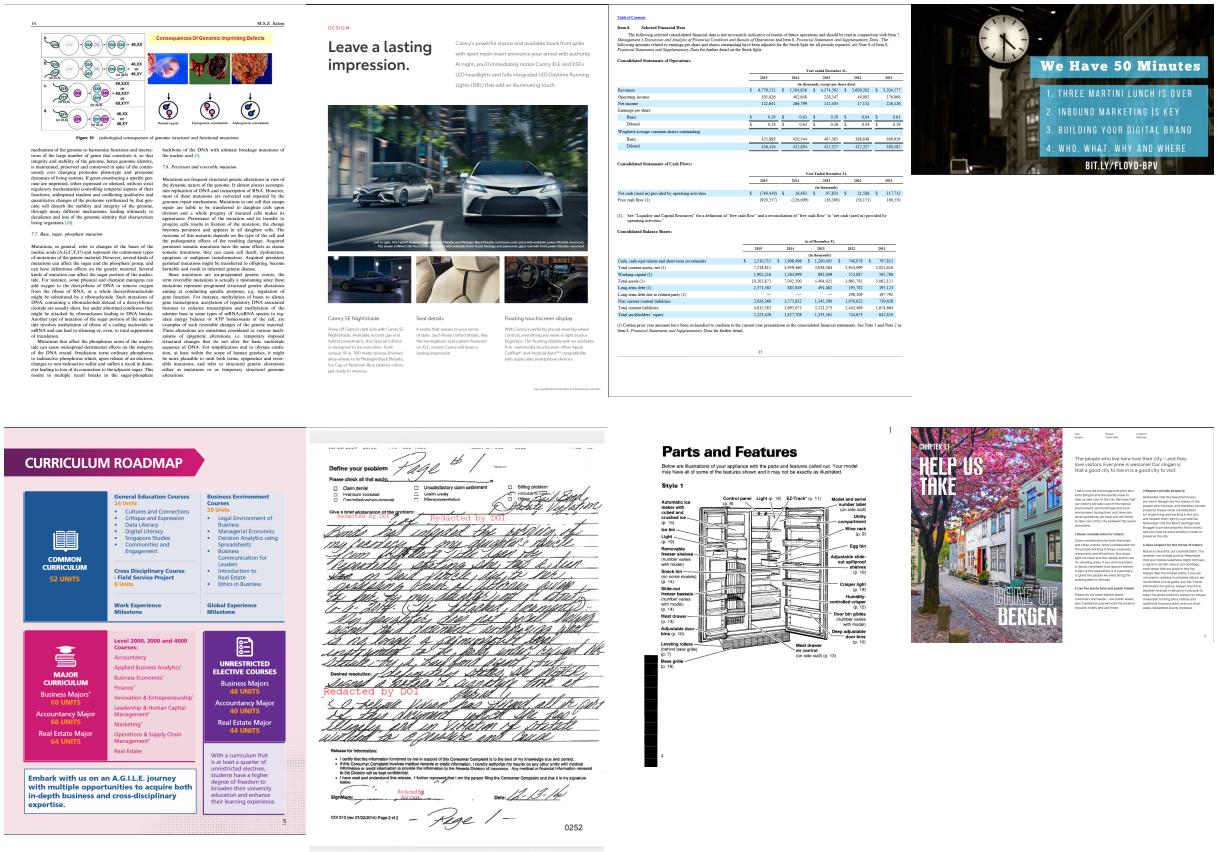


Figure 1.1 Representative pages from MMLongBench-Doc.

Automated document understanding predates the era of instruction-tuned language models and originally addressed a much narrower range of document types than the heterogeneous pages encountered today (Figure 1.1). Early systems were brittle and task-specific, treating document understanding as information extraction over fixed format documents such as forms and scanned business documents, typically pairing OCR with rule-based or shallow machine-learning classifiers to recover fields such as names, dates, and totals [17, 18]. LayoutLM [1] established in 2020 that layout, the two-dimensional position of text on a page, could be pretrained jointly with textual content rather than attached afterwards, moving the field from task-specific classifiers to a general pretrained backbone trained end to end. DocVQA [5] reframed the task the following year as natural-language question answering over a document image, so that a single model could answer arbitrary questions instead of extracting a predetermined set of

fields. Both, however, remained confined to a single page. MP-DocVQA [6] then extended the setting to multiple pages in 2023, pairing a benchmark of multi-page documents with the end-to-end Hi-VT5 baseline; its questions were still answerable from one page and its documents relatively short, but it marked the point at which multi-page visually rich document understanding (MP-VRDU) became a distinct problem. Multimodal large language models (MLLMs), emerging over the same period, replaced task-specific encoders with general backbones that consume text, layout, and page images together and answer in natural language [19, 20]. A backbone reads only what fits its context window, however, while training specialised models end to end becomes increasingly costly at document scale.

1.3.2 General-Domain MLLMs and MP-VRDU

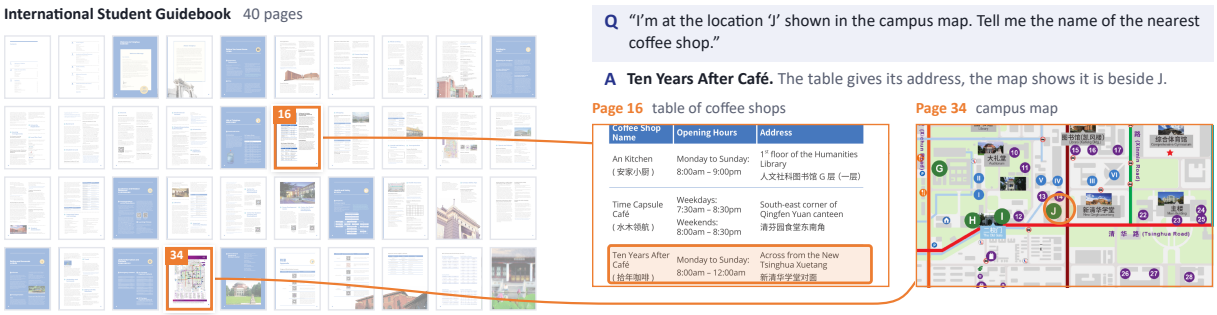


Figure 1.2 Example of a multi-hop question over a long visually rich document.

General-domain MLLMs include both frontier closed-source systems and openly released, open-weight models. Although these models are trained for broad multimodal tasks rather than document understanding specifically, they can read rendered pages directly and interpret textual, visual, and layout information within a common backbone, for example in Figure 1.2. This has made them a natural foundation for modern VRDU systems, particularly as their context windows and visual capabilities have improved. In the multi-page setting, however, the backbone alone is rarely sufficient: as document length grows, systems must decide how pages are represented, which evidence is retained or retrieved, and how information from different parts of the document is organised for reasoning.

An alternative is to pretrain or continue-train models specifically for document understanding, a common approach in SP-VRDU but considerably less prevalent in MP-VRDU, where document-scale pretraining data remain scarce (§2.7.1). Most MP-VRDU systems therefore inherit a pretrained general-domain MLLM and concentrate their design choices on the document-specific techniques surrounding it. This includes multimodal representation, evidence selection and delivery, inference-time reasoning, and, in some cases, additional task-specific training. The boundary is not absolute, since some methods also modify the backbone itself, but the dominant MP-VRDU design space lies in how a general multimodal model is adapted to operate over long documents [11]. Chapter 2 examines these choices systematically through architectures, multimodal representation, inference-time adaptation, training, and datasets.

1.3.3 Document Understanding in Practice

Consumer-facing LLM assistants provide one of the most visible examples of document-understanding research being translated into widely used products. Systems such as ChatGPT, Claude, and NotebookLM incorporate techniques developed in the field, including multimodal page processing, retrieval, and tool-based access, to answer questions over uploaded documents and collections that may be too

large to process directly in a single context [21–23]. From the user's perspective, this often appears as a simple question-answering interface, but the system may need to identify relevant pages, inspect them in greater detail, and combine evidence before producing an answer. These products therefore demonstrate how document-understanding techniques can support general-purpose assistants, but they represent only a small part of the broader range of document-processing problems encountered in practice.

In real-world and industrial settings, document understanding may instead involve classifying, summarising, extracting structured information from, or otherwise processing thousands or millions of documents. These workloads can involve sensitive medical, financial, legal, scientific, or operational records that cannot be sent to third-party services because of privacy or data-residency requirements, while sufficiently large workloads can also make commercial API usage prohibitively expensive. Organisations may therefore need to deploy open-weight models locally under fixed memory and compute budgets, sometimes adapting them to specialised document types or tasks through additional training. Under these conditions, document understanding is not only about whether a system can answer correctly, but also about how to make document processing faster, more accurate, and more efficient.

1.3.4 Single-Page VRDU Overview

Although MP-VRDU introduces document-scale problems, much of the machinery used to process each individual page is inherited directly from single-page VRDU. This includes how text, layout, and visual information are represented, as well as the document-oriented encoders and preprocessing techniques used before information is combined across pages. The single-page setting is also comparatively mature: document-specialised systems such as ORCA [24] approach human performance on the DocVQA [5] benchmark, reaching 0.9729 ANLS against a human baseline of 0.9811, with comparable progress on infographics [3], charts [4], and scanned forms. The discussion below therefore follows the organisation of Ding et al.'s survey of MLLM-based VRDU [12], restricted to the page-level techniques that multi-page systems build upon.

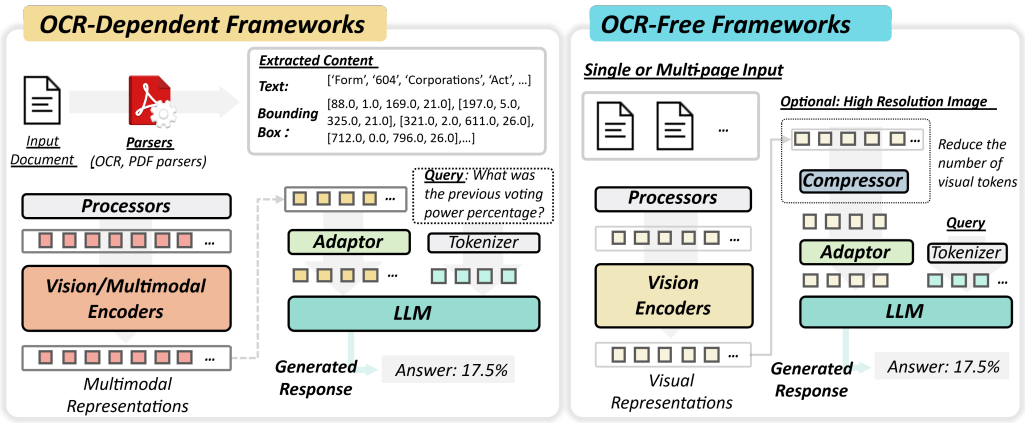


Figure 1.3 OCR-dependent and OCR-free framework architectures for MLLM-based VRDU. Adapted from Ding et al. [12].

MLLM-based VRDU Architectures. At the architectural level, single-page systems divide into two families, shown in Figure 1.3. OCR-dependent frameworks parse the page first, running off-the-shelf OCR or a PDF parser to recover text alongside its bounding-box coordinates. These parsed signals are then encoded, either injected as text into the language model prompt [25] or passed through a vision

or multimodal encoder such as LayoutLMv3 that binds text, layout, and image into joint multimodal representations [26, 27], which an adaptor and tokeniser convert into the token sequence the language model consumes. Relying on the parser lets these systems capture structural information without heavy visual pretraining, but ties their accuracy to the parser's, most visibly on handwritten or low-quality scans where recognition degrades. OCR-free frameworks discard the parser and let a vision encoder read the rendered page directly from pixels, producing visual representations that the language model is trained to interpret through objectives such as text recognition and grounding [20, 28]. Because fine-grained text is legible only at high resolution, these frameworks consume large images and often insert a compression step to reduce the resulting visual token count before the language model [29]; in exchange, they train end to end and handle handwriting and irregular layouts that OCR struggles with.

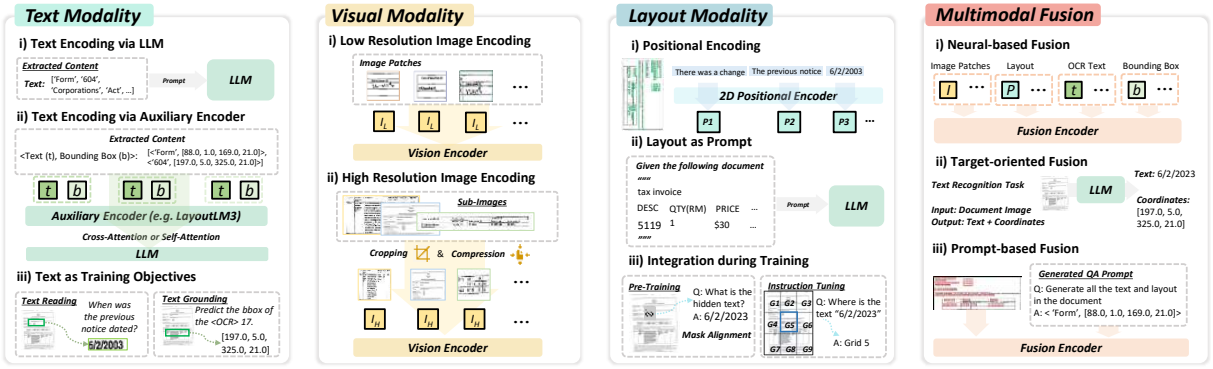


Figure 1.4 Multimodal feature representation and fusion mechanisms. Adapted from Ding et al. [12].

Single-page Multimodal Representation. Within both families, a system must still decide how to represent text, layout, and visual content, and how to fuse them (Figure 1.4). Text is recovered as injected OCR strings [25], through joint text-layout-vision encoders [27, 30], or as a pretraining objective for the vision encoder itself [31]. Layout is encoded as positional features attached to text tokens [1, 30], as coordinates verbalised in the prompt [32], or through dedicated layout tokens [33]. Visual content is managed through low-resolution patch embeddings [34], high-resolution shape-adaptive cropping [19, 20], or token compression that cuts the visual sequence by an order of magnitude [29, 35]. Fusion draws these together through dedicated multimodal encoders [26], shared training objectives [20, 31], or prompt-based interleaving of text, layout, and images. Single-page systems make these choices assuming the entire document fits in a single backbone input; the multi-page setting (Chapter 2), challenges that assumption, reframing each choice around which pages or regions the system should attend to in the first place.

Chapter 2

Survey of Architectures and Techniques

This chapter is adapted from **Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding** [11], by **Lewei Xu**, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, and Wei Liu, published in the *Main Conference of EMNLP 2026*. The first author was responsible for constructing the surveyed literature corpus, including paper identification and selection; developing the architectural taxonomy used to organise MP-VRDU systems; and preparing the technical content, including the survey analysis, appendix material, and tables. **This chapter additionally serves as the Literature Review of this thesis.**

2.1 Introduction

MP-VRDU has seen rapid growth in recent years, with new architectures, training strategies, and inference methods emerging. Existing work has explored joint multi-page encoding [6], adaptation of pretrained MLLMs [36], retrieval-augmented pipelines that decouple evidence selection from generation [37], and adaptive controllers that gather evidence iteratively at inference time [38], often combined with task-specific training or prompted reasoning loops [39, 40]. However, this body of work has developed along largely parallel lines, with limited cross-family comparison and inconsistent terminology, and few works explicitly distinguish what is genuinely multi-page-distinctive from what is inherited from the single-page setting. As a result, the field lacks a unified view of the design choices that matter most in multi-page settings, motivating the systematic organisation this chapter provides.

Prior surveys have charted VRDU from complementary angles. Subramani et al. [17] survey the OCR-centric deep-learning pipelines of the pre-MLLM era, covering text detection, recognition, and layout analysis as separate stages. Ding et al. [41] and Gbada et al. [42] both treat information extraction over visually rich documents, while a more recent survey by Ding et al. [12] covers the shift to MLLM-based architectures, training strategies, and benchmarks. Closest to the multi-page setting, Gao et al. [13] survey multimodal retrieval-augmented generation as a means of scaling document understanding to the multi-page setting. Several of these include multi-page systems among the work they cover, but none treats multi-page understanding as a distinct problem with its own design axes, addressing it only in passing within a broader single-page or extraction-focused scope. [1]

Our contributions are as follows: **(1)** We define MP-VRDU as a distinct document-understanding problem and organise the field through an evidence-management perspective. **(2)** We introduce a new architectural taxonomy for MP-VRDU, grouping existing systems into four families according to how they manage document-scale evidence (§2.3). **(3)** We provide a systematic survey of the architectures, multimodal representations, inference-time and training strategies, datasets, and evaluation practices used in

[1] A detailed survey comparison table is provided in Appendix A.1.

Model	Venue	OCR	Vision Encoder	LLM Backbone	Mod.	Search	Reasoning	Agentic
Page-to-Document Encoding Models								
Arctic-TILT (2025)	ACL	Yes	U-Net	T5-Large	T, V, L	-	-	-
GRAM (2024)	CVPR	Yes	DocFormerv2	DocFormerv2	T, V, L	-	-	-
Hi-VT5 (2023)	PR	Yes	DiT	T5-Base	T, V, L	-	-	-
RM-T5 (2024)	DAS	Yes	DiT	T5-Base	T, V, L	-	-	-
MLLM-Centric Adaptation Models								
DocR1 (2026)	AAAI	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	-	CoT, HD	-
DocSeeker (2026)	CVPR	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	-	CoT, HD	-
Leopard (2025)	TMLR	No	SigLIP-SO400M	Mult. (La/Mi)	V	-	CoT	-
mPLUG-DocOwl2 (2025)	ACL	No	ViT-L/14	mPLUG-Owl2	V	-	-	-
URaG (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B	V	-	HD	-
Docopilot (2025)	CVPR	No*	InternViT-300M	Mult. (In)	V	-	-	-
Texthawk2 (2024)	preprint	No*	SigLIP-SO400M	Qwen2-7B	V	-	-	-
DocSLM (2026)	CVPR	Yes	SigLIP2	Qwen2.5-1.5B	T, V, L	-	-	-
InstructDoc (2024)	AAAI	Yes	CLIP	FlanT5, BLIP2	T, V, L	-	-	-
LayTokenLLM (2025)	CVPR	Yes	-	Mult. (La/Qw)	T, L	-	-	-
CoR (2026)	preprint	Yes*	Qwen2.5-VL	Qwen2.5-VL-7B	T, V	-	CoT, PE, HD	-
Retrieval-Augmented Pipelines								
AVIR (2026)	MMAsia	No	Pix2Struct	Qwen2.5-VL-3B	V	D	-	-
DFVC (2026)	MDPI	No	VisRAG, ColQwen	Qwen2-VL-2B	V	D	-	-
DREAM (2025)	MM	No	InternVL2 ViT	InternVL2-4B	V	J	HD	-
M3DocRAG (2024)	preprint	No	ColPali, ColQwen	Mult. (Qw/In/Id)	V	D	-	-
MM-R5 (2025)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	D	CoT	-
MoLoRAG (2025)	EMNLP	No	ColPali	Mult. (Qw/DS/Lv)	V	D, Nav	-	1A
Self-Attention Scoring (2024)	ICDAR	No	Pix2Struct	Pix2Struct	V	D	-	-
SLEUTH (2025)	CVPR	No	ColPali, Qwen3-VL ViT	Mult. (Qw/GL/Ge)	V	D	HD	4A
SV-RAG (2025)	ICLR	No	InternVL2, Phi-3-vision	Mult. (In/Pa/Ph)	V	D	-	-
VDocRAG (2025)	CVPR	No*	Phi3V-4.2B	Phi3V-4.2B	V	D	-	-
CREAM (2024)	ACM	Yes	Pix2Struct	LLaMA2-7B	T, V	J	HD	-
KGP (2024)	AAAI	Yes	-	Mult.	T, L	S, IR, Nav	-	1A
MHier-RAG (2026)	preprint	Yes	Qwen-VL-Plus	Mult. (Qw/DS/GPT)	T, V, L	J	CoT, HD	-
MLDocRAG (2026)	preprint	Yes	Qwen2.5-VL	Mult. (Qw)	T, V, L	D	CoT, HD	-
MultiDocFusion (2026)	ACL	Yes	-	Mult. (Qw/La/Mi)	T, V, L	D	HD	-
M2RAG (2025)	NeurIPS	Yes*	VisRAG-Ret	Mult. (Qw)	T, V	J	-	3A
MDocAgent (2025)	preprint	Yes*	ColPali, ColQwen2	Mult. (La/Qw)	T, V	J	-	5A
PDF-WuKong (2026)	IJCV	Yes*	IXC2-VL-4KHD	IXC2-VL-4KHD	T, V	J	-	-
RAG-DocVQA (2025)	ICDAR	Yes*	DiT, Pix2Struct	Qwen2.5-VL-7B	T, V, L	D	-	-
VisDoMRAG (2025)	NAACL	Yes*	ColPali / ColQwen2	Mult. (Qw/Ge/GPT)	T, V	J	CoT	3A, SC
Adaptive-Trajectory Pipelines								
Doc-V* (2026)	ACL	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	D, IR, QR, Nav	ReAct, HD	1A
DocReact (2025)	ACL	No	ColPali, VisRAG	GPT-4o	V	D, IR, QR	ReAct, QD	1A
MACT (2025)	preprint	No	Multiple	Mult. (Qw/MM/In)	V	D	CoT, PE	4A, SC, SV
VRAG-RL (2025)	NeurIPS	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B	V	D, IR, QR	ReAct, HD	1A
DocLens (2026)	ACL	Yes	–	Mult. (GPT/Ge/Cl)	T, V, L	J	CoT	4A, SA
DMAP (2026)	WWW	Yes*	ColPali	Mult. (GPT/Qw)	T, V, L	J	HD	2A, SR
DocAgent (2025)	EMNLP	Yes*	–	Mult. (GPT/Ge/Cl)	T, V, L	J, IR, Nav	ReAct, HD	3A
DocDancer (2026)	preprint	Yes*	Qwen3-VL-235B	Qwen3-4B/30B	T, V, L	J, IR, QR	ReAct	1A
MM-Doc-R1 (2026)	ACL	Yes*	Qwen2.5-VL ViT	Qwen3-4B/8B	T, V	S, IR, QR, Nav	ReAct, HD	3A
SimpleDoc (2025)	EMNLP	Yes*	ColQwen-2.5	Mult. (Qw)	T, V	J, IR, QR	-	1A, SR, SV
ViDoRAG (2025)	EMNLP	Yes*	NV-Embed-V2, ColQwen2	Mult. (Qw/La/GPT)	T, V	J, IR	ReAct, HD	3A, SR, SV

Table 2.1 Survey of MP-VRDU models grouped by architecture. T = text; V = visual; L = layout. CoT = Chain of Thought; ReAct = Reason + Act; PE = Plan-Execute; QD = Question Decomposition; HD = Hierarchical / Coarse-to-Fine. S = Sparse; D = Dense; J = Joint; IR = Iterative Retrieval; QR = Query Reformulation; Nav = Navigation. nA = n agents; SR = Self-Refine; SV = Self-Verify; SC = Self-Consistency; SA = Sampling-Adjudication. OCR column: Yes* = not OCR-dependent; No* = OCR-derived supervision explicitly used during training. *Mult.* denotes a system evaluated on multiple backbones: Qw = Qwen; Ge = Gemini; GL = GLM; Cl = Claude; DS = DeepSeek; In = InternVL; Id = Idefics; La = LLaMA; Lv = LLaVA; Mi = Mistral; MM = MiMo; Ph = Phi; Pa = PaliGemma. '–' indicates not reported.

MP-VRDU (§2.4-§3.4.6). **(4)** We identify the principal open challenges that define the current frontier of the field with broader implications and future directions (deferred to Chapter 4).

2.2 Methodology and Scope

This chapter covers MP-VRDU systems published from 2023 (starting from MP-DocVQA's Hi-VT5 [6]) across major venues such as CVPR, ACL, EMNLP, and ICLR. Candidates were identified through keyword search over multi-page terms ("long document", "cross-page reasoning", "page-level retrieval") combined with task and method terms ("document visual question answering", "document

RAG", "long-context vision-language model"), supplemented by forward-citation sweeps anchored on established benchmarks such as MP-DocVQA, MMLongBench-Doc, and LongDocURL. We then manually screened for substantive multi-page contributions such as novel architectures, retrieval mechanisms, or training strategies. The dataset catalogue covers datasets used by the surveyed systems, plus a small number of relevant but underused ones. Rather than a systematic scoping review aiming at exhaustive retrieval, we prioritise representative coverage of the design space, since MP-VRDU is evolving rapidly and its terminology remains in flux. Full surveyed corpus details, search terms and inclusion criteria are available in Appendix A.2.

2.3 Multi-Page Architectural Overview

The defining problem in MP-VRDU is evidence management, and how a system manages cross-page evidence is fixed by its architectural design, which determines **how evidence is selected, preserved, and exposed for reasoning**. We therefore organise the field by architecture rather than by the OCR dependency used in prior surveys [12], into four families. *Page-to-document encoding models* jointly encode pages with cross-page representations (§2.3.1). *MLLM-centric adaptation models* adapt MLLMs to multi-page inputs within context and token budgets (§2.3.2). *Retrieval-augmented pipelines* retrieve relevant evidence before generation (§2.3.3). *Adaptive-trajectory pipelines* iteratively inspect evidence using an LLM controller (§2.3.4). Reported performance across the four families on principal MP-VRDU benchmarks is provided in Appendix A.3.1 (Table A.3).

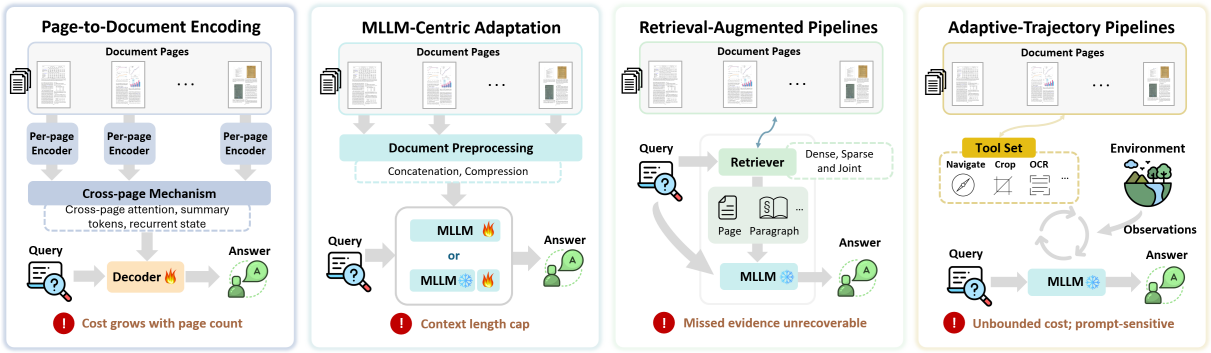


Figure 2.1 Illustration of MP-VRDU architectures. Snowflake indicates typically frozen; Fire indicates typically trained; Both indicates even split.

2.3.1 Page-to-Document Encoding Models

Page-to-document encoding models process each page through a shared layout-aware document encoder such as LayoutLMv3 [26] that fuses OCR text, layout, and visual features at the page level, then connect per-page representations through a document-level exchange mechanism within a single forward pass. The mechanism may take the form of per-page summary tokens [6], document-global tokens with local-global attention [44], recurrent memory states passed across pages [45], or sparse cross-page attention over long OCR sequences [43]. By preserving page identity, local layout, and reading order in the architecture itself, this design provides a strong inductive bias for multi-page structure and is well suited to documents with regular templates such as multi-page forms and business reports [6, 43]. The principal limitation is intrinsic to the design: every page is still encoded jointly in a single forward pass, and the per-page compression step that enables document-level exchange discards fine-grained evidence

as inputs scale. As the context windows of pretrained MLLMs have widened, the motivation for a dedicated cross-page encoder has weakened, and subsequent families absorb cross-page information into the backbone or select evidence before reading.

2.3.2 MLLM-Centric Adaptation Models

MLLM-centric adaptation models remove the dedicated cross-page encoder entirely and process the document inside a pretrained MLLM backbone whose context window absorbs cross-page information directly, with adaptation ranging from training only lightweight modules to fine-tuning the backbone itself. At the lightweight end, *projectors* that map per-modality features into the backbone's token space [34] and *adapters* that insert small trainable modules into a frozen backbone for parameter-efficient fine-tuning [33] keep most weights fixed, while visual-token compression further constrains token growth on long inputs to keep multi-page documents within the context budget [29]. At the heavier end, the MLLM backbone itself is unfrozen for continued pretraining or instruction tuning on document corpora, trading parameter efficiency for stronger document-specific grounding [36], with long-context tuning further improving performance on documents of moderate length [20, 35]. The family inherits the foundation MLLM's instruction-following and reasoning, avoids the engineering overhead of external retrieval or orchestration, and remains parameter-efficient to train. The cost of this design is that performance degrades on longer documents [47]: as most models in this category are OCR-free, reliable text recovery depends on high-resolution input that aggressive token compression readily compromises, and document structure such as section hierarchy and figure-caption linkage must be recovered implicitly from pixels rather than supplied through dedicated tokens [36, 49]. Combined with the attention-dilution problem [8], these limitations motivate subsequent families that move evidence selection out of the backbone altogether.

2.3.3 Retrieval-Augmented Pipelines

Retrieval-augmented pipelines decouple evidence selection from answer generation: a retriever first selects relevant evidence units, such as textual chunks, pages, or graph nodes, and the generator then produces an answer conditioned on this compact subset [37, 59]. This separation allows the pipeline to scale to multi-page inputs that exceed what MLLM-centric and page-to-document encoding models can hold in a single context, since inference cost depends on the retrieved content rather than on document length. The retriever may be a sparse lexical scorer, a dense embedding model, or a structure-aware index over a parsed representation of the document (§2.5.1), and the generator is typically a pretrained MLLM. The family spans single-pass retriever-generator systems [37] to multi-component pipelines that compose multiple LLM-instantiated "agents" in specialised roles such as modality-specific reading branches [40] or modality-fusion synthesis [65], but in either case the control flow is set at design time. Modular decomposition lets the retriever and generator be swapped independently, the retrieved evidence provides a partial audit trail for the answer, and the same interface extends to multi-document corpora without architectural change. Because the flow is fixed, evidence missed or poorly ranked at retrieval is unrecoverable at generation, questions that require aggregating multiple weakly relevant pages frequently fail [55], and upstream errors in chunking, OCR, or modality alignment propagate directly into the answer.

2.3.4 Adaptive-Trajectory Pipelines

Adaptive-trajectory pipelines instantiate the agent paradigm characterised by Plaat et al. [79] and Wang et al. [80], in which one or more LLM controllers reason, act on an external environment, and condition each subsequent action on observed outcomes rather than executing a plan fixed in advance [38, 69].

Unlike the fixed-role agents of §2.3.3, the controller's trajectory is not predetermined: it selects each evidence-gathering action at inference time, with actions surfacing as explicit calls to external retrievers, parsers, or OCR engines [71, 78], or as structured responses the framework interprets as a re-retrieval or as navigation over a pre-built document representation [77]. A controller might issue a refined query when the first retrieval returned irrelevant pages, request a higher-resolution view of a page surfaced by an earlier pass, or follow a cross-reference from one section to another. ReAct [81] is the most common controller realisation (§2.5.2), though other variants also appear. Because the trajectory is constructed rather than fixed, compute scales with question difficulty rather than document length, and the full action sequence becomes an inspectable audit trail. The cost of that flexibility is that inference cost is difficult to bound in advance, controllers are sensitive to prompt phrasing and may commit early to wrong evidence and continue confidently down the wrong trajectory without surfacing the error, stochastic generation complicates reproducibility, and the system still inherits the recall ceiling of every retriever, parser, or scoring module it consults.

2.4 MP-VRDU Multimodal Representation

Every evidence-management decision is enacted over a specific modality, and its cost depends on how that modality is represented. While Section 2.3 categorises MP-VRDU systems by where cross-page evidence is processed, this section analyses the problem from a modality perspective. MP-VRDU systems must locate sparse evidence across long multimodal contexts, forcing each modality to trade *page coverage* (how much of the document the system can attend to) against *reading fidelity* (how accurately each attended region is interpreted), since long-context models attend poorly to evidence buried in the middle of long inputs [8]. The single-page mechanisms inherited from prior work, including OCR-text injection, per-image patch encoding, and per-token layout signals, are reviewed in Section 1.3.4; here we concentrate on the multi-page pressures that text (§2.4.1), visual (§2.4.2), and cross-page structural (§2.4.3) evidence introduce.

2.4.1 Text Modality

Text in MP-VRDU presents four bottlenecks. **Volume** across pages routinely exceeds what the MLLM can process in full. This is addressed either by compressing all pages into a fixed budget through page summaries, memory tokens, or long-context encoders [29, 43, 45], or by indexing chunks and passing only selected text to the reader [37, 60, 66]. The first approach fixes compression before the question is seen, so evidence absent from the compressed representation cannot be recovered later in the pipeline; the second makes evidence missed at retrieval unrecoverable at the reading stage, and broadening the retrieved set merely shifts the cost from omission to a related failure mode known as *attention dilution*, in which the model spreads its attention across too many candidate passages and loses focus on the answer-bearing ones [8]. **Continuity** breaks when paragraphs, tables, and section context span page boundaries. This is preserved by recurrent state propagation that carries page context forward across the document [45], structure-aware chunking that groups text under recovered section headings rather than along sliding-window boundaries [64], and adjacent-page expansion at retrieval time that pulls in neighbouring pages when a candidate is surfaced [52]. Each restores local cross-page context at the cost of broader query relevance, since the additional context is selected by proximity rather than by question. **Noise** accumulates as OCR and parsing errors compound across pages: a recognition error rate that is tolerable on a single page multiplies over a long document, and because extracted text is the retrieval index in most pipelines, these errors degrade retrieval as well as reading, surfacing wrong

pages or burying the right ones. It is dampened upstream through layout-aware parsing pipelines that reduce recognition error, or downstream by treating low-confidence tokens as soft cues the reader can discount rather than as ground truth [49], yet how noise propagates through multi-page retrieval indexes remains largely uncharacterised. **Heterogeneity** in text density varies sharply across pages within a single document, yet receives the least attention: most systems apply fixed token budgets and fixed top-k uniformly across pages and queries, over-compressing dense pages while wasting capacity on sparse ones. Adaptive selection that scales the returned set with the per-query score distribution [51] only begins to address this variability, and no current system varies its per-page text budget within a single document.

2.4.2 Visual Modality

Visual evidence preserves signals that text extraction loses, but its cost structure introduces three bottlenecks of its own. **Resolution** cannot be maintained at full document length, since token cost scales with page count multiplied by per-page resolution, forcing a choice between compression and selective high-fidelity reading. Token-reduction modules and adaptive sub-image budgets [29, 35, 47] keep every page available in context but leave fine-grained content, including small text, table cells, and chart annotations, vulnerable to downsampling. Inspecting only a retrieved subset at high resolution [73, 77] preserves detail at the cost of inheriting the selector's recall ceiling: pages not surfaced by the retriever are never read, regardless of the relevance of their content. **Multi-signal competition** arises because the same visual budget must support text-bearing pixels, non-textual evidence such as charts, figures, and handwriting, and layout cues within a single page. Visual RAG systems, which retrieve over rendered page images rather than over extracted text, encode pages directly so all signals share one representation [37, 59]; the page becomes a single evidence unit that cannot be decomposed at retrieval time. Element-level cropping of tables, figures, and charts [73] preserves fine structure at the cost of detector dependence and separation from the surrounding explanatory text on which interpretation often depends. A third route delegates text recovery to an OCR pipeline so the visual budget can prioritise non-textual evidence [49], which avoids the budget competition at the cost of inheriting OCR fragility on handwriting and complex layouts. **Heterogeneity** in visual complexity varies sharply across pages within a single document, yet receives the least attention: most systems apply uniform resolution and sub-image budgets regardless of whether a page is a sparse cover sheet or a dense table-laden appendix, and evidence-guided allocation that varies the visual budget by content [46] only begins to address per-page and per-query variability.

2.4.3 Cross-Page Structure Modality

Structure turns isolated page observations into document evidence. Per-token layout remains relevant at the multi-page scale, with specialised tokens encoding spatial geometry within each page [33], but the MP-distinctive difficulty is relationships whose scope crosses pages. **Hierarchy and reading order** is the foundational problem: sections, lists, columns, and captions routinely span page boundaries, and the same evidence may serve different reasoning paths depending on which parent section it sits under. Explicit parsers expose this structure as outlines, trees, or hierarchical indices that retrieval can navigate [38, 64], while OCR-free backbones attempt implicit recovery from page images through layout-aware pretraining objectives [29, 35]. The unresolved gap is robustness: parser errors in heading detection or table-boundary segmentation propagate directly into every downstream retrieval and navigation step, and implicit recovery depends on high-resolution input that the compression discussed in §2.4.2 may have already removed. **Cross-page references** demand resolving pointers such as "see Figure 3" or "as shown in Section 2" to their referents on other pages, rather than ranking pages by surface simil-

arity. Graph-based methods represent these references explicitly as edges between document elements, through knowledge graphs over passages and structural nodes [61] or page graphs with semantic-logical edge scoring [55], improving navigability at the cost of edge quality bounded by extraction heuristics or LLM judgement at construction time. **Spanning elements** are the structural form of the text-continuity problem from §2.4.1: multi-page tables, section-spanning arguments, and theorem-proof or footnote-reference linkages require evidence units larger than a page but smaller than the document. Hierarchical retrieval that routes through section-level units before drilling into chunks [62] and cross-page retrieval that preserves parent context [52] address this partially, but the boundary between useful context and irrelevant neighbours remains hard to set without per-query supervision.

2.5 Inference-Time Adaptation Strategies

Inference-time adaptation improves MP-VRDU performance through changes to the inference pipeline rather than to model parameters: a frozen LLM or vision-language backbone is paired with off-the-shelf retrievers, structural parsers, prompted reasoning loops, or multi-agent orchestrations, with no gradient updates required. These techniques are predominantly used in the retrieval-augmented and adaptive-trajectory frontier, though prompting strategies such as chain-of-thought reasoning also apply to MLLM-centric and page-to-document models that operate over the assembled document directly. We organise these strategies by the evidence-management decision they implement: selecting evidence through retrieval and navigation (§2.5.1), reasoning over assembled evidence (§2.5.2), and iteratively acquiring evidence or distributing acquisition across multiple controllers (§2.5.3).

2.5.1 Retrieval and Navigation

Retrieval narrows long documents to compact evidence units for downstream reasoning. Two broad families have emerged, distinguished by whether passages are scored independently against the query or navigated through explicit document structure.

Similarity-based retrieval. Similarity-based methods score each passage against the query independently and return the top-ranked subset. Sparse lexical signals such as BM25 persist as baselines or auxiliary channels, while dense retrieval in MP-VRDU has consolidated around vision-language late-interaction encoders such as ColPali [82], which extends the late-interaction paradigm of ColBERT [83] to page images by producing token-level query vectors and patch-level page embeddings, then scoring a query-page pair by taking, for each query token, the maximum cosine similarity over all page patches and summing the result (Figure 2.2). This *MaxSim* aggregation preserves fine-grained alignment between specific query terms and specific page regions, critical when a single question token may need to align with one figure caption, table cell, or paragraph buried in a long document. Granularity varies from chunks to whole pages, with adaptive selectors adjusting the per-query retrieval count when evidence density is uneven across the document [51]. A second reranking stage commonly refines an initial dense recall through LLM-based scoring over candidate passages or page summaries [60, 77], optionally under persistent working memory that carries cross-query context across iterations [77]. Independent scoring handles paraphrase and cross-modal queries well, but cannot follow cross-page references or multi-hop dependencies that no single passage carries.

Relation-aware retrieval. Relation-aware methods retrieve along explicit document structure rather than scoring passages independently. Tree-based variants extract section hierarchies and outlines and

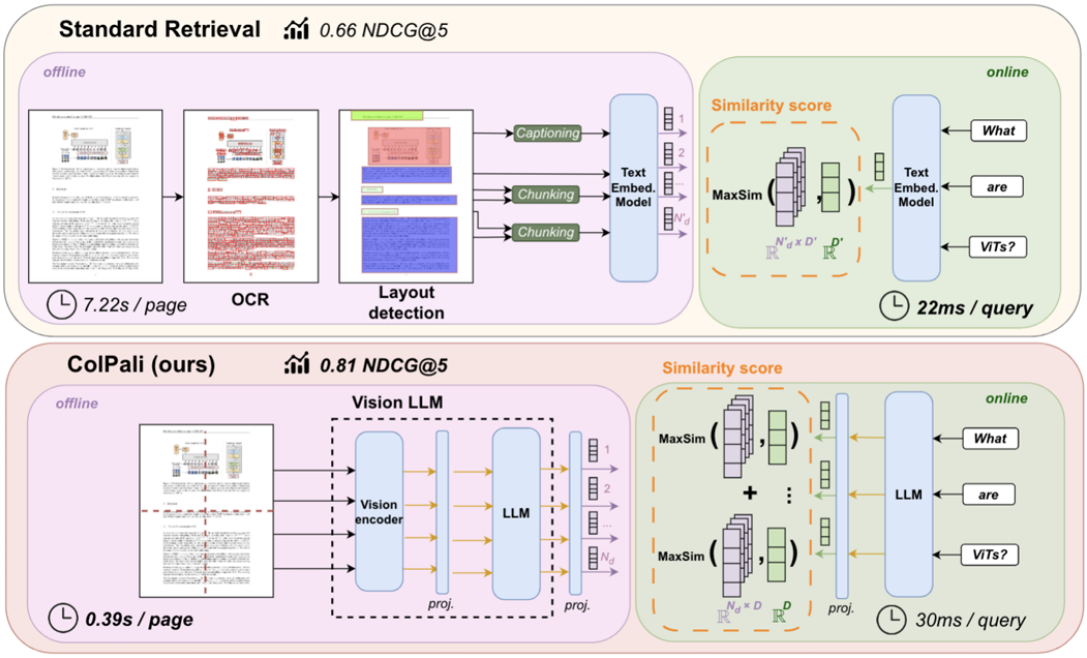


Figure 2.2 ColPali late-interaction retrieval for visually rich document pages. Query-token embeddings are matched against patch-level page embeddings using *MaxSim*, producing a relevance score that preserves fine-grained alignment between query terms and local page regions.

group passages along semantic boundaries so retrieved chunks preserve parent context, with some routing queries through dual in-page and cross-page indices to recover both local and global evidence [38, 62, 64]. Graph-based variants treat headings, entities, and cross-references as graph nodes with typed edges, so controllers or retrievers can traverse from one passage to related ones rather than relying on similarity alone; extensions add logical relevance scoring or query-conditioned graph construction to bias traversal toward inferentially connected nodes [55, 61, 63]. Relation-aware methods recover references that similarity scoring misses, at the cost of construction overhead, dependence on parser quality, and traversal cost that grows with the multi-hop count required to answer the question.

2.5.2 Reasoning Strategies

Reasoning strategies act over a single fixed evidence buffer assembled prior to the reasoning call. The buffer may be the full document representation, a retrieved subset, or the output observed at one step of an agentic loop; the unifying property is that the reasoning step itself does not acquire new evidence, which makes these strategies independent of architectural family. They range from purely prompted interventions to strategies requiring changes to decoding, sampling, or control flow, but none updates model parameters. We cover the mainstream reasoning paradigms below.

Sequential Reasoning. A single reasoning thread proceeds linearly, optionally revising itself as it goes. *Chain-of-Thought* (CoT) prompting [84] elicits intermediate reasoning before the final answer, which in MP-VRDU binds evidence from different pages step by step, mitigates attention dilution over long inputs, externalises the reasoning trace for verification, and decomposes multi-hop questions into single-hop steps [39, 47]. Because it operates only over the assembled buffer, its contribution lies in how faithfully that evidence is used rather than in recovering evidence the buffer omits: missing evidence cannot be reasoned back into existence. *Self-Reflection* extends the single thread with iterative revision,

having the model audit or refine its own answer against the buffer, optionally conditioned on feedback from a verifier or critic step [77, 78, 85]. This reduces single-pass error at the cost of additional rounds and a self-evaluation signal whose reliability bounds the gains.

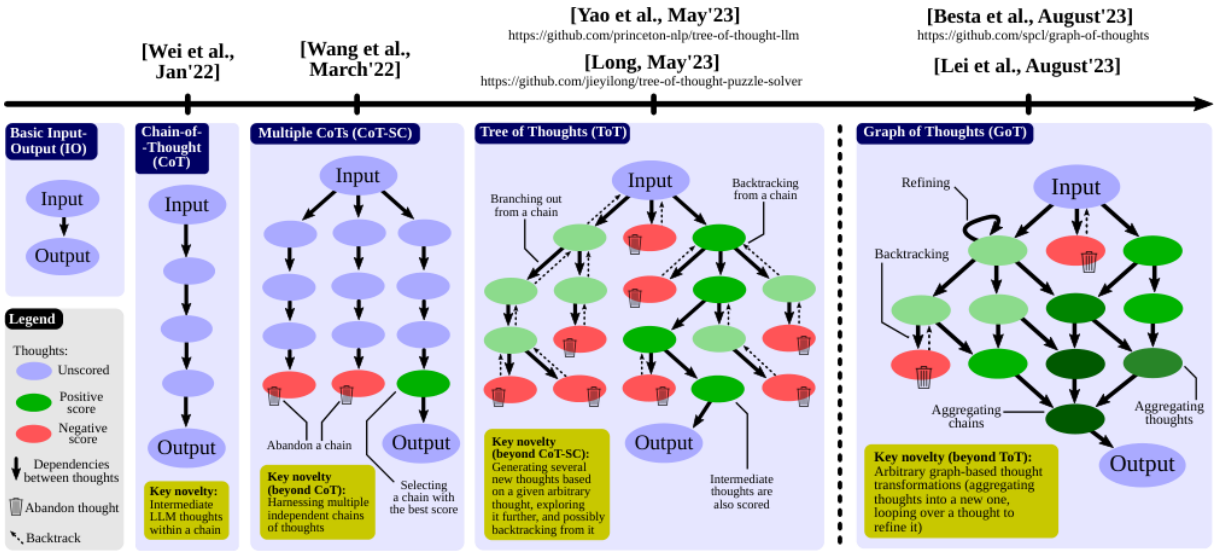


Figure 2.3 Evolution of reasoning topologies used in prompting schemes. Extracted from Besta et al. [86]. Only CoT is used pervasively in MP-VRDU; this diagram is for illustration purposes only.

Parallel Exploration. Several reasoning paths are produced independently and combined into one answer. *Self-Consistency* [87] samples multiple chains and selects the majority answer, trading inference cost for robustness against single-chain failures, with sampling-adjudication variants replacing the majority vote with an LLM judge that compares reasoning paths and selects the most consistent conclusion [73], and parallel-pipeline variants enforcing consistency across modality-specific chains rather than across samples of one chain [68]. *Tree-of-Thoughts* (ToT) [88] instead explores reasoning branches with backtracking, but has seen limited uptake in MP-VRDU, where evidence sparsity rather than search-space branching is the dominant difficulty. These patterns are independent of architectural family: they can wrap a single retrieval-augmented forward pass or operate inside one step of an adaptive-trajectory loop, in which case the loop and the within-step pattern compound their effects on robustness and cost.

2.5.3 Agentic Methods

Agentic methods distribute the MP-VRDU workflow across one or more LLM-instantiated controllers, either iterating evidence acquisition under inference-time control or coordinating across specialised roles. We note that *agentic* is used ambiguously in the MP-VRDU literature, covering both systems whose controller selects actions at inference time, following the framing by Plaat et al. and Wang et al., and systems composed of multiple LLM-instantiated agents in fixed roles; we treat both senses here, following the architectural distinction in §2.3.3–§2.3.4.

ReAct and Tool-Augmented Reasoning. ReAct prompting [81] augments the action space with language thoughts, interleaving thought-action emissions in a single stream whose observations feed the next step, letting a single controller acquire evidence during inference rather than reason over a fixed buffer. The action space takes two forms in MP-VRDU. Document-side tool calls invoke external retrievers, parsers, or OCR engines, with each round concentrating on a chosen page or region [38, 70],

lifting the single-step recall ceiling of retrieval-augmented pipelines at the cost of tool latency and compounding error from early wrong actions. Structured navigation instead operates over a pre-built document representation, with the controller selecting which node, section, or page to inspect next [69], which trades dependence on external tool fidelity for dependence on the quality of the representation built at index time. Adaptive-trajectory systems that lack the thought-action schema, such as those built on reflection-style iterative refinement [77], share the iterative control regime without being ReAct in the strict sense.

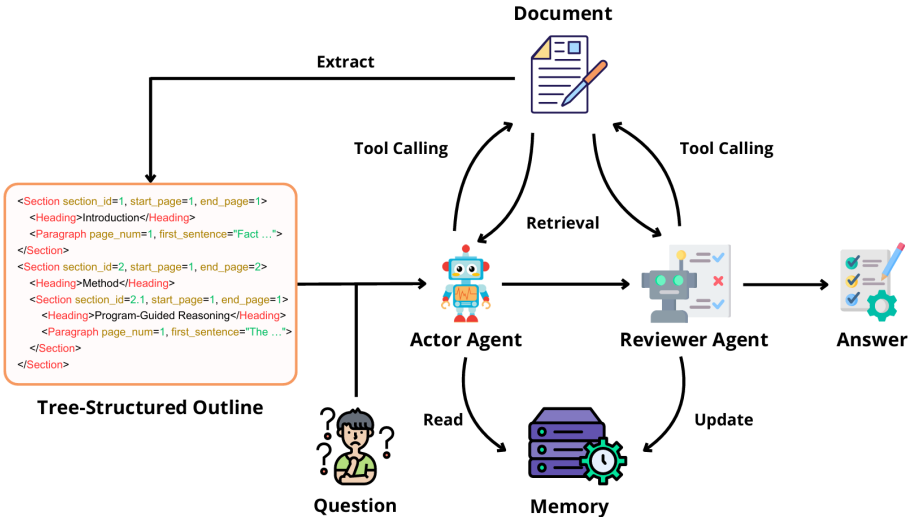


Figure 2.4 Architecture of DocAgent. Included for illustrative purposes. Extracted from Yang et al. [38]. "We first construct the document outline as a compact XML tree structure. An actor agent processes the outline and question as input, leveraging retrieval tools to extract relevant content for answering the question. Next, an evaluator agent reviews the initial response. If a correction is necessary, the actor agent's memory is updated through reflection, facilitating knowledge transfer across tasks."

Agentic Decomposition. Agentic decomposition distributes the workflow across multiple controllers whose interaction pattern, rather than a single monolithic controller, drives evidence gathering and answer synthesis. Each controller operates on a narrower input and clearer objective than a single backbone tasked with the entire workflow, mitigating the cognitive overload that arises when one model must locate, read, verify, and answer over a long multimodal document [71]. Two patterns recur. *Modality-specialised* decomposition assigns separate controllers to text and visual evidence and adds a synthesis controller that reconciles their outputs, preserving single-modality reading fidelity at the cost of a reconciliation step that lacks access to the original document [40, 65]. *Role-specialised* decomposition assigns controllers to procedural stages such as planning, execution, verification, and synthesis, with explicit routing between stages when a downstream controller detects an error [38, 71]. Separating judgement from generation reduces the cognitive blind spots of self-correction, but introduces inter-controller compatibility costs and a recall ceiling bounded by upstream stages.

2.6 Training Strategies

VRDU training follows a stepwise pipeline established in the single-page setting [12]: self-supervised pretraining instils document reading through objectives such as masked information modelling and cross-modality alignment, instruction tuning aligns outputs with user queries, and supervised fine-tuning adapts

the model to target tasks. MP-VRDU inherits this pipeline but concentrates its effort at the later stages, since document-scale pretraining corpora remain scarce and most systems adopt backbones already pretrained on single-page data, with few exceptions targeting cross-page objectives directly [29, 36]. Fine-tuning instead carries the weight, scaled by LVLM-synthesised supervision (§3.4.6), with reinforcement learning increasingly layered above it. Training therefore internalises the evidence-management decisions of §2.5 into model parameters, supplying capabilities that prompting cannot: cross-page evidence selection (§2.6.1), document-scale reasoning over assembled evidence (§2.6.2), and reliable agentic acquisition (§2.6.3). Two cross-cutting techniques recur across all three (§2.6.4).

2.6.1 Trained Retrieval Components

Off-the-shelf encoders such as ColPali and ColQwen [82] score each page independently against the query, so retrieved results are dominated by local query-page similarity rather than by document-level context. Retriever training has therefore explored three complementary directions. *Modality-alignment* methods learn visual document embeddings that remain aligned with OCR-derived text, enabling page images to be retrieved without the storage and latency cost of the late-interaction patch-level scoring discussed in §2.5.1 [59]. *Cross-page-fusion* methods enrich each page representation with neighbouring-page context, typically through a lightweight head trained with parameter-efficient methods such as LoRA [89], a tuning approach that adds small trainable matrices to frozen weights; the result makes evidence distributed across adjacent or related pages retrievable as a unit rather than as a set of independently-ranked passages [52]. *Logical-relevance* methods train graph-based scorers to rank pages not only by embedding similarity but also by their role in cross-page reasoning paths, turning retrieval into a learned multi-hop walk rather than a single-step similarity match [55]. The three directions are not interchangeable: modality-alignment improves retrieval throughput, cross-page-fusion improves recall on adjacent evidence, and logical-relevance improves recall on inferentially-connected evidence.

2.6.2 Trained Reasoning over Assembled Evidence

Prompted reasoning leaves quality sensitive to prompt design and base-model behaviour, motivating approaches that train the reasoning component directly so the model produces structured reasoning trajectories on its own rather than relying on a prompt schema imposed at inference time. Three patterns recur. *Trace-imitation* variants supervise the backbone on demonstration chains wrapped in fixed schema tags such as `<think>`, `<evidence>`, and `<answer>`, optionally extended across procedural stages such as plan, execute, integrate, and verify; staged training pipelines further refine through Direct Preference Optimisation (DPO), a method that trains on pairs ranked by preference rather than scalar rewards to prefer faithful chains over hallucinated shortcuts [50]. *Reward-driven* variants optimise the reasoning policy directly with Group Relative Policy Optimisation (GRPO), a reinforcement-learning method that compares grouped rollouts to compute relative rewards; the composite reward typically combines answer accuracy, evidence-page recall, and format compliance, and the policy is often warm-started from a trace-imitation stage rather than learned from scratch [39]. The two are therefore sequential rather than alternative, with reward-driven training refining what imitation alone cannot supply. *Uncertainty-calibration* variants take a different route, training an aggregation head that selects across candidate outputs by calibrated uncertainty rather than internalising the reasoning chain itself, which yields a learned self-consistency signal that scales to long documents under tight parameter budgets and avoids the cost of sampling multiple chains at test time [49].

2.6.3 Trained Agentic Methods

Prompt-only agentic control is unreliable on smaller open-weight backbones, which cannot consistently follow ReAct schemas, select among tool calls, or terminate trajectories reliably. Two patterns recur. *Trajectory-distillation* variants clone agentic trajectories from a stronger teacher model through supervised fine-tuning on think-action-observation turns, with teacher rollouts collected in closed-loop interaction and filtered for format validity, answer correctness, and evidence coverage; this trains the smaller model to match the trajectory shape of the larger teacher rather than learning the underlying policy from scratch, with some variants additionally conditioning tool selection on a learned representation of the agent's current goal so that action choices reflect the trajectory's evolving intent rather than only its most recent observation [75]. *Reinforcement-learned* variants optimise the agentic policy directly against trajectory-level rewards that combine answer correctness, evidence recall, and tool-use shape, often pairing a supervised warm-start with GRPO over a constrained action space such as retrieval and page-fetch calls; learned termination is a recurring concern, with entropy-based stopping letting the trained model commit to an answer once its prediction confidence stabilises rather than running for a fixed number of rounds [69]. Multi-agent variants extend this further by allocating sample budgets per controller across procedural roles such as planner, executor, judge, and answerer rather than running every controller at a fixed budget, spending more compute on the role that the question's difficulty most stresses [71]. The surveyed evidence does not cleanly separate distillation from reinforcement learning, since reported comparisons confound the training objective with backbone, reward design, and trajectory budget.

2.6.4 Cross-Cutting Training Techniques

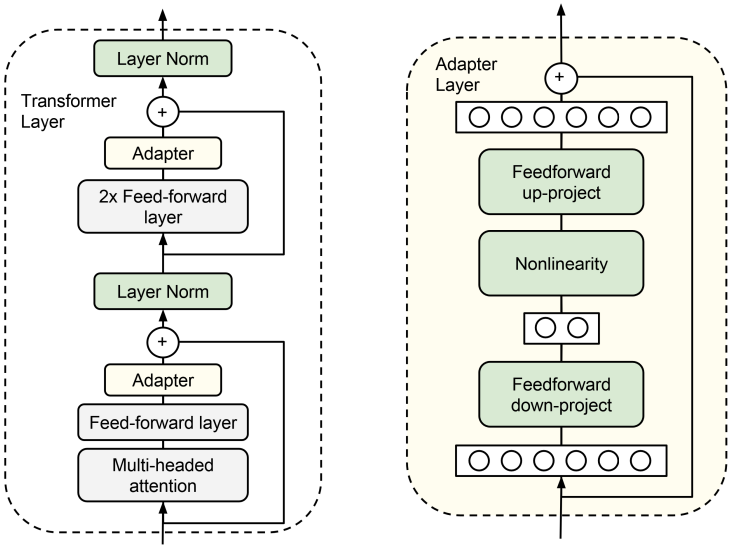


Figure 2.5 Adapter-based parameter-efficient tuning. Included for illustrative purposes. Two adapter modules are inserted into each transformer layer (left); each is a small bottleneck that down-projects to a low dimension, applies a nonlinearity, up-projects back, and adds the result to its input through a residual connection (right). During training the backbone weights are frozen and only the adapter modules and layer norms are updated. Adapted from Houlsby et al. [90].

Two techniques recur across retriever, reasoner, and orchestration policy training, independent of which component is being optimised. **Parameter-efficient tuning** is the dominant adaptation strategy: full backbone fine-tuning is prohibitive at long-document context lengths and risks erasing pretrained

capabilities, so most systems freeze the backbone and train lightweight modules through LoRA matrices, projection heads, or task-specific adapters [33, 52, 60, 75]. The adapted module varies by target: layout-token modules inject structural information [33], projection heads align frozen retrievers with downstream readers [52], and LoRA matrices adapt reasoning or trajectory behaviour without touching backbone weights [60, 75]; the choice between parameter-efficient and full fine-tuning is more often a compute-budget decision than a target-component one. **Multi-stage curriculum training** addresses a different concern: several systems converge on curricula that start on single-page or local objectives before introducing document-scale supervision, exploiting the fact that text reading and layout understanding transfer from single-page data while cross-page integration must be learned separately [29, 49]. Later stages typically add streaming, abstention, or evidence-grounding objectives once foundational reading capabilities are established, with the longest schedules layering five or more stages from image-OCR alignment through to streaming-abstention calibration. Their recurrence across otherwise dissimilar systems indicates that cross-page integration does not emerge from single-page supervision alone.

2.7 Datasets

Multi-page documents impose a structural constraint absent from the single-page setting: per-page annotation does not scale to the page counts typical of MP-VRDU, and multi-page corpora are scarce because long PDFs are harder to source, license, and parse than single pages. This constraint shapes the three roles a dataset can play, pretraining (PT), supervised fine-tuning (SFT), and benchmarking (BM), differently from SP-VRDU, and several datasets occupy more than one role across different systems.

2.7.1 Pretraining

Pretraining is the exception rather than the norm in MP-VRDU. Only a handful of page-to-document and MLLM-centric systems pretrain or continue-train a backbone at all; retrieval-augmented and agentic pipelines inherit pretrained vision-language backbones and skip the stage entirely. Two factors explain its scarcity. Document-scale pretraining is computationally expensive, and genuinely multi-page corpora are few, so most systems that pretrain fall back on single-page archives or repurpose single-page data. Early work drew on single-page OCR archives such as IIT-CDIP [113] and OCR-IDL [114], which supply text-recognition supervision but no cross-page structure. Genuinely multi-page corpora have appeared only recently: DocStruct4M [20] mixes single- and multi-page samples with structure-aware parsing supervision, while the purely multi-page MP-DocStruct1M [29], Doc-750K [36], and PaperPDF [66] support continued pretraining over genuine multi-page inputs. Where multi-page pretraining does occur, its objective has shifted from text recognition toward structure-aware supervision, but the corpus as a whole shows that most cross-page capability is instilled at fine-tuning rather than pretraining.

2.7.2 Supervised Fine-Tuning

Because per-page human annotation does not scale, supervised fine-tuning has converged on two strategies, often combined within one system. The first aggregates publicly released question-answer collections into broad instruction mixes spanning heterogeneous question and answer types [29, 34, 36, 47]. The second curates a bespoke corpus for the system under study, typically synthesised by a strong vision-language model that writes question-answer pairs, reasoning traces, or agent trajectories over scraped PDFs [50, 55, 59, 66, 75]. Synthetic generation has become the dominant scaling path for fine-tuning data, and its underlying PDF pools are typically drawn from the same scientific archives, slide

Dataset	Pages	Type	Venue	Domain	# Docs	Avg Pages	# Q/A	Multi/S Hop
DocVQA (2021)	SP	PT+BM	WACV	Industry	12767	–	50,000	S
FeTaQA (2022)	SP	PT+BM	TACL	Wikipedia tables	10330	–	10,330	M
InfographicVQA (2022)	SP	PT+BM	WACV	Infographics	5,485	–	30,035	M+S
PDF-VQA Task A (2023)	SP	PT+BM	ECML-PKDD	Biomedical	–	–	81,085	S
PDF-VQA Task B (2023)	SP	PT+BM	ECML-PKDD	Biomedical	–	–	53,872	S
TabFact (2020)	SP	PT+BM	ICLR	Wikipedia tables	16573	–	118,275	M+S
TextVQA (2019)	SP	PT+BM	CVPR	Scene-text images	28408	–	45,336	S
VisualMRC (2021)	SP	PT+BM	AAAI	Webpages	10197	–	30,562	M+S
WikiTableQuestions (2015)	SP	PT+BM	ACL	Wikipedia tables	2108	–	22,033	M
ChartQA (2022)	SP	BM	ACL	Charts	20882	–	32,719	M+S
DocGenome (2024)	MP	PT+BM	NeurIPS	Academic Paper	500000	13	3000	M+S
DuReaderVis (2022)	MP	PT+BM	ACL	Cross-domain	158000	–	15,000	S
Kleister-Charity (2021)	MP	PT+BM	ICDAR	Financial	2778	22.19	21,612	S
Kleister-NDA (2021)	MP	PT+BM	ICDAR	Legal	540	5.98	2,160	S
M-LongDoc (2025)	MP	PT+BM	EMNLP	Cross-domain	180	210.8	851	S
MMVQA (2024)	MP	PT+BM	IJCAI	Biomedical	3146	9.6	262,928	M+S
MP-DocVQA (2023)	MP	PT+BM	PR	Industry	5928	8.27	46,176	S
MultiHiertt (2022)	MP	PT+BM	ACL	Financial reports	2513	~2.5	10,440	M
PaperPDF (2026)	MP	PT+BM	preprint	Academic Paper	–	–	1,110,000	M+S
PDF-VQA Task C (2023)	MP	PT+BM	ECML-PKDD	Biomedical	1147	~10	5,653	M
QASPER (2021)	MP	PT+BM	NAACL	Academic Paper	1585	~8	5,049	M
SlideVQA (2023)	MP	PT+BM	AAAI	Presentations	2619	20	14,484	M+S
MP-DocStruct1M (2025)	MP	PT	ACL	Cross-domain	–	–	1,113,259	S
OpenDocVQA (2025)	MP	SFT+BM	CVPR	Cross-domain	–	–	–	M+S
CoR-Dataset (2026)	MP	SFT	preprint	Cross-domain	–	–	26,088	M+S
Doc-750K (2025)	MP	SFT	CVPR	Academic Paper	–	–	758,000	M+S
MHDocVQA (2025)	MP	SFT	CVPR	Cross-domain	–	9.5	38,020	M
MP-DocReason51K (2025)	MP	SFT	ACL	Cross-domain	–	–	51,754	M+S
BRIDGE (2026)	MP	BM	preprint	Academic Paper	262	–	11,857	M
DocBench (2024)	MP	BM	preprint	Cross-domain	229	–	1,102	S
DUDE (2023)	MP	BM	ICCV	Cross-domain	5019	5.72	41,541	M+S
FetaTab (2024)	MP	BM	NeurIPS	Wikipedia Articles	871	~11	1,016	S
LongDocURL (2025)	MP	BM	ACL	Cross-domain	396	85.6	2,325	M+S
M3DocVQA (2025)	MP	BM	CVPR	Wikipedia Articles	3368	12.2	2,441	M+S
MM-NIAH (2024)	MP	BM	NeurIPS	Synthetic	–	–	16,486	M+S
MMDocIR (2025)	MP	BM	ACL	Cross-domain	313	65.1	1,658	M+S
MMLongBench (2025)	MP	BM	NeurIPS	Cross-domain	–	–	13,331	M+S
MMLongBench-Doc (2024)	MP	BM	NeurIPS	Cross-domain	135	47.5	1,082	M+S
PaperTab (2024)	MP	BM	NeurIPS	Academic Paper	307	~11	393	M+S
PaperText (2024)	MP	BM	NeurIPS	Academic Paper	1087	~11	2,804	M+S
ViDoSeek (2025)	MP	BM	EMNLP	Cross-domain	–	–	1,200	M+S
VisDoMBench (2025)	MP	BM	NAACL	Cross-domain	1277	19.11	2,271	M+S
VisR-Bench (2025)	MP	BM	ICLR	Cross-domain	226	–	471	M+S
DocStruct4M (2024)	MP+SP	PT+SFT	EMNLP	Cross-domain	–	–	4,000,000	M+S
TAT-DQA (2022)	MP+SP	PT+BM	ACM	Financial reports	2758	1.33	16,558	M
InstructDoc (2024)	MP+SP	SFT	AAAI	Cross-domain	–	–	–	M+S
Leopard-Instruct (2025)	MP+SP	SFT	TMLR	Cross-domain	–	–	925,000	M+S

Table 2.2 Datasets referenced in the MP-VRDU survey. **Pages**: document scope (MP = Multi-Page, SP = Single-Page, MP+SP = Mixed). **Type**: dataset role – PT (Pretraining), SFT (Supervised Fine-Tuning), BM (Benchmark), and combinations. **Hop**: question-hop type (Single, Multi, or both). '–' indicates not reported.

repositories, and benchmark sources that also supply evaluation. Layered above this, a growing body of work applies reinforcement or preference learning with rewards that credit correct evidence-page citation, faithful chain-of-thought structure, or appropriate tool use rather than final-answer accuracy alone [39, 46, 50]. These reinforcement pools remain small relative to the supervised mixes that precede them, indicating that reinforcement signals shape behaviour, instilling evidence grounding, output formatting, and multi-step reasoning, rather than scaling the data itself.

2.7.3 Benchmarking

Benchmarks have evolved through several tiers that accumulated rather than replaced one another. Single-page extractive evaluation on DocVQA [5], InfographicVQA [3], and ChartQA [4] persists as a per-page baseline, and mid-length corpora such as MP-DocVQA [6], DUDE [7], and SlideVQA [104] extend the same extractive formats into a few-page setting. Long-document multimodal benchmarks then introduced documents whose total length routinely exceeds backbone context windows and

whose evidence requires cross-page integration over mixed text, tables, charts, and figures, including MMLongBench-Doc [9], LongDocURL [10], DocBench [106], and M-LongDoc [100]. Retrieval-oriented benchmarks such as UDA-Benchmark [107] and DocLens [73] additionally score evidence-page selection and layout-level localisation alongside answer accuracy, and the most recent additions push into cross-document settings, including M3DocRAG [37], ViDoRAG [78], and VDocRAG [59]. Despite this proliferation, evaluation in practice has consolidated around a small set of long-document and mid-length benchmarks that recur in most reported comparisons. That the same scarce pool of source PDFs supplies both synthetic fine-tuning data and evaluation is a contamination risk we return to in §4.2.4.

2.8 Conclusion

This chapter established MP-VRDU as a distinct document-understanding problem organised around evidence management. It introduced a taxonomy of four architectural families and surveyed the multimodal representations, inference-time strategies, training approaches, datasets, and evaluation practices used across the field. Across these dimensions, the literature consistently shows that long-document performance depends not only on backbone capability, but on how evidence is preserved, selected, compressed, and integrated across pages. The broader challenges and future research directions that emerge from this survey are discussed in Chapter 4.

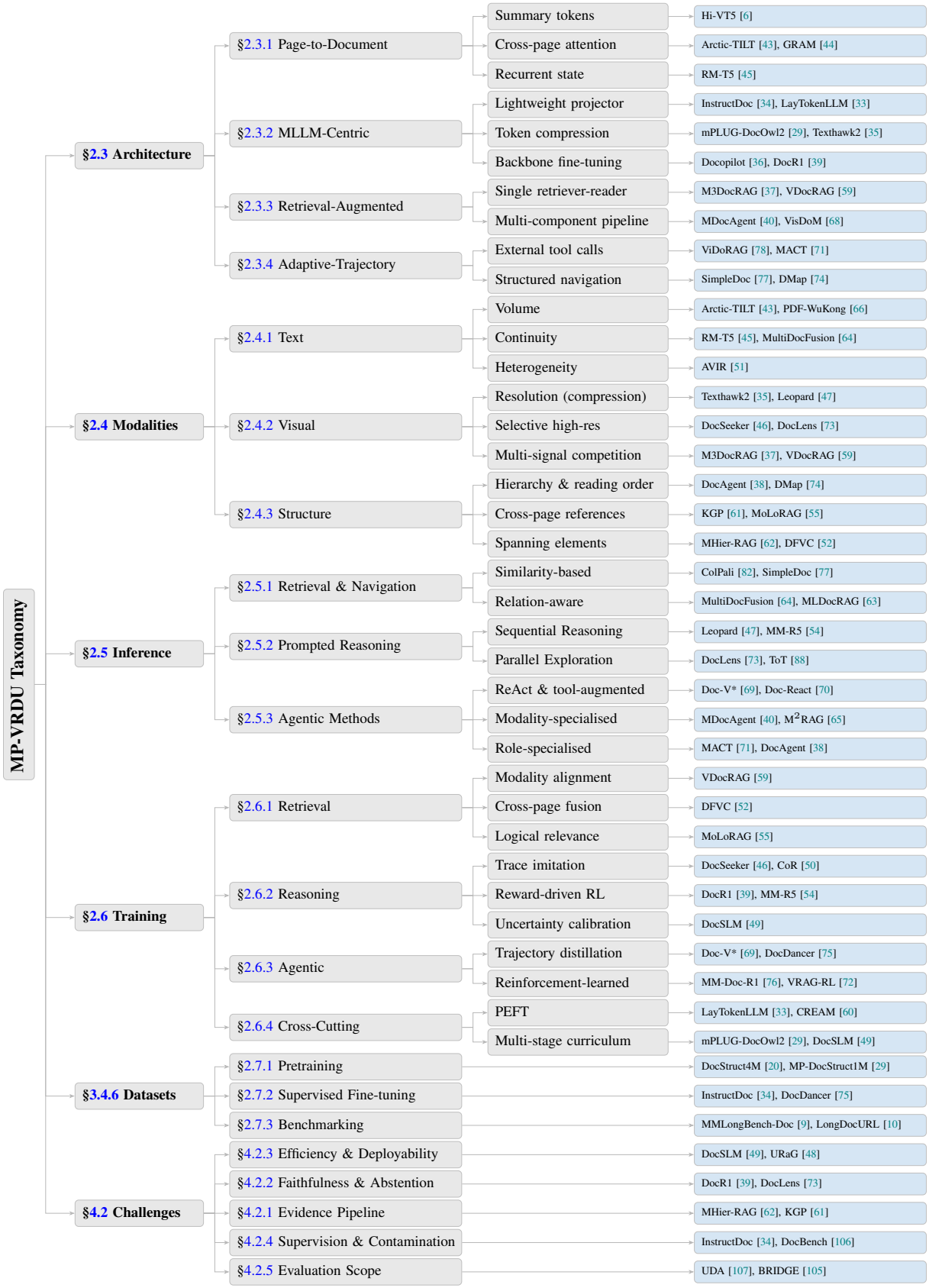


Figure 2.6 MP-VRDU taxonomy.

Chapter 3

Empirical Attribution of Failure Modes

This chapter is adapted from **Locating Failure in Multi-Page Visually Rich Document Understanding Systems: An Empirical Attribution** [115], by **Lewei Xu**, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, and Wei Liu, *currently under review at ACL ARR*. The first author was responsible for formulating the failure taxonomy and attribution framework; designing the experimental methodology; implementing and running all experiments; analysing and interpreting the results; and preparing the full manuscript, including the main text, methodological details, empirical analysis, figures and tables, and appendix material. **Related work and motivation from the paper is provided in §3.2; full methodology details are reported §3.4 per honours thesis guidelines.**

3.1 Introduction

Building effective multi-page visually rich document understanding (MP-VRDU) systems requires understanding not only how well they perform, but also where and why they fail, particularly when privacy and compute constraints shape system design [9, 73]. Different approaches may achieve similar overall accuracy while succeeding and failing on different questions, and the same incorrect answer may arise from distinct problems within the system. Examining these failures provides guidance beyond aggregate performance, helping identify which components require improvement and which interventions are likely to be effective.

Existing studies show that MP-VRDU performance varies with input modality, parser, retrieval setting, and supplied evidence [9, 116, 117]. Mechanism-level studies also examine OCR errors and distractors, often in text-based RAG and one mechanism at a time [14, 118, 119]. These analyses establish important findings, but their effects are measured using different backbones, document collections, and experimental conditions. Differences in reported gains or losses may therefore reflect the evaluation setup as well as the mechanism being studied, making it difficult to compare their consequences for a given system. A common testbed allows us to examine these choices under consistent conditions while holding other parts of the pipeline fixed. We organise this analysis around three failure loci: *representation*, *selection*, and *reasoning*, each encompassing distinct failure mechanisms.

We aim to explain why MP-VRDU systems fail, in terms that can directly inform system design. The main analysis uses MMLongBench-Doc [9] with Qwen3-VL [120], with complementary checks on LongDocURL [10] and Gemma-3 [121]. Our results show that complementary inputs reduce conversion sensitivity: adding images shrinks the accuracy spread across parsers from 15.7 to 4.2 points on born-digital documents, while embedded text with images matches parser text with images using 28% fewer input tokens. For selection, retrieval rank does not reliably indicate which required evidence can be omitted, and additional context can both repair and break answers despite complete annotated coverage. For

reasoning, scaling improves overall accuracy while the gap between single- and multi-hop questions persists. Abstention instructions reduce unsupported answering but increase false refusals; adding reasoning largely recovers answerable accuracy for Qwen3-VL but offers little benefit for Gemma-3. We connect these findings to qualitative examples and accuracy–cost comparisons, considering memory, latency, and compute constraints relevant to local processing of privacy-sensitive documents. Together, the analyses motivate selective representation, evidence-set management, targeted reasoning, and backbone-specific calibration rather than uniformly increasing input richness, retrieval depth, or model capacity.

Our contributions are as follows: **(1)** We organize MP-VRDU failures into three architecture-independent loci: representation, selection, and reasoning (§3.3). **(2)** We investigate these failure modes through controlled interventions and stratified comparisons in a unified pipeline, with experiments across document collections and independently developed model families (§3.4–§3.5). **(3)** We connect the findings to qualitative examples and accuracy–cost comparisons to derive practical guidance for representation, retrieval, and reasoner choices under task requirements and resource constraints (§3.6).

3.2 Related Work

3.2.1 Dataset and Diagnostic Studies

Benchmark analyses reveal persistent difficulties in using document evidence. MMLongBench-Doc [9] finds that performance declines on cross-page questions and evidence located later in documents; supplying oracle evidence pages improves accuracy but leaves substantial errors. LongDocURL [10] shows that preserving table structure improves text-input performance and identifies perception as the largest category in its sampled error analysis. DocScope [117] finds that correct answers frequently lack complete supporting evidence chains, while oracle interventions identify faithful perception and fact extraction as major bottlenecks. Complementing these benchmark analyses, Ak et al. [116] show that the benefits of agentic access depend on model capacity and benchmark, and that strong pipelines succeed on different questions. These findings leave open how specific design choices affect failure behaviour under deployment constraints; we investigate representation, selection, and reasoning through controlled interventions, qualitative examples, and accuracy–cost comparisons.

3.2.2 Individual Component Studies

Component studies link input quality, context composition, and prompting to reliability. OHRBench [14] finds that semantic OCR noise degrades retrieval and generation, whereas formatting sensitivity varies across components. The Power of Noise [118] shows that highly ranked non-answer passages can harm accuracy, while random additions can improve it in their experiments. Jin et al. [119] find that accuracy initially improves and then declines as retrieval depth increases, identifying hard negatives as a contributor and demonstrating benefits from retrieval reordering and fine-tuning. Context-faithful Prompting [122] shows that opinion-based prompts and counterfactual demonstrations improve adherence to supplied evidence. Sufficient Context [123] finds that stronger models often answer incorrectly with insufficient context, while weaker models may hallucinate or abstain even with sufficient context; sufficiency-guided selective generation improves accuracy among answered questions. Beyond document-specific OCR analysis and primarily text-based selection and calibration studies, we examine reciprocal compensation between text and images, how additional context can both improve and degrade answers despite complete annotated evidence coverage, and backbone-dependent interactions between reasoning and abstention instructions in MP-VRDU.

3.2.3 System Design and Deployment

Existing MP-VRDU methods demonstrate the effectiveness of different approaches to evidence management. M3DocRAG [37] combines visual page retrieval with multimodal answer generation, while MDocAgent [40] coordinates specialised agents to process textual and visual evidence. SimpleDoc [77] uses retrieval, summarisation, and reranking to iteratively gather evidence, whereas DocTrace [124] develops evidence localisation and hierarchical reasoning capabilities through supervised fine-tuning and reinforcement learning. Although these methods demonstrate improvements on document-understanding benchmarks, computational costs and inference trajectories are not consistently reported alongside accuracy, limiting comparisons of their deployment practicality. Such considerations include GPU memory, latency, token consumption, and the number of pages or reasoning steps required to produce an answer. We complement this performance-oriented literature by examining accuracy–cost trade-offs across representation, retrieval, and reasoning choices, and evaluating the performance and computational characteristics of a complete MP-VRDU pipeline.

3.3 Attribution Framework

We decompose MP-VRDU into three loci defined by function rather than by architecture: **representation**, **selection**, and **reasoning**. Figure 3.1 summarises the loci and the two mechanisms through which each can fail. Because these loci describe functions rather than fixed pipeline stages, they may recur throughout a system, particularly in adaptive or agentic pipelines (§2.3.4). The techniques surveyed in Chapter 2 therefore address failures at one or more loci rather than forming separate categories. Their logical dependency remains: evidence lost during representation cannot be selected, and evidence never selected cannot be reasoned over.

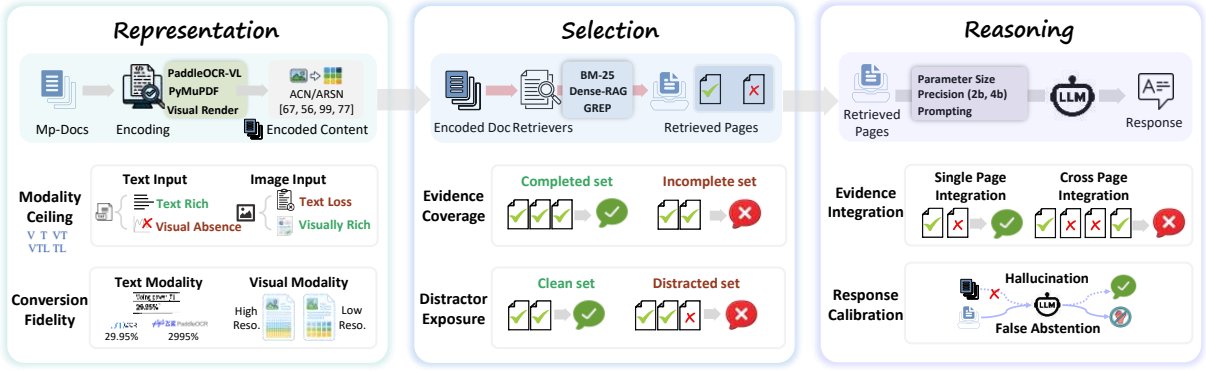


Figure 3.1 The three failure loci and their mechanisms.

3.3.1 Representation

Representation converts document content into the artifact consumed by later operations, such as an embedded text layer, parser markdown, a rendered page, a cropped visual region, or a combination of these. It may occur once during preprocessing or repeatedly during inference, as when an agent re-renders a region in response to an earlier observation. This locus concerns document conversion; operations performed by the backbone after the artifact is tokenised, such as projection, resampling, or compression, belong to reasoning because they are properties of how the model processes its input. The chosen representation determines both which document signals can be preserved and the cost of

producing them. Representation failure occurs through two mechanisms: **(1) Modality Ceiling.** A representation limits which signals are accessible at its available granularity. Text cannot express non-linguistic visual information, while images expose fine detail only when sufficient resolution and token budget are available. **(2) Conversion Fidelity.** Even when a modality can represent a signal, conversion may degrade or corrupt it. A parser may misread a numeric value or flatten a table's structure, leaving later operations to work from an incorrect but seemingly reliable artifact [14].

3.3.2 Selection

Selection determines which represented content is placed before the model within the available context budget. Similarity retrieval is one implementation, but structural navigation, adaptive reranking, and agentic inspection perform the same function, in a single pass or through repeated acquisition (§2.5.1, §2.5.3). Selection may operate over pages, sections, tables, or visual crops. Most systems select whole pages, matching common benchmark annotations and late-interaction retrievers. Selection failure occurs through two mechanisms: **(1) Evidence Coverage.** Coverage measures the proportion of required evidence units included in the selected context. Omitting required evidence leaves the model without information needed for the answer, bounding accuracy regardless of reasoning capability. **(2) Distractor Exposure.** Exposure measures irrelevant or misleading content accompanying the required evidence. Distractors do not make the evidence insufficient, but they increase the burden of identifying and using the relevant information even when coverage is complete.

3.3.3 Reasoning

Reasoning maps the supplied content and instruction to a response, which may be a final answer, a structured output, or, in an agentic system, a tool call that directs further representation or selection. When representation preserves the required evidence and selection supplies it, remaining failures concern how the model interprets and acts on that evidence. These failures may arise from inference-time behaviour or deeper limitations in the model's underlying capacity and training, and may be addressed through prompting, model scale, architecture, or additional training (§2.5.2, §2.6.2). Reasoning can fail in many ways, such as arithmetic errors over correctly supplied values. We focus on two mechanisms particularly relevant to MP-VRDU: **(1) Evidence Integration.** A model may process each evidence unit correctly in isolation yet fail to combine them across pages or sources, including content whose meaning spans page boundaries. Long-context position effects can amplify this difficulty [8]. **(2) Response Calibration.** The decision to answer should reflect the support provided by the available evidence. Failure occurs in two directions: *hallucination*, where the model answers without sufficient support [125], and *false abstention*, where it refuses despite sufficient support. These outcomes must be assessed jointly, since interventions that increase willingness to answer can reduce false abstention while increasing unsupported answers [126].

3.4 Methodology

We test the six failure mechanisms through controlled interventions in a common document question answering pipeline. This section describes how the pipeline supports attribution (§3.4.1), the experiments for each locus and their generalisation to a second model family (§3.4.2–§3.4.5), and the datasets (§3.4.6) and evaluation protocol (§3.4.7).

3.4.1 Attribution by Construction

We instantiate the attribution framework in a **single-pass retriever–generator** pipeline whose three stages correspond to the three loci. Each stage runs once per question, in order:

1. **Encoding (representation).** Each page is converted into the artifacts consumed by later stages: its embedded text, parser markdown, a rendered page image, or a combination of these.
2. **Retrieval (selection).** A retriever scores every page of the document against the question and returns the highest-ranked pages, at page granularity to match the benchmarks' annotations.
3. **Inference (reasoning).** A vision-language model receives the selected pages, the question, and a task instruction, and produces an answer in a single pass without tools or follow-up turns. It has access only to the evidence delivered by retrieval.

The experiments isolate each failure mechanism by varying one condition while holding the others fixed. Representation and reasoning experiments use annotated gold pages to eliminate selection errors, while selection experiments manipulate evidence coverage and distractor exposure by removing or adding pages. Retrieval rankings determine which pages are manipulated, while unanswerable questions use retrieved context because no gold evidence is available. Each intervention is compared against its corresponding baseline on the same questions.

```
You are answering a question about a document.
{instruction}
Question:
{question}
Document evidence:
{context}
Answer:
```

Figure 3.2 Reasoner prompt template. `{instruction}` is empty in the default configuration. `{context}` is the selected pages in document order, as parser markdown, page images, or both according to the rung.

Default Configuration. Pages are rasterised at 200 DPI, and their embedded text layers are extracted with PyMuPDF [127]. Parser markdown is produced by PaddleOCR-VL [128]. Page images are downsampled to the `med` preset before reaching the reasoner (Table 3.2). Unless an experiment varies the component under study, pages are supplied at **TLV** (Table 3.1) to Qwen3-VL-8B-Instruct [120] in `bf16`. Where a condition depends on rank, pages are ordered by ColQwen3, a ColPali-style late-interaction visual retriever [82]; unanswerable questions receive the top three pages ranked by BM25 [129]. Selected pages are restored to their original document order, whatever their rank. Figure 3.2 specifies the reasoner prompt template. Decoding is greedy (`do_sample=False`) with no temperature, top-p, top-k, or repetition penalty.

3.4.2 Representation Experiments

Representation experiments supply the annotated gold pages of answerable questions, holding selection fixed while varying how those pages are encoded. Each page can contribute a text channel, a visual

channel, or both. We examine questions grouped by evidence source and document characteristics.

Rep.	Text channel	Image	Cost	Role
T	Embedded text (PyMuPDF)	–	Extraction only	Text floor; empty on scanned pages
TL	Parser markdown	–	Parser pass	Text-only ceiling; recovers reading order and table structure
TV	Embedded text (PyMuPDF)	Page image	Extraction + vision tokens	Parser-free contrast to TLV
TLV	Parser markdown	Page image	Parser pass + vision tokens	Combined encoding; default
V	–	Page image	Vision tokens	Vision only

Table 3.1 Modality ceiling experiments. Representation ladder ordered by fidelity and, as a consequence, by cost.

Modality Ceiling. We vary which signals reach the reasoner across four representations in Table 3.1: embedded text (**T**), parser markdown (**TL**), parser markdown with the page image (**TLV**), and the page image alone (**V**). These configurations span different processing costs and allow us to examine the information accessible through text alone, vision alone, or their combination. The text-based representations progress from inexpensive extraction to richer parsing, while adding images introduces visual information at greater inference cost. A fifth representation, **TV**, pairs embedded text with the page image without a parsing pass, isolating the benefit of parsing when vision is already available. No explicit layout coordinates are passed to the reasoner; layout is conveyed only through markdown structure or page images.

Parser (text channel)			*Resolution* (image channel)			
Parser	Params	Rungs	Preset	Tok./page	Pixel cap	Rungs
PaddleOCR-VL	0.9B	TL, TLV	low	285	313,600	TLV, V
MinerU 2.5	1.2B	TL, TLV	med	456	501,760	TLV, V
Unlimited-OCR	3.3B	TL, TLV	high	744	752,640	TLV, V

Table 3.2 Conversion fidelity experiments. PaddleOCR-VL and `med` are the defaults; the two halves vary independently. Tokens are Qwen3-VL's median visual tokens per page (one token per 32×32 px).

Conversion Fidelity. We vary the quality of each representation channel while holding its modality configuration fixed (Table 3.2). For the text channel, we compare PaddleOCR-VL with MinerU 2.5 [130] and Unlimited-OCR [131] within **TL** and **TLV**, with embedded text serving as a parser-free reference. This tests how parsing quality affects reasoning and whether the visual channel compensates for extraction errors. For the visual channel, we sweep the `low`, `med`, and `high` image presets within **V** and **TLV**, examining whether explicit text reduces sensitivity to image resolution. Each axis holds the other at its default, except for an additional text-source comparison at `low` resolution to test whether coarse images already compensate for parsing errors. We examine these conditions separately for scanned and born-digital documents, where text extraction presents different challenges.

3.4.3 Selection Experiments

Selection experiments construct page sets from annotated gold pages (Table 3.3), using ColQwen3 to rank gold and non-gold pages separately by relevance to the question. All conditions are evaluated at **TLV**.

Condition	Pages supplied	Question pool
`drop_top`	All gold pages except the highest-ranked	≥ 2 gold pages
`drop_bottom`	All gold pages except the lowest-ranked	≥ 2 gold pages
`keep_top`	The highest-ranked gold page only	≥ 2 gold pages
`keep_bottom`	The lowest-ranked gold page only	≥ 2 gold pages
$+k$ distractors	All gold pages plus the k highest-ranked non-gold pages, $k \in \{3, 5, 7, 10, 13, 15, 18\}$	≥ 1 gold page
$+k$ lowest-ranked	All gold pages plus the k lowest-ranked non-gold pages, same k	≥ 1 gold page

Table 3.3 Gold-set perturbations for the selection experiments, each compared with the complete gold set on the same questions.

Evidence Coverage. We vary how much required evidence reaches the reasoner by withholding gold pages from answerable questions with multiple annotated evidence pages. Removing either the highest- or lowest-ranked gold page tests whether retrieval rank predicts the importance of missing evidence, while retaining only one of them examines more severe losses of coverage. These conditions distinguish the effect of omitting individual evidence pages from restricting the reasoner to a single page, where complementary information may be unavailable. For questions with exactly two gold pages, `drop_top` and `keep_bottom` supply the same page, as do `drop_bottom` and `keep_top`; the conditions differ only for larger gold sets.

Distractor Exposure. We vary the amount of competing context by adding the k highest-ranked non-gold pages to the complete gold set, for k from 0 to 18. These pages are selected for their similarity to the question, making them plausible competing evidence while annotated coverage remains fixed. This constructs an idealised high-recall retrieval setting, allowing us to examine how additional context affects reasoning even when all required evidence is available. The experiment captures the combined effects of context content, length, and evidence position rather than isolating each factor. We further examine distractor exposure by adding the k *lowest*-ranked non-gold pages, testing whether additional context remains disruptive when it is less similar to the question. The highest-ranked sweep is also repeated on a second document collection (§3.4.6).

3.4.4 Reasoning Experiments

Reasoning experiments supply evidence directly at **TLV**, holding representation and selection fixed while varying the reasoner or its instruction (Table 3.4). We examine evidence integration across single- and multi-hop questions, and response calibration across answerable and unanswerable questions.

Evidence Integration. We supply the annotated gold pages of answerable questions and compare single-hop and multi-hop questions across model scale, using Qwen3-VL-Instruct at 2B, 4B, 8B, and 32B, all in `bf16` under the uninstructed `none` prompt. Scaling the reasoner while keeping evidence and

instructions fixed tests whether greater model capacity improves its ability to integrate supplied information, particularly when multiple evidence units must be combined. Single- and multi-hop questions form distinct pools, so their comparison characterises differences in task difficulty rather than isolating integration from locating individual facts. We additionally examine questions grouped by evidence source to assess whether capacity requirements differ across textual, structural, and visual evidence.

Mode	Instruction
`none`	(empty)
`grounded`	Use only the provided document evidence and keep the answer concise.
`abstain`	If the evidence does not contain the answer, answer exactly: Not answerable.
`abstain_balanced`	`abstain`, plus: If the evidence does contain the answer, give it: do not decline a question the evidence supports.
`cot`	Think step by step before answering. Then write your final answer on a new line beginning with 'Answer:'.
`extract_cot`	A verbatim-evidence extraction step, with visual evidence described in enough detail to answer from the description alone, then reasoning constrained to what was extracted, then the `Answer:` line.
`abstain_cot`	`abstain`, plus: Think step by step before answering. Then write your final answer on a new line beginning with 'Answer:'.

Table 3.4 Reasoning prompt modes for the `{instruction}` field. The main experiments use `none`, `cot`, `abstain`, and `abstain_cot`; the remaining modes are reported in §3.4.5.

Response Calibration. We compare four prompt modes on Qwen3-VL-8B-Instruct and Gemma-3-12B: `none`, `cot`, `abstain`, and `abstain_cot`, which combines reasoning and abstention instructions. These conditions separate the effects of explicit reasoning and abstention guidance, while their combination tests whether reasoning changes how the model applies an abstention instruction. Answerable questions receive their gold pages, while unanswerable questions receive the top three pages ranked by BM25, providing plausible but non-supporting context. This allows us to examine whether the model distinguishes genuinely insufficient evidence from answerable questions that require more involved reasoning. Under prompts requesting reasoning, the text after the last `Answer:` marker is extracted as the final answer before judging.

3.4.5 Additional Experiments

Beyond the attribution experiments, we test whether the main findings extend to a second model family and examine practical choices that affect how the findings translate into system design and deployment. These results support the discussion in §3.6; those not reported in the main text are given in Appendix B.1.

Generalisation. We use Google's Gemma-3 [121] as an independently developed alternative to Alibaba's Qwen3-VL, with distinct training data and model-development pipelines, to examine whether our findings extend beyond a single model family. We repeat the distractor-exposure and response-calibration experiments with Gemma-3-12B, and the evidence-integration experiment across Gemma-3 at 4B, 12B, and 27B. Each repetition follows the protocol of the corresponding experiment in §3.4.3–§3.4.4, changing only the reasoner and retaining the corresponding question pools.

Retrievers. We compare six page-level retrievers spanning lexical, dense-text, and late-interaction visual retrieval (Table 3.5). Text retrievers index both the embedded text layer and default-parser output, while visual retrievers embed pages rendered at 200 DPI. We additionally evaluate joint retrieval as the deduplicated union of a matched text and visual retriever. Retrieval is examined independently of answer generation against annotated evidence pages at depths $k \in \{1, 3, 5, 7, 10\}$.

Arm	Retriever	Params	Index and cost
Text	BM25	–	Term postings per document; no encoding pass
Text	BGE-M3	568M	One vector per page
Text	Qwen3-Embedding-4B	4B	One vector per page; sequence cap 4096
Vision	ColModernVBERT	250M	One vector per image patch; requires rendering
Vision	ColQwen2.5	3B	One vector per image patch; requires rendering
Vision	ColQwen3	4B	One vector per image patch; requires rendering

Table 3.5 Selection: retriever exerpiments. Rrdered by cost within each arm. Late-interaction scoring stores many vectors per page rather than one, so the vision arm carries a larger index and a more expensive scoring pass than its parameter count alone suggests.

Prompt Modes. Beyond the four modes used in the calibration experiments, we evaluate `grounded`, `abstain_balanced`, and `extract_cot` (Table 3.4). These vary how strongly the reasoner is grounded in the supplied context, how abstention is framed, and whether evidence is extracted explicitly before answering, while holding the supplied evidence fixed. We examine questions grouped by evidence source and representation, with full results in Appendix B.1.3.

Reasoner Variations. We vary the reasoner across scale, weight precision, model family, and reasoning variant (Table 3.6). Within Qwen3-VL-8B, we compare `bf16` with 8-bit and 4-bit weights, and a 4-bit 32B model against the `bf16` 8B model at a similar memory budget. We also compare Qwen3-VL-8B with alternative families at a similar scale, and the Instruct variant with its Thinking counterpart. Quantised variants use bitsandbytes weights, and the Thinking variant receives a 2048-token output limit. These comparisons inform the deployment trade-offs in §3.6, with full results in Appendix B.1.

Axis	Checkpoints
Scale	Qwen3-VL-Instruct at 2B, 4B, 8B, 32B; Gemma-3-IT at 4B, 12B, 27B
Precision	Qwen3-VL-8B at `bf16`, 8-bit, 4-bit; Qwen3-VL-32B at 4-bit
Family	InternVL3-8B, GLM-4.6V-Flash, MiniCPM-V-4.5, Llama-3.2-11B-Vision
Reasoning variant	Qwen3-VL-8B-Thinking

Table 3.6 Reasoner variations. All share the prompt template and decoding configuration of the default pipeline.

Multi-Pass Strategies. We additionally evaluate two inference-time reasoning strategies beyond the single-pass configurations. Self-Consistency [87] samples 40 reasoning paths at temperature 0.7 and returns the majority answer, testing whether aggregating independent reasoning attempts improves reliability. Reflexion [85] allows up to three attempts, in which the reasoner evaluates its previous answer

and carries a memory of up to three reflections into subsequent attempts, testing whether iterative self-correction offers similar benefits. Both use Qwen3-VL-8B with 4-bit weights on a single 16 GB GPU, running zero-shot under a step-by-step instruction without the abstention sentence. We examine single- and multi-hop questions alongside answerability, using the single-pass `none` and `cot` prompts as reference conditions.

3.4.6 Datasets

MMLongBench-Doc. MMLongBench-Doc [9] is the primary benchmark (Table 3.7). It is the only multi-page benchmark we are aware of that annotates, per question, the evidence pages, the evidence source, and the document domain, while also supplying questions unsupported by the document. Evidence pages define the oracle evidence set and make coverage measurable, evidence sources support analysis by question requirements, and unanswerable questions allow response calibration to be examined. Questions are created or revised by expert annotators, and multi-page questions are constructed to require each annotated page, so we treat gold pages as necessary rather than redundant evidence, while recognising that human-annotated sets may not be minimal in every case.

The following labels and question pools are derived from the annotations and held fixed across experiments:

- **Answerability:** unanswerable questions have the gold answer exactly *Not answerable* (244 questions). Two answerable questions have no annotated evidence page and are excluded from oracle conditions.
- **Evidence source:** multi-label, so a question tagged with two sources contributes to both source groups.
- **Evidence structure:** a question is *single-hop* when it requires only one evidence source, and *multi-hop* when answering requires integrating information from multiple evidence sources.
- **Scan status:** a document is scanned when none of its sampled pages yields at least 20 extractable characters, giving 104 born-digital and 31 scanned documents.

Property	Value	Evidence source	Q	Gold pages	Q	Answer format	Q
Questions	1,091	Pure-text (plain)	298	1 page	485	Int	290
Answerable	847	Figure	293	2 pages	247	Str	250
Unanswerable	244	Table	218	3+ pages	113	None	244
Documents	135	Chart	178	0 pages	246	Float	160
Born-digital	104	Generalized-text (layout)	119			List	147
Scanned	31	(none)	254				
Mean pages/doc	48.3						

Table 3.7 MMLongBench-Doc composition and question labels. vidence source is multi-label and sums past the question count; answer format is exclusive.

LongDocURL. LongDocURL [10] contains 2,325 questions over 396 born-digital documents of 51–149 pages. We use it to test whether the modality ceiling depends on the document collection, and whether the effect of added context holds on a second collection. Its documents frequently expose answer-bearing content through the text layer, even when the annotated evidence is a figure, and its

LVLM-generated, human-verified questions further favour evidence recoverable this way. Its annotated evidence pages also need not form a minimal sufficient set, so the evidence-coverage results on it are reported only in Appendix B.1, and we do not repeat the conversion-fidelity or reasoning analyses on it. The distractor-exposure sweep uses a 1,000-question sample.

3.4.7 Evaluation

Judge. Answers are scored by an LLM judge rather than exact match, because correct responses may differ from the gold answer in units, rounding, or formatting. Gemini-2.5-Flash judges each response at temperature zero with JSON-only output, using the rubric in Figure 3.3. The judge receives only the question, the gold answer, an unanswerable flag derived from the gold answer, and the extracted model answer; it never sees document content, evidence annotations, or the experimental condition, which keeps the scoring criterion fixed across conditions and prevents the judge from conditioning on the intervention being evaluated. It returns *correct*, *incorrect*, or *abstained*. An abstention is credited as correct only for an unanswerable question; on an answerable question it is scored incorrect and additionally counted as a false refusal. An extension of the rubric that assigns each incorrect response an error category is used for the error analysis in Appendix B.3.

```
You judge answers to document questions.
Return only JSON with keys:
- verdict: one of correct, incorrect, abstained
- extracted_answer: the answer extracted from the model response, or
empty string
- rationale: a short reason

Mark correct when the model answer is semantically equivalent to the
gold answer.
For unanswerable questions, mark correct only when the model abstains.

Return the verdict "abstained" whenever the response declines to
answer because it judges the evidence insufficient, however it is
worded. "Not answerable", "the document does not provide this",
"there is no document evidence to determine this" and "it is not
possible to determine this from the given information" are all
abstentions. A response that notes a limitation and then commits to
an answer anyway is NOT an abstention: judge the answer it commits
to.
```

Figure 3.3 Judge rubric.

Judge Validation. We validate the judge's correctness verdicts in three ways. Against GPT-4o, an independent model family, on 500 stratified responses, agreement is 96.0% with Cohen's $\kappa = 0.918$ (95% CI [0.881, 0.951]), and remains stable across representations (0.938–0.969). Against a human annotator on a 120-response subset, agreement is 98.3% with $\kappa = 0.966$ (95% CI [0.915, 1.000]). Because κ is deflated when one verdict dominates, we also report the prevalence index of the validation pool over correct and not-correct verdicts, 0.156, indicating near-balanced verdicts. Finally, we compare the rubric against an otherwise identical binary version without the abstention rule. Across 19,983 responses scored under both, the full rubric is uniformly 0.8–2.9 points stricter across representations, datasets and models.

Metrics. Accuracy is the share of the evaluated question pool judged correct. For paired conditions evaluated on the same questions, verdict transitions report the share moving from correct to incorrect (**C→I**) or incorrect to correct (**I→C**) relative to the baseline. In distractor sweeps, transitions compare consecutive tested depths rather than every depth against the complete-gold baseline. Refusal rates are computed separately on answerable and unanswerable pools, distinguishing false refusal from correct refusal. Retrieval is evaluated independently of answer generation through page-level precision, recall, and F1 against annotated evidence pages at the specified depths. The practical analysis additionally considers inference latency, token consumption, and GPU memory alongside predictive performance.

Statistical Uncertainty. We estimate 95% confidence intervals for the principal accuracy contrasts using 10,000 document-level bootstrap resamples, pairing conditions on questions completed by both. Documents rather than individual questions are resampled to account for dependence among questions from the same document. Consequently, paired differences may differ slightly from differences between separately reported accuracies, which can use different completed question pools. Intervals that exclude zero indicate effects distinguishable from sampling variation, while those containing zero are treated as ties. The full contrasts and confidence intervals are reported in Appendix B.2.

3.5 Empirical Findings

This section reports the controlled results for each locus under the setup of §3.4.

3.5.1 Representation

	Abstention				Accuracy			
Evidence	**T**	**TL**	**TLV**	**V**	**T**	**TL**	**TLV**	**V**
MMLongBench-Doc								
Text	23.0	11.5	7.3	8.0	39.2	47.8	**55.3**	44.3
Layout	20.3	10.2	0.8	4.2	28.0	37.3	**50.8**	44.1
Table	15.9	7.2	6.7	**9.1**	40.6	**48.4**	**48.4**	37.8
Chart	32.6	**28.1**	7.3	6.2	18.5	19.1	**45.5**	43.3
Figure	**36.6**	24.8	**7.6**	7.9	15.5	23.8	**44.8**	40.3
All	26.3	17.2	6.2	7.1	31.9	38.8	**52.5**	45.6
LongDocURL								
Text	4.6	3.5	**1.5**	**1.6**	75.3	79.5	**82.3**	74.0
Layout	2.1	1.6	0.7	0.9	48.7	52.4	**56.3**	52.2
Table	5.0	3.2	1.1	0.8	61.2	**69.5**	68.2	57.3
Figure	**6.7**	**8.6**	**1.5**	1.3	62.5	65.7	**73.4**	65.2
All	4.7	4.3	1.3	1.3	63.7	68.0	**71.9**	63.1

Table 3.8 Abstention and accuracy by evidence source and representation.

Modality Ceiling: Evidence Accessibility Depends on Modality. A modality can only support questions whose required evidence it preserves. On MMLongBench-Doc, this effect is clearest for chart and figure questions (Table 3.8): **T** and **TL** have abstention rates of 24.8–36.6%, whereas adding vision brings abstention below 10% across evidence sources. This is accompanied by substantial accuracy gains, with chart questions improving from 19.1% under **TL** to 45.5% under **TLV**. The converse

also holds: **V** remains less accurate than **TLV**, particularly for text and table questions, where explicit text supports precise extraction and interpretation. Text and vision therefore preserve complementary evidence. LongDocURL shows the same dependency in a different document setting. Its born-digital documents preserve more evidence in text, so adding vision to **TL** improves overall accuracy by only 3.9 points, while figure questions gain 7.7 points. Table questions even perform slightly better under **TL**, illustrating that additional modalities do not necessarily improve every task. The modality ceiling therefore depends on whether the evidence required by a question is accessible in the chosen modality.

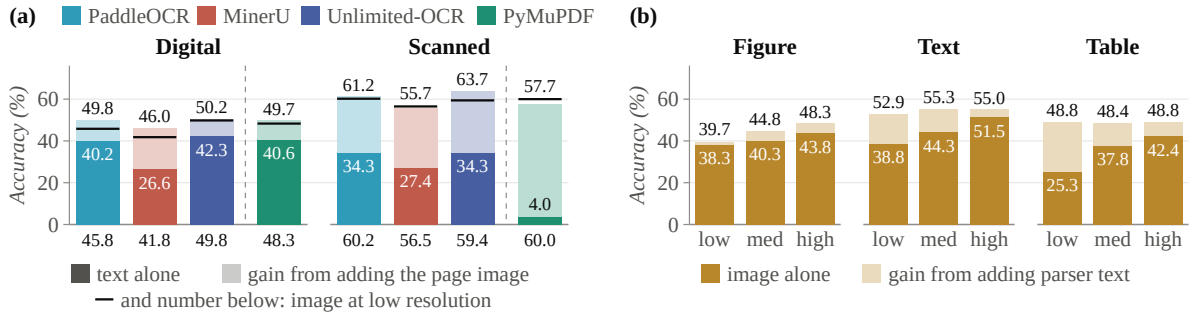


Figure 3.4 Representation: Conversion Fidelity. (a) Accuracy by text source and scan status. The rule on each bar and the number below it give the accuracy with the page image at `low` resolution. (b) Accuracy by image resolution, with and without parser text.

Conversion Fidelity: Complementary Modalities Reduce Conversion Sensitivity. Conversion errors have less impact when the affected information remains accessible through another modality. Page images can compensate for parsing errors, while parsed text can reduce dependence on image resolution. On born-digital documents, adding page images at `med` reduces the accuracy spread across text sources from 15.7 to 4.2 points (Figure 3.4a). Even `low`-resolution images provide substantial compensation, narrowing the spread to 8.0 points and bringing each text source within 4.2 points of its `med` accuracy. The effect is particularly pronounced on scanned documents, where adding a `low`-resolution image raises accuracy with embedded text from 4.0% to 60.0%. Conversely, increasing image resolution produces much larger gains under **V** than under **TLV**, especially for text and table questions (Figure 3.4b). For tables, accuracy increases by 17.1 points under **V** from `low` to `high`, but remains unchanged under **TLV**. This compensation is not universal: scanned documents remain sensitive to parser quality, while figure questions still benefit from higher visual resolution. Overall, conversion fidelity matters most when degraded information cannot be recovered through another modality, and even a relatively coarse complementary representation can substantially reduce this sensitivity.

3.5.2 Selection

Evidence Coverage: Omission Causes Large Accuracy Losses. Accuracy is highly sensitive to missing required evidence. On MMLongBench-Doc, removing a single gold page reduces accuracy from 39.2% to 18.5% or 15.9%, depending on whether the lowest- or highest-ranked page is removed (Figure 3.5). Paired verdicts show that correct answers are broken far more often than incorrect ones are repaired, indicating that the remaining evidence rarely compensates for the omission. The similar losses also suggest that retrieval rank does not reliably predict the importance of a required page. LongDocURL shows a different pattern: removing the highest-ranked gold page reduces accuracy from 69.3% to 28.1%, compared with 53.3% when removing the lowest-ranked page. This difference reflects the benchmarks' evidence annotations. MMLongBench-Doc constructs multi-page questions requiring every annotated

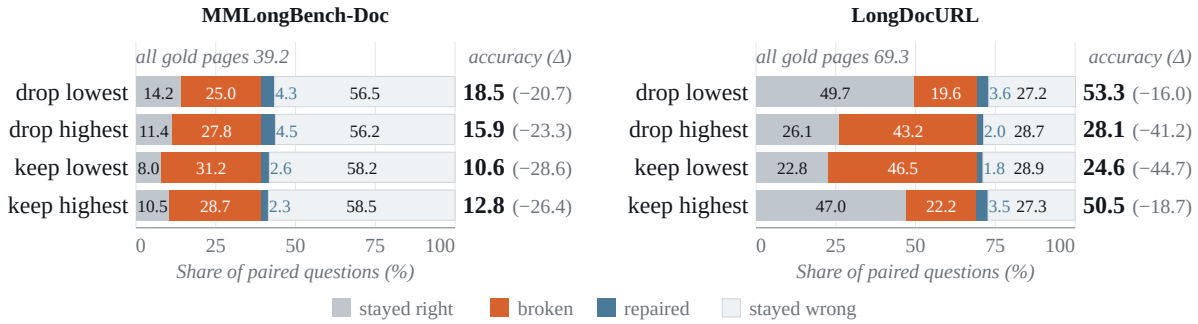


Figure 3.5 Selection: Evidence Coverage. Paired verdict transitions and accuracy after removing or retaining gold pages, on MMLongBench-Doc and LongDocURL.

page, whereas LongDocURL's evidence sets need not be minimal (§3.4.6). Consequently, one page may contain most of the answer-bearing information, with the remaining pages providing supporting context. Retrieval rank is therefore more informative on LongDocURL, but the central finding remains that coverage failures depend on whether the evidence necessary to answer the question is retained.

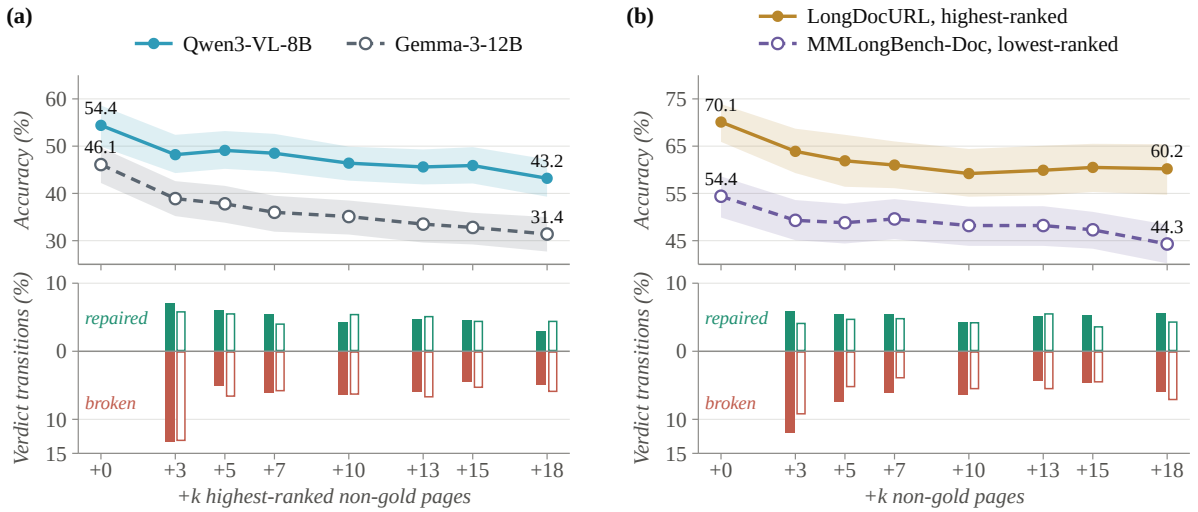


Figure 3.6 Selection: Distractor Exposure. Accuracy and paired verdict transitions as non-gold pages are added: (a) the highest-ranked pages on MMLongBench-Doc for two reasoners; (b) Qwen3-VL-8B on LongDocURL, and with the lowest-ranked pages on MMLongBench-Doc.

Distractor Exposure: Added Context Causes Gradual but Noisy Degradation. Adding non-gold pages gradually reduces accuracy, but individual answers remain unstable even when all annotated evidence is retained. On MMLongBench-Doc, Qwen3-VL-8B accuracy declines from 54.4% to 43.2%, while Gemma-3-12B falls from 46.1% to 31.4% (Figure 3.6a). Both experience their largest drop after the first three pages, followed by smaller fluctuations. Paired transitions reveal that each step both breaks correct answers and repairs incorrect ones, with these effects largely offsetting for Qwen3-VL-8B but more consistently favouring degradation for Gemma-3-12B. The pattern extends to LongDocURL, where accuracy falls from 70.1% to 60.2%, with similar rates of broken and repaired answers across subsequent depths (Figure 3.6b). Interestingly, adding the lowest-ranked pages on MMLongBench-Doc also produces comparable accuracy losses to adding the highest-ranked ones. Although the first three lowest-ranked pages cause fewer transitions (9.3% broken and 4.2% repaired, compared with 13.3% and 7.0%), subsequent steps show broadly similar instability. These results suggest that distractor effects are not limited to

pages highly relevant to the question: additional context can both disrupt and repair answers regardless of retrieval rank, while increased input length and changes in evidence position may also contribute to the observed degradation.

3.5.3 Reasoning

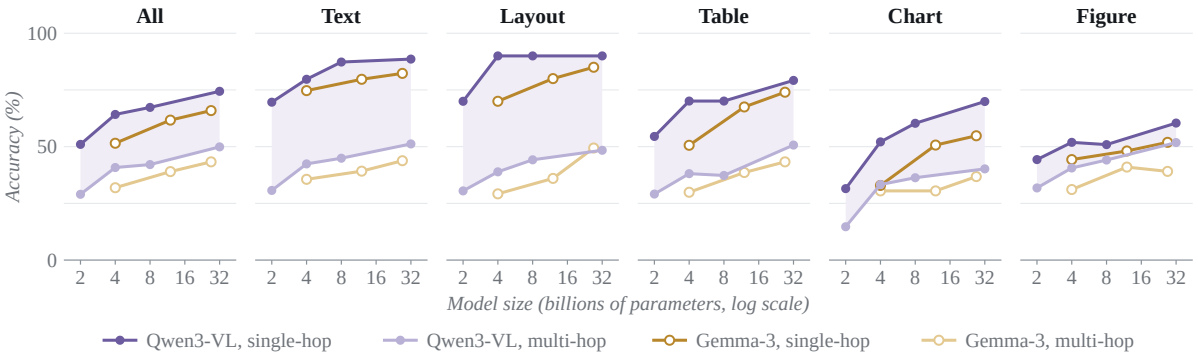


Figure 3.7 Reasoning: Evidence Integration. Accuracy on single- and multi-hop questions across model scales.

Evidence Integration: Scaling Helps, but the Multi-Hop Gap Persists. Larger reasoners improve both single- and multi-hop accuracy, but do not close the gap between them (Figure 3.7). For Qwen3-VL, multi-hop accuracy rises from 29.0% at 2B to 49.9% at 32B, while single-hop accuracy increases from 51.0% to 74.4%, leaving the gap largely unchanged (21.9 to 24.5 points). The gap is widest for text and layout questions, where single-hop performance approaches 90% at 32B while multi-hop accuracy remains near 50%. Conversely, figure questions have the narrowest gap, partly because interpreting individual figures is itself challenging. Gemma-3 exhibits a similar separation between single- and multi-hop questions, but its scaling gains are less consistent. Multi-hop chart accuracy remains at 30.5% from 4B to 12B, while figure accuracy declines slightly from 12B to 27B, suggesting that greater model capacity does not uniformly improve evidence integration. These differences may reflect variations in the model families' training objectives, architectures, or multimodal capabilities, although the experiments cannot isolate their contributions. Nevertheless, both families exhibit the widest gaps on text and layout questions and the narrowest on figures, suggesting that the additional difficulty of combining evidence persists across backbones. Scaling therefore improves overall evidence use without necessarily resolving the distinct challenges of cross-page integration.

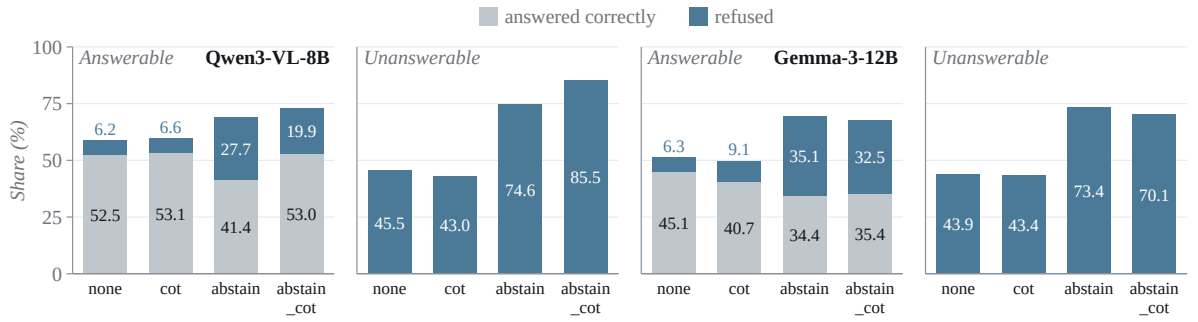


Figure 3.8 Reasoning: Response Calibration. Answerable accuracy, answerable refusal, and unanswerable refusal across four prompt modes.

Response Calibration: Reasoning Before Abstaining Limits False Refusal. Abstention prompting substantially increases refusal on unanswerable questions, but it also makes the reasoner more likely to refuse answerable ones (Figure 3.8). This trade-off appears for both backbones. For Qwen3-VL-8B, `abstain` raises unanswerable refusal from 45.5% to 74.6%, but also increases false refusal from 6.2% to 27.7% and reduces answerable accuracy from 52.5% to 41.4%. Gemma-3-12B shifts the same way, with unanswerable refusal increasing from 43.9% to 73.4% and false refusal from 6.3% to 35.1%. Adding CoT changes this balance differently for the two models. For Qwen3-VL-8B, `abstain_cot` reduces false refusal to 19.9%, raises answerable accuracy to 53.0%, and improves unanswerable refusal further to 85.5%. For Gemma-3-12B, false refusal falls to 32.5%, but answerable accuracy improves by only 1.1 points and unanswerable refusal declines to 70.1%. Reasoning before abstaining can therefore recover supported answers, but the benefit depends on the backbone, and false refusal remains substantially above the uninstructed baseline for both models.

3.6 Practical Discussion

We now discuss what the findings of §3.5 imply for designing and deploying document question-answering systems, drawing on qualitative examples (Figure 3.9), performance–cost comparisons (Table 3.9), and results of additional reasoning strategies (Figure 3.10).

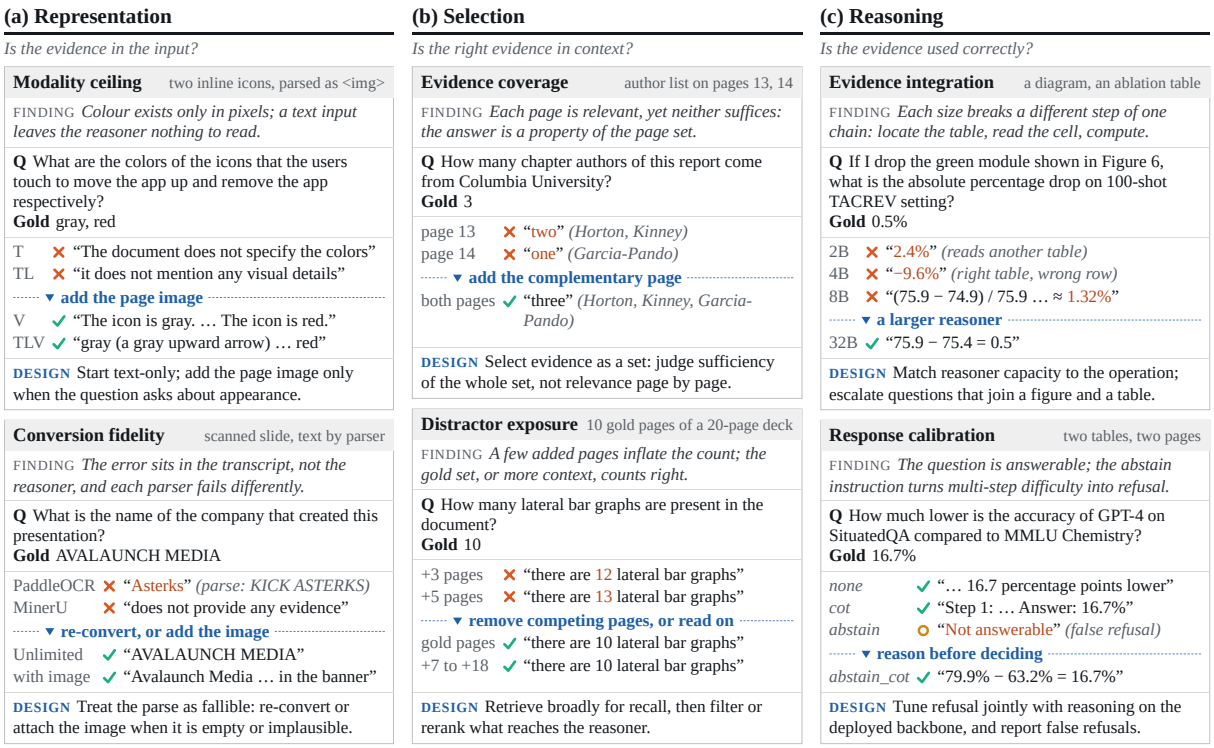


Figure 3.9 Qualitative examples of representation, selection, and reasoning failures and the design response to each.

3.6.1 Representation

Representation failures arise when the input omits or distorts task-relevant information, consistent with prior findings on OCR noise, structural information loss, and perception bottlenecks [14, 117]. Our results further show that text and images can compensate for each other's limitations when the required

(a) Representation	Acc.	Text	Vis.	Tokens	Prefill	Decode
T	31.9	501	0	501	1.1	5.4
TL	38.8	886	0	886	1.5	6.3
TV	51.6	500	459	959	24.2	6.6
TLV	52.5	886	459	1345	24.2	6.6
V	45.6	20	459	479	22.7	6.6

(b) Retriever	R@1	R@5	R@10	Idx. (s)	Qry. (ms)	GB
BM25	32.7	58.1	68.1	0.02	0.3	CPU
BGE-M3	32.8	61.3	72.4	0.01	47	1.2
Qwen3-Emb.	33.5	63.9	74.8	0.19	89	13.1
ColModernVBERT	47.1	74.0	82.7	0.91	41	5.1
ColQwen2.5	49.2	77.2	84.8	1.50	293	8.6
ColQwen3	53.7	81.0	87.9	0.57	110	9.4

(c) Reasoner	T	TL	TLV	V	GB	Time (s)
Qwen3-VL-8B, bf16	31.9	38.8	52.5	45.6	17.5	30.8
8-bit	32.5	37.9	53.6	45.7	10.3	43.5
4-bit	31.8	39.4	51.8	45.5	6.8	35.4
Qwen3-VL-32B, 4-bit	37.0	42.1	59.1	52.3	20.0	47.5
InternVL3-8B	19.3	25.3	32.6	24.1	15.9	5.3
MiniCPM-V-4.5	30.7	38.1	60.1	53.3	17.4	47.2
GLM-4.6V-Flash	25.9	32.9	42.7	30.9	20.6	19.8
Llama-3.2-11B-V	26.5	34.3	33.3	25.3	21.3	10.4
Gemma-3-12B	30.5	36.9	48.1	41.6	24.4	8.0

Table 3.9 Performance and cost of each representation, retriever and reasoner. Accuracy is for MMLongBench-Doc for (a) and (c); Recall is across both datasets in (b), and text retrievers use parser output.

information remains accessible through another modality. Conversion quality should therefore be assessed across the full representation rather than for each channel independently. Figure 3.9a illustrates the distinction between exposing otherwise unavailable information and recovering distorted content. The accuracy–cost comparisons in Table 3.9a reinforce this trade-off. With images available, additional parsing offers limited benefit: embedded text with images (**TV**) achieves accuracy comparable to parser text with images (**TLV**), 51.6% against 52.5%, while requiring fewer tokens per page (959 versus 1,345) and avoiding a parsing pass. Conversely, vision alone (**V**) achieves 45.6% accuracy with just 479 tokens per page, but loses information recoverable through explicit text. Adding images also introduces substantial processing cost, with prefill latency reaching 24.2 s under **TLV** compared with 1.5 s under **TL**, despite the relatively modest increase in input tokens. The low-resolution results suggest one way to contain this cost, since coarse images recover much of the information lost through imperfect text extraction. These findings favour selecting representations according to the evidence required, rather than uniformly applying the richest encoding to every page.

Deployment suggestions. Systems should use task-specific representations that preserve sufficient evidence within the available resource budget: rely on embedded text where it is adequate, add page images when visual information is needed or extraction is unreliable, and re-parse when the existing text is inadequate. For local deployment, modality choice, conversion fidelity, and evidence selection depth should be jointly configured to balance accuracy, memory, and inference latency.

3.6.2 Selection

Selection failures arise when necessary evidence is omitted or when additional retrieved context reduces the reliability of evidence use. Degradation from added context is consistent with prior observations on distracting passages and hard negatives in RAG [118, 119], but our paired analysis further shows that this degradation is noisy: successive additions both break correct answers and repair incorrect ones. Small changes in aggregate accuracy can therefore conceal substantial changes in individual answers. Additional document context may clarify relationships needed to interpret evidence, while plausible competing information can mislead the reasoner. Figure 3.9b illustrates these opposing effects alongside the need for complementary evidence. The lowest-ranked sweep adds a caution for retrieval design: pages unrelated to the question cost about as much accuracy as highly ranked ones, so filtering for relevance alone does not remove the cost of a longer context. Selection should therefore preserve collective evidence coverage while managing the amount and composition of the supplied context. A plausible an-

swer from a retrieved subset does not establish that further document context is unnecessary. Structured navigation, summary-based reranking, and iterative retrieval are possible ways to balance these requirements [38, 77]. These choices also involve resource trade-offs: visual retrievers offer higher recall at the cost of GPU memory and greater query latency (Table 3.9b).

Deployment suggestions. For frequently queried collections, stronger visual retrievers may justify their indexing and memory costs, whereas lightweight text retrievers remain attractive for resource-constrained or infrequently queried collections. Since retrieval and reasoning costs recur with every query, retrieval depth and refinement strategies should be evaluated by answer quality, stability, and latency rather than recall alone.

3.6.3 Reasoning

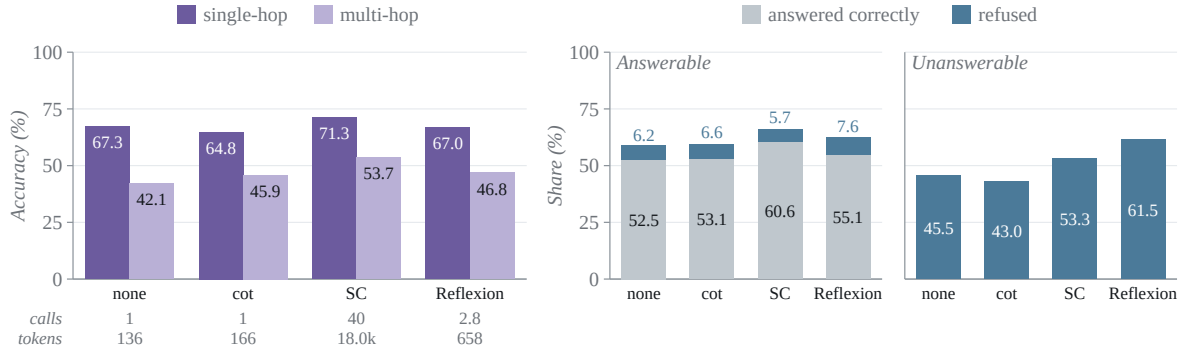


Figure 3.10 Comparison of Self-Consistency (SC), Reflexion, and single-pass prompting across single- and multi-hop accuracy, answerable accuracy, and refusal behaviour.

The reasoning findings reveal two distinct failures: models struggle to integrate evidence across multiple sources, and abstention instructions can suppress answers supported by the available evidence (Figure 3.9c). Scaling improves evidence use but leaves the multi-hop gap largely unchanged, suggesting that greater model capacity alone does not resolve the difficulty of combining information across pages. Meanwhile, abstention instructions reduce unsupported answering at the cost of false refusals, and adding explicit reasoning recovers more supported answers for Qwen3-VL than for Gemma-3. These failures call for targeted strategies for evidence integration and backbone-specific response calibration, depending on whether the model struggles to combine available facts or incorrectly judges the evidence as insufficient. Additional inference-time reasoning offers one approach (Figure 3.10). Self-Consistency [87] improves answerable accuracy by 8.1 points and narrows the single- versus multi-hop gap from 25.2 to 17.6 points, while Reflexion [85] provides a smaller accuracy gain of 2.6 points but increases correct refusal from 45.5% to 61.5%. However, these gains come at substantially higher inference costs, requiring 40 model calls for Self-Consistency and an average of 2.8 for Reflexion, compared with a single call under standard prompting. Together, these findings motivate evaluating reasoning and refusal behaviour jointly, while balancing additional inference and verification against deployment costs.

Deployment suggestions. Reasoning and abstention instructions should be tuned jointly for the intended task, considering supported-answer accuracy, correct refusal, and false refusal. Backbone selection should account for both task-specific performance and deployment constraints, as model scale, precision, and inference strategies introduce different accuracy–cost trade-offs (Table 3.9c). Quantisation can reduce memory requirements but may increase latency, while multi-pass reasoning can improve reliability at substantially greater computational cost. Additional reasoning or verification should therefore be applied selectively to difficult or uncertain questions rather than uniformly across all inputs.

3.7 Conclusion

Our controlled quantitative comparisons and qualitative examples connect MP-VRDU failures to practical design choices. First, we find that even low-resolution images can substantially compensate for parsing errors, a surprising result given their limited visual detail, while extracted text can reduce dependence on higher image resolutions. Second, added non-gold pages both repair and break answers despite complete annotated coverage, an unexpected finding that challenges the assumption that more document context is necessarily beneficial. This instability persists across document collections and even when the added pages are unrelated to the question, revealing noisy degradation beneath aggregate accuracy. Finally, reasoning and abstention instructions interact differently across backbones, requiring supported-answer accuracy and false refusals to be evaluated together, while multi-pass strategies improve both at substantially higher inference cost. These findings guide representation, evidence selection, and reasoner choices under resource constraints. Future work should investigate the underlying mechanisms and evaluate adaptive policies that acquire complementary inputs, refine evidence sets, and invoke further reasoning while accounting for their computational cost.

Modality ceiling — mmlongbench:000014 · chart · 1 gold page

Q According to the chart on page 14 how much time was spent with family and friends in 2010? **GOLD** 21%

T	○	"The document does not provide any evidence or chart on page 14, so it is not possible to determine [...]"
TL	○	"The document does not provide sufficient textual or numerical information to determine how much [...]"
V, TLV	✓	"According to the chart on page 14, in 2010, 21% of weekend time was spent with family and friends."

NOTE *The gold page reaches every rung, so nothing is missing from the context. Only the rungs carrying the page image can read the bar, and the text rungs decline rather than guess.*

Conversion fidelity — mmlongbench:000513 · plain text + table · 1 gold page

Q How many geographic market names are listed under Los Angeles area? **GOLD** 13

T (raw)	✓	"The document lists 13 geographic market names under the Los Angeles area. [...]"
TL (Paddle)	✗	"26 The document lists 26 geographic market names under the "Los Angeles Area" section [...]"
TL (MinerU)	✗	"26 The document lists 26 geographic market names under [...]"
TL (unlimited)	✓	"There are 13 geographic market names listed under the Los Angeles area. [...]"
TLV	✓	"Based on the provided document, there are 13 geographic market names listed under [...]"

NOTE *The embedded text layer already carried the count; two parsers restructure the list and return its last label instead. The reasoner cannot tell a corrupted number from a real one, and both a better parser and the page image recover it.*

Evidence coverage — mmlongbench:000082 · plain text · 2 gold pages

Q Which method integrates knowledge into white-box models via directive fine-tuning? **GOLD** PKG

oracle	✓	"Based on the provided document, the method that integrates knowledge into white-box models via directive fine-tuning is PKG. [...]"
drop top	✗	"Based on the provided document, the method that integrates knowledge into white-box models via directive fine-tuning is Fine-tuning Embedding Model. [...]"
drop bottom	✗	"The method that integrates knowledge into white-box models via directive fine-tuning is directive fine-tuning. [...]"

NOTE *Neither gold page is sufficient alone. Withholding either one does not make the reasoner abstain, it answers from the page that is left, so the missing evidence never announces itself.*

Distractor exposure — mmlongbench:000078 · table + plain text · 2 gold pages

Q What is the paper's full title that proposes the method that has a retrieval granularity of phrase? **GOLD** Copy is all you need.

gold only	✓	"Based on the provided document evidence, the method with a retrieval granularity of "Phrase" is CoG [29]."
+3	✗	"Based on the provided document, the paper's full title is Retrieval-Augmented Generation for Large Language Models [...]"
+5, +18	✗	"Based on the provided document, the full title of the paper is: Retrieval-Augmented Generation for Large Language Models: A Survey [...]"

NOTE *Both gold pages are in every row and nothing in them changes. One of the added pages is the survey's own title page, a plausible match for "the paper's full title", and the reasoner takes it instead of following the table entry to its reference.*

Evidence integration — mmlongbench:001015 · table · 2 gold pages

Q what is EBITDA for costco in FY2021? **GOLD** 8489

2B	✗	"Based on the provided document, EBITDA for Costco in FY2021 is $6,708 million."
4B	✗	"The EBITDA for Costco in FY2021 is $6,708 million."
8B	✗	"Based on the provided financial documents, EBITDA for Costco in FY2021 (52 weeks ended August 29, 2021) is $8,469 million."
32B	✓	"EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization) for Costco in FY2021 (52 weeks ended August 29, 2021) [...]"

NOTE *Every size of Qwen3-VL receives the same two statements. The two smallest stop at operating income on the first, the 8B reaches the second but adds the wrong lines, and only the 32B joins operating income to depreciation and amortization.*

Response calibration — mmlongbench:000418 · chart · 2 gold pages

Q How many people in India were using a debit card when the e-commerce sales reached 11 billion? **GOLD** 399 million

none	✓	"Based on the provided document, when e-commerce sales reached $11 billion (in 2014), the number of debit card users in India was 399 million."
abstain	○	"Not answerable"
abstain_cot	✓	"[...] For 2014, the number is 399 million."

NOTE *The answer is on no single page: one chart gives the year, the other the count. Told it may decline, the reasoner refuses a question it answers unprompted, and reasoning before deciding makes it find both values first.*

✓ judged correct ✗ judged incorrect ○ declined to answer

Figure 3.11 Additional worked examples, one question for each of the six failure mechanisms, with the reasoner's output under each condition.

Chapter 4

Method and Conclusion

Chapter 2 uses an evidence-management perspective to organise the rapidly developing MP-VRDU literature, while Chapter 3 examines where current systems fail through controlled attribution across representation, selection, and reasoning, deriving practical guidance for system design and deployment. Together, these chapters provide both a structured view of how MP-VRDU systems are built and empirical evidence on how design choices affect accuracy, reliability, and computational cost. This chapter brings those contributions together by translating them into a deliberately simple working pipeline, evaluating its performance and deployment characteristics, and considering the broader challenges and conclusions that follow.

4.1 A Simple Adaptive-Trajectory Method

We translate the evidence-management perspective of Chapter 2 and the empirical findings of Chapter 3 into a deliberately simple MP-VRDU pipeline. The aim is not to introduce a new specialised backbone or complex orchestration framework, but to test whether the preceding findings provide a useful basis for system design when combined in a complete working method. Where appropriate, the pipeline uses mechanisms already established in the literature, including iterative evidence memory [77] and answer verification [73]. We then evaluate whether this combination can remain competitive with substantially more specialised systems using similar open-weight models under a fixed hardware budget.

4.1.1 Method Overview

The pipeline separates evidence acquisition from final evidence presentation and reasoning. A retriever first ranks the document pages, after which three roles implemented by the same vision-language model operate in a fixed loop (Figure 4.1). The `reader` acquires and records evidence, the `synthesizer` produces a candidate answer from the accumulated evidence, and the `arbiter` either accepts that answer or requests further reading. This process continues until the answer is accepted or the reading budget is exhausted. Full implementation details, including the role-specific prompts and system configuration, are provided in Appendix C.

`Reader`: Evidence Acquisition and Recall-Oriented Selection. The `reader` traverses pages in retrieval-ranked order using complementary textual and visual representations, following the finding that these channels can compensate for conversion errors. Unlimited-OCR [131] supplies parser text, while the page image preserves information that may be lost or difficult to recover through parsing alone. Each turn distils the current page into a compact evidence record that is appended to a persistent evidence memory. Given the substantial cost of omitting required evidence, the `reader` flags pages that

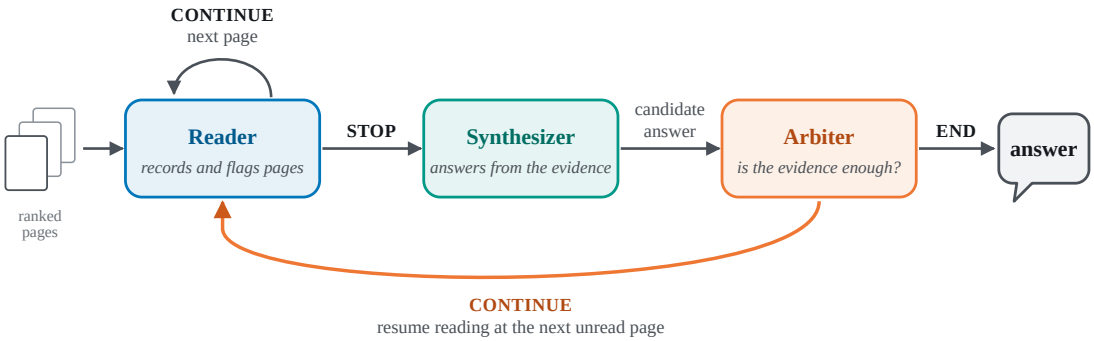


Figure 4.1 Architecture of the proposed method.

the answer may depend on, including uncertain but potentially relevant evidence. This separates broad evidence acquisition from selective evidence presentation: compact records retain potentially useful document context, while flagged pages are later reintroduced in full multimodal form. Once the minimum reading depth has been reached, the `reader` may propose `STOP` when it judges the accumulated evidence sufficient for answer synthesis.

`Synthesizer`: Evidence Integration and Premise Validation. The `synthesizer` receives the accumulated evidence in document order when reading stops. Flagged pages are restored in full through parser text and page images, while other useful pages remain represented by their compact evidence records. This retains surrounding document context that may help interpret the answer-bearing evidence while controlling the amount of full multimodal input. The `synthesizer` performs cross-page evidence integration to produce a candidate answer and explicitly validates the premises and presuppositions of the question against the supplied evidence. If the required support is absent, it must identify the unresolved evidence requirement before returning *Not answerable*. This separates answer generation from evidential sufficiency assessment, reflecting the Chapter 3 finding that evidence availability alone does not guarantee correct integration or calibration.

`Arbiter`: Evidential Verification and Adaptive Re-Acquisition. The `arbiter` receives the evidence memory, the candidate answer, and the same flagged pages in full textual and visual form. Access to the underlying multimodal evidence supports verification of the candidate against information that compressed records may omit. The `arbiter` cannot revise the answer itself; it separates verification from answer construction, either accepting the candidate or identifying an unresolved evidence requirement and requesting further acquisition. This gate reflects the selection finding that a plausible answer from a retrieved subset does not establish that further document context is unnecessary. On `CONTINUE`, control returns to the `reader` at the next unread page before a new synthesis attempt. The resulting loop supports adaptive evidence re-acquisition while retaining a deliberately constrained action space: it does not reformulate retrieval queries, invoke arbitrary tools, or construct a separate plan, but adapts only the depth of evidence acquisition.

4.1.2 Evaluation Setup

Deployment Parameters. The system is controlled by five parameters that bound evidence acquisition and generation. B caps the number of pages that may be read, p sets the number of pages processed per `reader` turn, k defines the minimum initial reading depth before a `STOP` decision is honoured, and m defines the minimum additional reading after an `arbiter` decision reopens the trajectory. We use $B =$

30, $p = 1$, $k = 6$, and $m = 4$. Motivated by the evidence coverage and distractor exposure findings in Chapter 3, the initial reading floor encourages sufficient evidence acquisition, while the reopening depth ensures that rejected answers trigger meaningful additional exploration. Processing one page per turn limits per-call memory requirements, while the overall reading and completion budgets bound trajectory length and generation cost. These parameters are practical deployment choices rather than optimised values.

Configuration. All three roles share a single `Qwen3-VL-8B-Instruct` model using 4-bit NF4 quantisation, FP16 compute, and greedy decoding, deployed on a single NVIDIA V100 16 GB GPU. ColQwen3.5 ranks pages once per question. The `reader` receives Unlimited-OCR text together with high-resolution page images, while flagged pages supplied to the `synthesizer` and `arbiter` use the same text with medium-resolution images. These choices follow the accuracy–cost trade-offs examined in §3.6: quantisation reduces model memory requirements without affecting answer quality, while selective evidence presentation and role-specific image resolutions manage visual-token consumption and inference cost without discarding potentially useful information during initial inspection.

Benchmarks. We evaluate on MMLongBench-Doc, LongDocURL, and DocBench using each benchmark's official scoring protocol. LongDocURL was used in Chapter 3 only for limited replication experiments, so it remained largely unseen during development of the complete pipeline, while DocBench was entirely unseen. Together, they provide additional tests of generalisation beyond the primary benchmark. We compare against published systems using the same Qwen3-VL-8B backbone and against a gold-page **TLV** baseline that directly supplies annotated evidence. We additionally report trajectory depth, role invocations, token usage, latency, and GPU memory to characterise deployment cost alongside accuracy.

4.1.3 Results

			MMLongBench-Doc				LongDocURL				DocBench
Method	**OCR**	**Trained**	SIN	MUL	UNA	**ACC**	UND	REA	LOC	**ACC**	**ACC**
Gold pages + VLM	✓	×	57.7	37.4	52.5	49.5	69.7	63.6	59.5	65.7	–
Whole document (via MARDoc)	×	×	–	–	–	41.9	–	–	–	–	64.6
M3DocRAG (via SLEUTH)	×	×	–	–	48.6	45.9	60.1	52.4	45.5	54.6	–
DocAgent (via MARDoc)	✓	×	–	–	–	46.6	–	–	–	–	67.1
MDocAgent (via SLEUTH)	✓	×	–	–	49.7	47.8	57.5	51.5	44.7	53.1	–
MoLoRAG (via SLEUTH)	×	×	–	–	51.7	48.8	63.4	52.7	49.3	57.6	–
MARDoc	✓	×[a]	–	–	–	52.7	–	–	–	–	**72.3**
SLEUTH	×	×	–	–	67.4	52.8	65.7	53.0	**53.6**	60.0	–
DocTrace	✓	✓	53.2	**41.1**	70.5	52.9	–	–	–	56.4	–
Ours (4-bit)	✓	×	**60.5**	39.2	**76.9**	**56.5**	**67.2**	**66.3**	47.9	**61.3**	71.6

Table 4.1 Accuracy (%) under each benchmark's official protocol (answer extraction + rule-based scorer) for systems built on Qwen3-VL-8B, the backbone our method uses. Values are as each paper reports them; "via" marks a result reproduced by another study, and "–" indicates not provided. [a]Its document outline uses visual-element descriptions from Qwen3-VL-235B.

Performance across benchmarks. Table 4.1 shows that our deliberately simple pipeline remains competitive with the strongest published Qwen3-VL-8B systems. On MMLongBench-Doc, it achieves 56.5% accuracy, exceeding DocTrace at 52.9% [124], SLEUTH at 52.8% [57], and MARDoc at 52.7% [132], as well as the 49.5% gold-page **TLV** baseline. This suggests that annotated evidence

pages are not necessarily an upper bound when additional document context can be acquired and selectively incorporated during reasoning. The system performs particularly well on single-page and unanswerable questions, although DocTrace retains an advantage on multi-page questions. On LongDocURL, it reaches 61.3%, slightly above SLEUTH at 60.0% but below the 65.7% gold-page baseline. Its stronger performance on reasoning questions but weaker localisation suggests that the benefits of adaptive evidence acquisition depend on task requirements. On the 1,065 completed DocBench questions, the system achieves 71.6% accuracy, approaching MARDoc's reported 72.3% and exceeding DocAgent's 67.1% and the whole-document baseline's 64.6%.

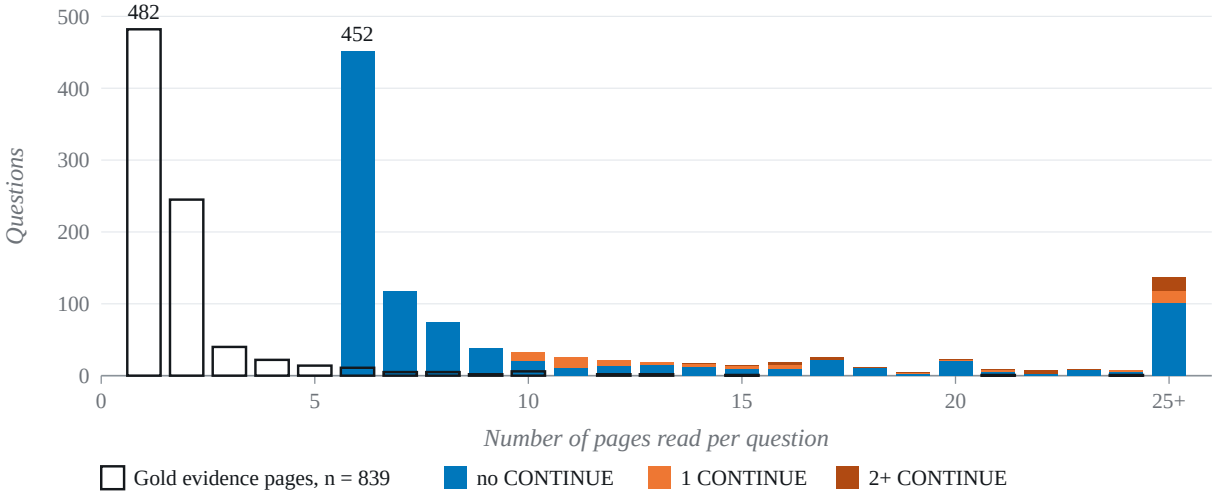


Figure 4.2 Pages read by the proposed pipeline compared with the number of annotated gold evidence pages on MMLongBench-Doc. Trajectories are separated by the number of times the `arbiter` requests further evidence acquisition.

Adaptive evidence acquisition. Figure 4.2 shows that the pipeline does not consume a fixed page budget for every question. The largest group terminates after six pages, the minimum initial reading depth, while progressively fewer questions continue further through the ranked document and a smaller tail reads 25 or more pages. Reopened trajectories account for part of this tail, with some questions requiring additional reading after an initial synthesis attempt. This behaviour is closely tied to retriever quality: when relevant pages are ranked early, sufficient evidence can be collected quickly and the `reader` can terminate near the minimum depth; when evidence is poorly ranked or distributed across several pages, deeper traversal becomes necessary. Adaptive acquisition therefore shifts compute toward questions whose evidence is harder to recover rather than assigning the same context budget to every query.

Inference cost. Table 4.2 shows that the pipeline averages 13.0 role turns per question, with most computation spent in the `reader`. Mean peak memory is only 8.4 GiB, keeping the system comfortably within the 16 GB deployment budget, but repeated sequential decoding makes latency the principal cost, with an average runtime of approximately 10.7 minutes per question. The method therefore trades low memory use for additional sequential inference and broader adaptive evidence acquisition.

Limitations. The deliberately simple design leaves several clear weaknesses. LongDocURL localisation tasks require the system to map topic descriptions or paragraph summaries to exact section titles,

		Tokens			Latency (s)		Peak
	Turns	Input text	Input visual	Output	Prefill	Decode	VRAM (GiB)
`reader`	10.8	5,031	817	747	1.8	50.2	7.2
`synthesizer`	1.1	9,124	2,293	438	4.0	33.9	8.3
`arbiter`	1.1	7,154	2,082	455	3.2	34.1	8.1
Per question, mean	13.0	72,350	13,689	9,041	27.5	615.8	8.4
Per question, median	9.0	38,845	9,643	5,776	14.6	363.6	8.1

Table 4.2 Inference cost of the pipeline per agent turn and question across both datasets.

identify named tables from descriptions, or jointly interpret figures and tables. Our pipeline is not explicitly designed for this kind of structured localisation: it follows a fixed page ranking, stores free-form evidence records, and has no section-aware indexing, table-level retrieval, or mechanism for matching descriptions against document structure. Multi-page reasoning remains difficult for a related reason. Evidence from most inspected pages is compressed into `reader` records, leaving the `synthesizer` to combine these summaries with a smaller set of flagged pages in a single reasoning step. More sophisticated extensions could incorporate structure-aware retrieval over section titles and tables, finer-grained evidence units, relation-aware navigation, or hierarchical cross-page reasoning. These are left for future work, since the present method is intended primarily as a proof of concept that the preceding design findings can already support a competitive system with a deliberately simple pipeline.

4.2 Open Challenges and Future Work

The survey of Chapter 2 identifies five challenges that define the open MP-VRDU frontier. They arise from a consistent pattern across the surveyed systems: progress comes from moving evidence handling out of the backbone through compression, retrieval, or iterative orchestration, but each shift relocates the bottleneck rather than removing it. Three challenges concern system capability, namely the evidence pipeline, faithfulness, and efficiency, while two concern data and evaluation, namely supervision integrity and evaluation scope.

4.2.1 Evidence Pipeline Bottleneck

MP-VRDU systems must reconcile high-resolution per-page reading with document-scale evidence aggregation, an objective that exceeds any MLLM's compute and context budget [9]. Every architectural family takes a lossy compromise. Compression can discard fine-grained content and is typically fixed before the question is seen. Retrieval bounds answer accuracy when required evidence is missed, while broader retrieval can introduce distracting context and attention dilution. Iterative retrieval can recover from some initial misses, but still inherits the limitations of the retrievers, parsers, and other tools it invokes. Cross-page structure compounds the problem further, since none of the approaches in §2.4.3 scales cleanly. Future work should explore query-conditional compression, retrieval that follows cross-references and logical relations, and learned cross-page structural representations.

4.2.2 Faithfulness and Abstention

Long documents amplify hallucination risk. When retrieval misses the answer-bearing region or compression discards it, current systems may still produce confident answers. Few surveyed systems support

abstention, calibrated uncertainty, or citation-grounded generation, leaving users with limited signals for distinguishing supported answers from unsupported ones. Future work should integrate evidence-grounded decoding, learned abstention for unanswerable or unretrievable questions, and trajectory-level faithfulness audits that verify whether the evidence used by the system actually supports the final answer.

4.2.3 Efficiency and Deployability

Agentic and retrieval-augmented pipelines can issue many model calls per question, with token and latency costs scaling with both document length and trajectory depth. The surveyed literature reports comparatively little inference-cost analysis: end-to-end latency, token budgets per query, memory requirements, and accuracy–cost trade-offs are often absent despite their importance for practical deployment. Future work should treat compute as a first-class evaluation axis, develop efficient inference techniques for long documents such as cross-round caching, speculative decoding, and adaptive depth, and explore compact MP-VRDU models [49] that operate under realistic on-device or privacy constraints.

4.2.4 Supervision Scarcity and Contamination

Per-page annotation does not scale to documents averaging tens of pages, so most systems rely on LVLM-synthesised QA over scraped PDFs [34, 39, 75, 77]. Compounding this, benchmark construction often reuses documents from earlier suites: MMLongBench-Doc draws documents from DUDE, SlideVQA, and ChartQA [9], while MP-DocVQA is built on DocVQA [5, 6]. Together, MLLM-generated QA data [10] and benchmark-reused fine-tuning sets [36, 39] risk contamination: models may train on benchmark-derived documents and be evaluated on overlapping data, sometimes judged by the same MLLM family. Future work should adopt held-out documents, declare cross-suite overlap, and separate generators from evaluators.

4.2.5 Evaluation Methodology and Scope

Beyond contamination, two further evaluation gaps persist. Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation alongside VQA-style answers [73, 107], but the practice is not yet standardised, and adaptive pipelines can produce variable trajectories that further complicate reproducibility. Task, language, and domain scope also remain narrow. Real-world MP-VRDU spans summarisation and structured extraction beyond VQA, multilingual layouts beyond English, and domain-specialised settings such as medical, legal, and financial documents, as well as streaming inputs that arrive page-by-page. These dimensions are inherited from SP-VRDU [12] rather than introduced by the multi-page setting and remain underexplored in both regimes. Future work should therefore standardise trajectory-level evaluation with retrieval and grounding metrics, report variability where adaptive inference is used, and broaden coverage along these inherited dimensions.

4.3 Concluding Remarks

The empirical analysis in Chapter 3 partially addresses several of the open challenges identified above. It clarifies where failures arise across representation, selection, and reasoning, examines the trade-off between supported answering and abstention, and characterises how design choices such as retrieval depth, model scale, and numerical precision affect both accuracy and deployment cost. These results

do not fully resolve the broader challenges in evidence handling, faithfulness, or efficiency, but provide more concrete guidance for navigating them. Section 4.1 then offers a limited system-level validation by showing that these findings can be combined into a deliberately simple competitive pipeline.

More broadly, this thesis contributes three complementary views of MP-VRDU: a survey that organises the design space through an evidence-management perspective, an empirical framework for locating failure within that space, and a working system that translates those findings into practice. Together, these contributions provide a clearer account of how modern MP-VRDU systems are built, where they fail, and how those failures can inform subsequent design.

Word Count and Appendices

The main text is approximately 16,000 words, excluding the abstract, acknowledgements, tables, captions, and the appendix. The appendix contains supplementary detail including the full survey methodology, surveyed corpus details, extended result tables and full method details; these not essential for reading this thesis.

Statement of LLM Use

LLM assistants were used to correct grammatical inaccuracies and suggest phrasing revisions in the manuscript and to assist with implementation of the experimental pipeline. All generated code was read and tested by the author and the author takes full responsibility for the content of this thesis. LLM assistants did not participate in research ideation.

Additional Artefacts

- Survey code and surveyed corpus: https://github.com/LeweiXu/MP-VRDU-Survey
- Empirical attribution and method code: https://github.com/LeweiXu/MP-VRDU-Analysis
- Original project proposal: https://www.overleaf.com/read/ntdnhjhxkgqt#386d92

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Appendix A

Additional Survey Details

A.1 Comparison with Prior Surveys

Survey	Venue	Scope	Multi-Page Treatment	Organising Framing
A Survey of Deep Learning Approaches for OCR and Document Understanding Subramani et al.	arXiv 2020	Pre-MLLM OCR pipelines	Not covered	Pipeline stage (detection, recognition, layout)
Deep Learning Approaches for Information Extraction from Visually Rich Documents: Datasets, Challenges and Methods Gbada et al.	IJDAR 2025	Information extraction over VRDs	Not covered	Extraction task and method
Deep Learning based Visually Rich Document Content Understanding: A Survey Ding et al.	AI Review 2026	Content understanding over VRDs	Not covered	Task formulation, architecture, pretraining
A Survey on MLLM-based Visually Rich Document Understanding: Methods, Challenges, and Emerging Trends Ding et al.	ACL 2026	MLLM-era VRDU	In passing	OCR dependency; architecture, training, benchmarks
Scaling Beyond Context: A Survey of Multimodal Retrieval-Augmented Generation for Document Understanding Gao et al.	ACL 2026	Multimodal RAG for long documents	Partial (RAG techniques only)	Retrieval-augmented generation
This survey	EMNLP 2026	**Multi-page VRDU**	**Distinct problem**	**Evidence management**

Table A.1 Comparison of this survey with prior VRDU surveys. Prior surveys either predate the multi-page literature or address it only within a broader single-page or extraction-focused scope. This survey is the first to treat multi-page VRDU as a distinct problem organised around evidence management.

Table A.1 contrasts this survey with the five prior VRDU surveys closest to our scope. The comparison is along three axes. **Scope** records the era and problem each survey targets, from pre-MLLM OCR pipelines to the recent MLLM and retrieval-augmented literature. **Multi-page treatment** records whether multi-page understanding is treated as a distinct problem with its own design axes, addressed only in passing within a broader scope, or left out. **Organising framing** records the lens each survey uses to structure the field. Prior surveys either predate the multi-page literature or treat it as one topic within a single-page or extraction-focused scope. This survey is the first to define MP-VRDU as a distinct problem, to organise it around evidence management, and to cover the four architectural families, modality pressures, inference-time and training strategies, and datasets specific to the multi-page setting.

A.2 Full Survey Methodology

A.2.1 Search keywords.

Queries combined a multi-page cluster with a task or method cluster. Multi-page terms included "multi-page document", "long document", "long-context document", "cross-page reasoning", "page-level retrieval", "document-scale evidence", and "hundred-page". Task and method terms included "document visual question answering", "visually rich document understanding", "document VQA", "document agent", "visual document retrieval", "document RAG", "hierarchical document transformer", and "long-context vision-language model". Benchmark-name probes ("MP-DocVQA", "MMLongBench-Doc", "LongDocURL", "DUDE", "SlideVQA", "M3DocVQA", "DocBench", "M-LongDoc") were used as anchors for forward-citation sweeps to capture systems that report against multi-page evaluation suites.

A.2.2 Inclusion and exclusion criteria.

A paper was retained if it (i) proposed a multi-page model, framework, system, or pipeline rather than a benchmark or dataset alone, and (ii) reported quantitative results on at least one multi-page benchmark, with cross-page evidence handling as an explicit design contribution. Excluded categories were single-page-only systems benchmarked solely on DocVQA, InfographicVQA, ChartQA, FUNSD, or CORD; survey, position, and opinion papers; generic OCR or table-extraction tools without VQA or reasoning evaluation; long-context LLM benchmarks where documents are one input type among many; and dataset-only contributions, which are catalogued separately in §3.4.6. Peer-reviewed venue publications and arXiv preprints were assessed against the same substantive criteria, since recent MP-VRDU progress has appeared predominantly in preprint form; venue status is recorded per entry in Table 2.1. Table A.2 lists the surveyed systems grouped by architectural family, with the multi-page contribution on which each was included.

A.2.3 Knowledge extraction and verification.

Each retained paper was converted to markdown and processed through LLM-assisted key-information extraction with evidence-span location, populating per-model fields covering architectural family, OCR dependency, vision encoder, LLM backbone, adaptors and projectors, page-rendering resolution, training stages, datasets used per stage, and reported scores on each benchmark. Every extracted field was manually verified against the source paper before being added.

A.2.4 Dataset catalogue.

The dataset catalogue in §3.4.6 was built primarily from the pretraining, fine-tuning, and benchmarking datasets named by every paper in the surveyed corpus, supplemented with a small set of datasets the corpus does not yet adopt but that cover MP-VRDU-relevant settings such as long scientific documents, multi-page tables, and cross-page reasoning suites, included to surface datasets whose adoption may strengthen future cross-system comparison.

A. ADDITIONAL SURVEY DETAILS

Model	Venue	Multi-Page Contribution
Page-to-Document Encoding Models		
Arctic-TILT (2025)	ACL	Layer-wise tensor-product text-image fusion with chunked sparse attention, serving up to 389k tokens on a single GPU
GRAM (2024)	CVPR	Interleaves page-level and document-level attention sub-layers with learnable global Doc tokens, avoiding quadratic concatenation cost
Hi-VT5 (2023)	PR	Learnable [PAGE] tokens summarise each page; decoder consumes the concatenated page tokens, with answer-page prediction for explainability
RM-T5 (2024)	DAS	Recurrent memory tokens propagate context page-by-page; per-page memories concatenated for decoding
MLLM-Centric Adaptation Models		
CoR (2026)	preprint	Four-stage Chain-of-Reading paradigm with a caption-reconstruction objective for figure and table grounding
Docopilot (2025)	CVPR	Native multi-page MLLM trained on Doc-750K cross-page proxy tasks with context-extension engineering
DocR1 (2026)	AAAI	EviGRPO adds an evidence-page F1 reward over structured think/evidence/answer outputs
DocSeeker (2026)	CVPR	Analysis-Localization-Reasoning paradigm with page-identifier grounding and evidence-guided resolution allocation
DocSLM (2026)	CVPR	Hierarchical compressor fixes 576 tokens per page; streaming segments with entropy-based abstention for edge deployment
InstructDoc (2024)	AAAI	Document-former fuses OCR, layout, and visual features into learnable tokens, with pages encoded in parallel
LayTokenLLM (2025)	CVPR	Per-segment layout token shares its segment's position ID, adding layout without consuming context
Leopard (2025)	TMLR	Adaptive high-resolution multi-image encoding under a global token budget with pixel-shuffle compression
mPLUG-DocOwl2 (2025)	ACL	DocCompressor cross-attends global low-resolution queries to sub-image features, compressing each page to 324 tokens
Texthawk2 (2024)	preprint	Two-step resampling and rearrangement for 16× visual-token compression with a coordinate-token detection head
URaG (2026)	AAAI	Cross-modal retrieval module at an early Transformer layer discards non-retained pages before deeper layers
Retrieval-Augmented Pipelines		
AVIR (2026)	MMAsia	Adaptive page selector uses score-distribution clustering to set per-query retrieval depth
CREAM (2024)	ACM	Coarse-to-fine retrieval with iterative reranking, paired with attention-pooled multi-page visual features
DFVC (2026)	MDPI	Adapter predicts gating weights over neighbouring-page embeddings to fuse cross-page context into page representations
DREAM (2025)	MM	Learning-to-rank fusion of embedding similarity and keyword-guided MLLM scores, with cross-page mixture-of-experts routing
KGP (2024)	AAAI	Knowledge graph over passage and structural nodes traversed by an LLM agent to gather supporting facts
M2RAG (2025)	NeurIPS	Dual-tower sparse and dense retrieval with text, visual, and fusion agents performing consistency checking
M3DocRAG (2024)	preprint	ColPali late-interaction page retrieval with approximate indexing, scaling to 40k+ pages
MDocAgent (2025)	preprint	Five-stage pipeline with parallel text and image retrieval and specialised modality agents
MHier-RAG (2026)	preprint	Combines flattened in-page indexing with a recursively summarised cross-page tree for multi-granularity retrieval
MLDocRAG (2026)	preprint	Chunk-Query Graph of LVLM-generated queries links chunks for retrieval by query-to-query similarity
MM-R5 (2025)	AAAI	Two-stage reranker training with structured reasoning chains and rank-discounted GRPO rewards
MoLoRAG (2025)	EMNLP	Traverses a page graph combining semantic and VLM-judged logical relevance to reach weakly-similar evidence
MultiDocFusion (2026)	ACL	LLM-constructed document hierarchy tree drives depth-first hierarchical chunking
PDF-WuKong (2026)	IJCV	End-to-end sparse sampler jointly trained with the LLM selects top-K text and image evidence
RAG-DocVQA (2025)	ICDAR	Systematic comparison of textual and visual RAG variants for multi-page DocVQA
Self-Attention Scoring (2024)	ICDAR	Self-attention scoring module ranks independently encoded pages, extending evaluation to 793-page documents
SLEUTH (2025)	CVPR	Training-free clue discovery and page screening keep effective context fixed as retrieval depth grows
SV-RAG (2025)	ICLR	Dual LoRA adapters over one MLLM backbone serve both late-interaction page retrieval and answering
VDocRAG (2025)	CVPR	Compresses each page image to a single dense embedding via self-supervised compression pretraining tasks
VisDoMRAG (2025)	NAACL	Parallel textual and visual RAG chains reconciled by consistency-constrained modality fusion
Adaptive-Trajectory Pipelines		
DMAP (2026)	WWW	Hierarchical document map traversed by tri-path retrieval with a reflective loop for insufficient evidence
Doc-V* (2026)	ACL	Thumbnail overview then alternating retrieval and high-resolution page fetch, trained by SFT then GRPO
DocAgent (2025)	EMNLP	Hierarchical XML outline navigated by an actor agent with five tools, plus a reviewer and persistent memory bank
DocDancer (2026)	preprint	End-to-end trained agent with only search and read tools, learned from synthesised exploration trajectories
DocLens (2026)	ACL	Page navigator and element localiser zoom to fine-grained evidence; sampling-adjudication selects among candidates
DocReact (2025)	ACL	Frames iterative retrieval as information-gain maximisation, halting when residual entropy stops decreasing
MACT (2025)	preprint	Four specialised agents with independent judgment routing and agent-wise adaptive test-time scaling
MM-Doc-R1 (2026)	ACL	Planner-seeker-answer workflow trained by similarity-weighted policy optimisation for multi-turn RL
SimpleDoc (2025)	EMNLP	Dual-cue retrieval over embeddings and page summaries, with a memory-carrying reasoner that reissues refined queries
ViDoRAG (2025)	EMNLP	Gaussian-mixture adaptive top-K retrieval with seeker, inspector, and answer agents over multi-scale views
VRAG-RL (2025)	NeurIPS	Visual perception action space lets the agent crop and zoom regions, trained with retrieval-efficiency rewards

Table A.2 The 46 surveyed MP-VRDU systems. Grouped by architectural family, with the multi-page contribution motivating each system's inclusion. Venue indicates the publication venue at time of writing.

		MP-DocVQA	DUDE	LongDocURL	MMLongBench-Doc		SlideVQA	
Model	**Variant**	**ANLS**	**ANLS**	**Accuracy**	**Accuracy**	**F1**	**EM**	**F1**
General Domain MLLM								
Claude-3-Opus	Claude-3-Opus	–	–	–	17.4	18.1	–	–
Gemini-1.5-Pro	Gemini-1.5-Pro	–	–	50.9	28.2	20.6	–	–
GPT-4o	GPT-4o (LVLM mode)	–	–	64.5	42.8	44.9	–	–
GPT-4V	GPT-4V(ision)	–	–	–	32.4	31.2	–	–
InternVL-Chat-v1.5	InternVL-Chat-v1.5-26B	–	–	–	14.6	13.0	–	–
LLaVA-OneVision-7B	LLaVA-OneVision-Chat-7B	–	–	25.0	–	–	–	–
MiniCPM-Llama3-V2.5-8B	MiniCPM-Llama3-V2.5	–	–	–	8.5	8.6	–	–
Qwen2-VL-7B	Qwen2-VL-7B	–	–	30.6	–	–	–	–
Page-to-Document Encoding Models								
Arctic-TILT	Arctic-TILT	81.2	58.1	–	25.8	–	55.1	–
GRAM	GRAM-859M	80.32	51.15	–	–	–	–	–
Hi-VT5	Hi-VT5 (multipage)	62.01	–	–	–	–	–	–
RM-T5	RM-T5	64.01	–	–	–	–	–	–
MLLM-Centric Adaptation Models								
DocR1	DocR1	87.45	54.39	–	–	–	–	71.96
DocSeeker	DocSeeker (full SFT + EviGRPO)	86.2	57.4	51.7	40.1	–	–	77.1
Leopard	Leopard-LLaVA-Pro	70.6	43.82	–	–	–	–	37.51
mPLUG-DocOwl2	DocOwl2-8B	69.42	46.77	–	–	–	–	–
URaG	URaG-7B	88.2	57.6	52.2	33.8	32.8	72.1	–
Docopilot	Docopilot-8B	81.3	–	–	28.8	23	–	–
Texthawk2	Texthawk2	–	–	–	–	–	–	–
DocSLM	DocSLM-2B	70	47.6	–	–	–	–	–
InstructDoc	InstructDoc (finetuned)	–	46.8	–	–	–	37.7	47.3
LayTokenLLM	LayTokenLLM-8B	74.31	52	–	–	–	–	–
CoR	Qwen2.5-VL-CoR-7B	–	–	51.5	37.4	36	–	–
Retrieval-Augmented Pipelines								
AVIR	AVIR	84.58	–	–	–	–	60.3	68.9
DFVC	DFVC (ColQwen backbone)	–	–	–	–	–	–	–
DREAM	DREAM† (InternVL2-4B + retrieved text)	85.72	52.14	–	27.3	28.6	–	–
M3DocRAG	M3DOCRAG (ColPali + Qwen2-VL 7B)	84.44	–	–	21	22.6	–	–
MM-R5	MM-R5 (Qwen2.5-VL-7B SFT+GRPO)	–	–	–	–	–	–	–
MoLoRAG	MoLoRAG+ (Qwen2.5-VL-7B)	–	–	51.85	41.01	–	–	–
Self-Attention Scoring	Pix2Struct-SA proposed	61.99	–	–	–	–	–	–
SLEUTH	SLEUTH (Qwen3-VL-8B + ColPali Top-5)	–	–	59.96	52.77	–	–	–
SV-RAG	SV-RAG-InternVL2† R5 (4B)	71.0	45.0	–	23.0	24.2	31.98	–
VDocRAG	VDocRAG	–	44	–	–	–	–	42
CREAM	CREAM	65.32	52.46	–	–	–	–	–
MHier-RAG	MHier-RAG (Qwen-turbo)	–	–	55.7	52.3	46	–	–
MLDocRAG	MLDocRAG	–	–	50.8	47.9	–	–	–
MultiDocFusion	MultiDocFusion	–	–	–	–	–	–	–
M2RAG	M[2]RAG top-5	–	–	57.68	39.9	–	–	–
MDocAgent	MDocAgent (top-4)	–	–	57.8	31.5	–	–	–
PDF-WuKong	PDF-WuKong	76.9	56.1	–	–	–	–	–
RAG-DocVQA	RAG-Qwen2.5-VL	73.7	39.9	–	–	–	–	–
VisDoMRAG	VisDoMRAG (ChatGPT-4o + ColPali + BGE)	–	–	–	–	–	–	67.22
KGP	KGP	–	–	–	–	–	–	–
Adaptive-Trajectory Pipelines								
Doc-V*	Doc-V* (GRPO)	86.2	64.5	56.3	42.1	–	–	77.2
DocReact	DocReact (w/ ColPali)	–	–	–	38.29	38.07	54.87	55.04
MACT	MACT-MiMo-VL-Series-28B	–	70.8	–	–	47.4	–	83.9
VRAG-RL	VRAG-RL (Qwen2.5-VL-7B-Instruct)	–	–	–	–	–	–	–
DocLens	DocLens (Gemini-2.5-Pro)	–	–	–	67.6	–	–	–
DMAP	DMAP Ours-Top4 (GPT-4o backbone)	–	–	60.7	43.2	–	–	–
DocAgent	DocAgent (Claude-3.5-Sonnet)	–	–	–	57.3	54.1	–	–
DocDancer	DocDancer (GPT-5.2)	–	–	–	57	56.8	–	–
MM-Doc-R1	MM-Doc-R1 (Qwen3-8B + SPO)	–	–	–	49.7	46.1	–	–
SimpleDoc	SimpleDoc (3.5 pages)	–	–	72.3	60.58	–	–	–
ViDoRAG	ViDoRAG	–	–	–	–	–	–	–

Table A.3 Performance of MP-VRDU models grouped by architecture category. Reported metrics follow the source papers and are not always directly comparable. '–' indicates not reported.

A.3 Additional Resources

A.3.1 Reported Performance by Architecture

Table A.3 summarises reported scores on the five most-adopted multi-page benchmarks, grouped by architectural family. It is included for coverage rather than ranking. The matrix is sparse and metric conventions diverge across MP-DocVQA, DUDE, LongDocURL, MMLongBench-Doc, and SlideVQA, so scores are not comparable across columns; for the same benchmark, differences in backbone scale,

proprietary-LLM dependency, evidence budget, and the training-versus-zero-shot distinction mean that a higher score does not by itself indicate a better architecture. Several of the strongest systems orchestrate closed-source frontier backbones of undisclosed scale, so their scores cannot be attributed to architecture alone. Read with these caveats, the table supports one robust cross-family observation: coverage tracks cost structure, with page-to-document and MLLM-centric systems confined to mid-length benchmarks and selective-evidence families dominating the long-document frontier.

A.3.2 General Domain MLLM for MP-VRDU

Frontier general-domain MLLMs handle short multi-page inputs well when the document fits in context, but the multi-page setting introduces three pressures absent in single-page VRDU: context-window overrun, attention dilution over sparsely distributed evidence [8], and a per-page reading fidelity versus whole-document coverage trade-off under a fixed token budget. Table A.3 shows that even the strongest frontier MLLMs trail specialised MP-VRDU systems by a wide margin on MMLongBench-Doc and LongDocURL, motivating the specialised architectures examined in §2.3.

A.3.3 Model Components Details

Table A.4 lists the concrete components of each surveyed system, organised by architectural family. The four families assemble these parts in characteristically different ways. Page-to-document transformers pair smaller LLM backbones (T5 variants, DocFormerv2) with custom layout-aware projectors. MLLM-centric adaptation models reuse standard vision-language stacks such as Qwen2.5-VL, mPLUG-Owl2, and SigLIP variants, and update only light projection or compression modules. Retriever-augmented pipelines decouple a frozen retriever (ColPali, ColQwen, VisRAG) from a frozen LVLM and train only the scorer, fuser, or reranker. Adaptive-trajectory pipelines mostly orchestrate frozen frontier VLMs, with parameter updates restricted to a few systems that fine-tune the controller via SFT or GRPO.

Table A.4 Per-model components of the surveyed MP-VRDU systems. Resolution lists the input image resolution or page-rendering setting; *dynamic* denotes encoders that accept variable-resolution input. '–' indicates not reported not applicable.

Model	Vision Backbone	Resolution	LLM Backbone	Adaptors and Projectors
Page-to-Document Encoding Models				
Arctic-TILT	U-Net	400k tokens / 500p	T5-Large	TILT fusion
GRAM	DocFormerv2	800 tok/page train, 8000 test	DocFormerv2	–
Hi-VT5	DiT	224x224 (DiT default), 20480 max tokens	T5	–
RM-T5	DiT	512 tok/page max	T5-base	–
MLLM-Centric Adaptation Models				
DocR1	Qwen2.5-VL	dynamic, max 1024 patches	Qwen2.5-VL-7B	Frozen Qwen MLP projector
DocSeeker	Qwen2.5-VL ViT	1024x784 eval, EGRA training	Qwen2.5-VL-7B-Instruct	–
Leopard	SigLIP-SO400M	364x364 / 980x980, max 50 sub-imgs	LLaMa-3.1-8B / Mistral-7B	MLP / Idefics2 resampler

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Table A.4 – continued from previous page

Model	Vision Backbone	Resolution	LLM Backbone	Adaptors and Projectors
mPLUG-DocOwl2	ViT-L/14	504x504, max 12 crops	mPLUG-Owl2	H-Reducer + DocCompressor
URaG	Qwen2.5-VL ViT	dynamic	Qwen2.5-VL-3B / Qwen2.5-VL-7B	Cross-modal retrieval module (2 linear + GELU) + LoRA
Docopilot	InternViT-300M	448 tile, max 24 tiles	InternLM2-1.8B / InternLM2.5-7B	2-layer MLP
Texthawk2	SigLIP-SO400M	224x224 base, max 72 sub-imgs	Qwen2-7B-Instruct	QPN + ReSA
DocSLM	SigLIP2	dynamic 384-1152, 576 tok/page	Qwen2.5-1.5B	OCR/Vision compressor + MLP
InstructDoc	CLIP	224x224	FlanT5 / BLIP2	Doc-former + proj
LayTokenLLM	–	–	LLaMA3-8B / Qwen1.5-7B	Layout tokenizer + LoRA
CoR	Qwen2.5-VL	dynamic, 32k max	Qwen2.5-VL-7B	Frozen Qwen MLP projector + LoRA
Retrieval-Augmented Pipelines				
AVIR	Pix2Struct	dynamic, 768-5120 patches	Qwen2.5-VL-3B-AWQ	Frozen Qwen MLP projector
DFVC	VisRAG, ColQwen	dynamic	Qwen2-VL-2B-Instruct	Cross-page MLP
DREAM	InternVL2 ViT	448 tile, max 24 tiles	InternVL2-4B	Cross-page MoE attention connector + LoRA
M3DocRAG	ColPali / ColQwen	144 DPI, ~1700px wide	Qwen2-VL-7B / InternVL2-8B / Idefics2-8B / Idefics3-8B	–
MM-R5	Qwen2.5-VL ViT	dynamic	Qwen2.5-VL-7B	–
MoLoRAG	ColPali	448x448	Qwen2.5-VL 3B/7B / Deepseek-VL-16B / LlaVa-Next-7B	Frozen base projector
Self-Attention Scoring	Pix2Struct	2048 patches	Pix2Struct	–
SLEUTH	ColPali, Qwen3-VL ViT	448x448	Qwen3-VL-8B / GLM-4.1V-Thinking-8B / Gemini-2.5-Flash	–
SV-RAG	InternVL2 / PaliGemma / Phi-3-vision	dynamic (InternVL2 default), 448x448	InternVL2-4B / PaliGemma-3B / Phi-3-vision-4B	Dual LoRA adapters (retrieval + QA) + Col-projection
VDocRAG	Phi3V-4.2B	336x336 dynamic, up to 1344x1344	Phi3V-4.2B	LoRA + MLP
CREAM	Pix2Struct	2048 patches	LLaMA2-7B	Attn pooling + MLP + LoRA
KGP	–	–	T5-large, LLaMa-7B, MDR (RoBERTa), ChatGPT	LoRA
MHier-RAG	Qwen-VL-Plus	dynamic	Qwen-VL-Plus / GPT-3-turbo / Qwen-turbo / GPT-4o / DeepSeek-chat / ERNIE-turbo	–
MLDocRAG	LVLM Qwen2.5-VL	dynamic	Qwen2.5-VL-32B-Instruct, Qwen2.5-VL-7B, Qwen2.5-72B	Frozen Qwen MLP projector

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Table A.4 – continued from previous page

Model	Vision Backbone	Resolution	LLM Backbone	Adaptors and Projectors
MultiDocFusion	–	–	Mistral-8B / Llama-3.2-3B / Qwen-2.5-3B / Qwen-2.5-7B	LoRA
M2RAG	VisRAG-Ret	dynamic	Qwen2.5-7B-Instruct, Qwen2.5-VL-7B-Instruct, Qwen2.5-VL-32B	Frozen Qwen MLP projector + LoRA + modal fuser
MDocAgent	ColPali / ColQwen2	144 DPI	LLaMA-3.1-8B-Instruct, Qwen2-VL-7B-Instruct	–
PDF-WuKong	IXC2-VL-4KHD	dynamic 336px-4K, max 16 tiles	IXC2-VL-4KHD	Sparse sampler
RAG-DocVQA	DiT / Pix2Struct	2048 patches	VT5 / Qwen2.5-VL-7B-Instruct	LoRA
VisDoMRAG	ColPali / ColQwen2	448x448 (ColPali), dynamic	Qwen2-VL-7B-Instruct / Gemini-1.5-Flash / GPT-4o	–
Adaptive-Trajectory Pipelines				
Doc-V*	Qwen2.5-VL ViT	1024x768 / 256x256 thumb	Qwen2.5-VL-7B-Instruct	Frozen MLP
DocReact	ColPali, VisRAG-Ret	448x448 (ColPali default) / dynamic	GPT-4o	–
MACT	Qwen2.5-VL-7B-Instruct / MiMo-VL-7B-SFT / InternVL3-9B	dynamic	Qwen2.5-VL-7B / MiMo-VL-7B-SFT / InternVL3-9B, Qwen2.5-7B/3B / MiMo-7B-SFT / InternVL3-8B/2B	–
VRAG-RL	Qwen2.5-VL ViT	dynamic	Qwen2.5-VL-3B-Instruct / Qwen2.5-VL-7B-Instruct	–
DocLens	frontier VLMs	dynamic	Gemini-2.5-Pro / Gemini-2.5-Flash / Claude-4-Sonnet	–
DMAP	ColPali	dynamic	GPT-4o / Qwen-plus	–
DocAgent	–	144 DPI	Gemini 2.0 Flash / GPT-4o / Claude Sonnet 3.5	–
DocDancer	Qwen3-VL-235B-A22B-Instruct	144 DPI	Qwen3-4B/30B-A3B	–
MM-Doc-R1	Qwen2.5-VL ViT	dynamic	Qwen3-4B / Qwen3-8B	–
SimpleDoc	ColQwen-2.5	dynamic	Qwen2.5-VL-32B-Instruct, Qwen3-30B-A3B	–
ViDoRAG	Nv-embed-V2, ColQwen2	dynamic	Qwen2.5-VL / LLaMA3.2-Vision / GPT-4o	–

Appendix B

Additional Attribution Details

This appendix extends the results of §3.5 and §3.6. Unless stated otherwise, results use Qwen3-VL-8B at **TLV** with gold pages on MMLongBench-Doc, and every accuracy is scored under the rubric of Figure 3.3.

B.1 Extended Result Tables

B.1.1 Representation

	PaddleOCR-VL				MinerU 2.5				Unlimited-OCR				PyMuPDF			
	Born-digital		Scanned		Born-digital		Scanned		Born-digital		Scanned		Born-digital		Scanned	
Source	Text	+Img.	Text	+Img.	Text	+Img.	Text	+Img.	Text	+Img.	Text	+Img.	Text	+Img.	Text	+Img.
Text	50.9	55.8	36.9	53.8	30.1	49.1	30.8	53.8	50.4	53.1	36.9	56.9	49.6	54.4	3.1	50.8
Layout	43.8	47.5	23.7	57.9	20.0	43.8	18.4	50.0	43.8	52.5	36.8	60.5	40.0	47.5	2.6	52.6
Table	45.9	44.3	62.5	71.9	36.8	42.7	56.2	65.6	49.7	48.1	59.4	71.9	47.6	44.9	0.0	59.4
Chart	23.8	42.3	6.2	54.2	17.7	38.5	14.6	47.9	23.8	42.3	2.1	58.3	25.4	37.7	0.0	54.2
Figure	17.9	38.0	34.0	56.6	12.5	38.6	25.5	50.9	20.7	38.0	34.0	58.5	20.1	41.3	7.5	52.8
All	40.2	49.8	34.3	61.2	26.6	46.0	27.4	55.7	42.3	50.2	34.3	63.7	40.6	49.7	4.0	57.7

Table B.1 Accuracy by text source, evidence source and scan status. Accuracy (%) on answerable MMLongBench-Doc questions with gold pages, with the text alone (Text) and with the page image added (+Img.).

Text Sources by Evidence Source. Table B.1 breaks the text-source comparison of Figure 3.4a down by evidence source; its **All** row holds the bars of that figure. The page image helps most where text extraction loses the evidence: chart and figure questions gain 12 to 26 points on born-digital documents under every text source, while table questions change by under 6 points and lose slightly under three of the four. On scanned documents embedded text answers almost nothing, and the image recovers 45 points or more for every source. The `low`-resolution runs marked in Figure 3.4a lost 35 (MinerU), 16 (Unlimited-OCR), and 12 (embedded text) of the 847 questions to memory limits, so those three marks are read on slightly fewer questions than the bars.

Resolution by Evidence Source. Table B.2 extends the resolution sweep of Figure 3.4b to all evidence sources. From `low` to `high`, **V** gains 12.4 points overall and 17.1 on tables, against 5.5 and 0.0 for **TLV**.

	TLV			V		
Source	low	med	high	low	med	high
Text	52.9	55.3	55.0	38.8	44.3	51.5
Layout	46.6	50.8	54.2	47.5	44.1	53.4
Table	48.8	48.4	48.8	25.3	37.8	42.4
Chart	38.2	45.5	48.3	34.3	43.3	48.3
Figure	39.7	44.8	48.3	38.3	40.3	43.8
All	49.2	52.5	54.7	37.8	45.6	50.2

Table B.2 Accuracy by image resolution and evidence source. Accuracy (%) on answerable MMLongBench-Doc questions with gold pages, with parser text (TLV) and without it (V).

Chart questions are the exception: they gain about 10 points under **TLV** as well, and parser text adds 2.2 points to them at `med` and nothing at `high`.

	Ladder @ med (%)				TLV		V		
Domain	**T**	**TL**	**TLV**	**V**	**low**	**high**	**low**	**high**	**ΔV**
Academic paper	36.4	33.1	**42.9**	33.8	37.0	46.8	19.5	38.3	−9.1
Admin / Industry	50.0	56.2	**67.2**	54.7	64.1	70.3	42.2	64.1	−12.5
Brochure	28.6	32.5	**45.5**	40.3	40.3	48.1	39.0	49.4	−5.2
Financial report	49.1	**52.8**	51.9	38.9	54.6	53.7	21.3	46.3	−13.0
Guidebook	35.8	40.0	**51.7**	45.0	48.3	51.7	38.3	47.5	−6.7
Research / Intro	22.6	27.4	**47.6**	45.3	42.9	50.0	38.2	47.6	−2.3
Tutorial / Workshop	14.3	48.2	**73.2**	67.9	71.4	74.1	74.1	70.5	−5.3
All	31.9	38.8	**52.5**	45.6	49.2	54.7	37.8	50.2	−6.9

Table B.3 Representation ladder by document domain and render resolution. Teal deepens with accuracy and bold marks each row's winning rung; the warm strip deepens with how far vision alone falls below the combined rung. Resolution lifts the domains whose evidence is pixel-limited and leaves the text-heavy ones where they were.

Representations by Document Domain. Table B.3 breaks the representations of Table 3.8 down by document domain, and sweeps the two image-carrying representations across resolution. Domain rows partition the pool, so each question is counted once. **TLV** is the most accurate representation in six of the seven domains, but variation across domains is considerably wider than variation across representations: at **TLV**, accuracy ranges from 42.9% on academic papers to 73.2% on tutorial and workshop material.

LongDocURL by Question Type. Table B.4 splits the LongDocURL results of Table 3.8 by question type. Accuracy varies far more across question types than across representations: `summary2tab` stays below 20% under every representation, while `extract` never falls below 74%. Parser text alone is the best representation for two of the locating types, and **V** is weakest on `calculate` and `compare`, which depend on exact values.

B.1.2 Selection

Retrieval Recall. Figure B.1 compares three text and three visual retrievers. Visual retrievers lead at every depth, and ColQwen3 is the strongest, which is why it orders pages in the selection experiments.

Task and question type	T	TL	TLV	V	n
Understanding (`extract`)	74.1	77.1	**83.4**	74.6	1,243
Locating	43.5	**51.6**	50.0	45.4	695
`extract_fig2tab`	58.4	**71.6**	62.3	55.4	231
`topic2title`	27.9	28.8	**35.1**	30.8	201
`summary2title`	67.2	**81.1**	77.2	74.3	137
`summary2tab`	14.9	18.3	**19.6**	19.0	126
Reasoning	66.8	67.5	**73.4**	58.3	387
`calculate`	61.0	66.7	**72.8**	50.7	145
`count`	57.9	56.2	**66.3**	58.0	117
`compare`	**81.1**	78.8	80.2	65.2	112
`summarize`	**84.6**	76.9	81.8	**84.6**	13
All	63.7	68.0	**71.9**	63.1	2,325

Table B.4 LongDocURL accuracy by task and question type. Accuracy (%) on answerable questions with gold pages; n is the number of questions.

Text retrieval over the embedded text layer collapses on scanned documents, and ranking parser text instead recovers much of the gap. Even ColQwen3 leaves gold pages outside the top ten, more often for questions with two gold pages.

	Precision					Recall					F1				
Retriever	**1**	**3**	**5**	**7**	**10**	**1**	**3**	**5**	**7**	**10**	**1**	**3**	**5**	**7**	**10**
MMLongBench-Doc (n=847)															
Text															
BM25	29.5	18.5	14.0	11.4	9.5	22.2	37.8	46.0	50.8	58.5	**24.2**	23.4	20.1	17.5	15.4
BGE-M3	33.8	18.7	14.7	12.0	10.0	25.5	38.8	48.2	53.2	61.1	**27.9**	23.9	21.2	18.4	16.3
Qwen3-Embedding	33.5	19.9	15.0	12.6	10.5	25.5	41.3	49.4	56.0	64.5	**27.9**	25.4	21.7	19.4	17.1
Vision															
ColModernVBERT	58.8	31.5	22.7	17.9	14.0	46.0	64.6	72.5	77.2	81.7	**49.6**	39.6	32.2	27.1	22.2
ColQwen2.5	63.3	33.8	23.6	18.5	14.4	48.8	68.3	74.9	79.2	83.8	**52.8**	42.5	33.5	27.9	22.9
ColQwen3	70.2	36.8	25.5	19.9	15.2	54.1	73.9	80.6	83.9	87.4	**58.6**	46.0	36.1	29.9	24.1
LongDocURL (n=2,317)															
Text															
BM25	49.6	27.0	19.0	14.8	11.3	35.6	53.9	61.1	65.6	69.5	**39.7**	34.3	27.7	23.3	18.7
BGE-M3	47.6	27.8	19.8	15.5	11.9	33.4	54.2	62.3	67.3	72.6	**37.6**	35.0	28.7	24.1	19.8
Qwen3-Embedding	48.9	28.8	20.6	16.4	12.5	34.1	56.0	64.8	70.6	75.5	**38.5**	36.3	29.9	25.5	20.7
Vision															
ColModernVBERT	64.7	33.5	23.2	17.9	13.6	47.5	67.3	74.5	78.6	83.1	**52.7**	42.7	33.9	28.0	22.5
ColQwen2.5	68.1	35.8	24.8	18.9	14.2	49.3	70.6	78.0	81.5	85.2	**54.9**	45.2	35.9	29.4	23.4
ColQwen3	73.3	38.2	25.9	19.8	14.8	53.6	74.5	81.1	84.6	88.1	**59.5**	48.1	37.5	30.7	24.4

Table B.5 Page-level retrieval across depth on both corpora. Amber deepens with the value on a per-metric scale and bold marks each retriever's best-F1 depth. Vision retrieval leads text at every depth on both corpora, and recall climbs with k as precision falls.

Retrieval Across Depth. Table B.5 reports page-level precision, recall, and F1 for all six retrievers at five depths on both datasets. F1 peaks at k=1 for every retriever, since precision falls faster than recall climbs, so a system optimising a single retrieval score should read shallowly while one optimising evidence coverage should read deeply. The two datasets are separate pools, so the table is read for the ordering of retrievers within a dataset and for the direction of the depth trend across both.

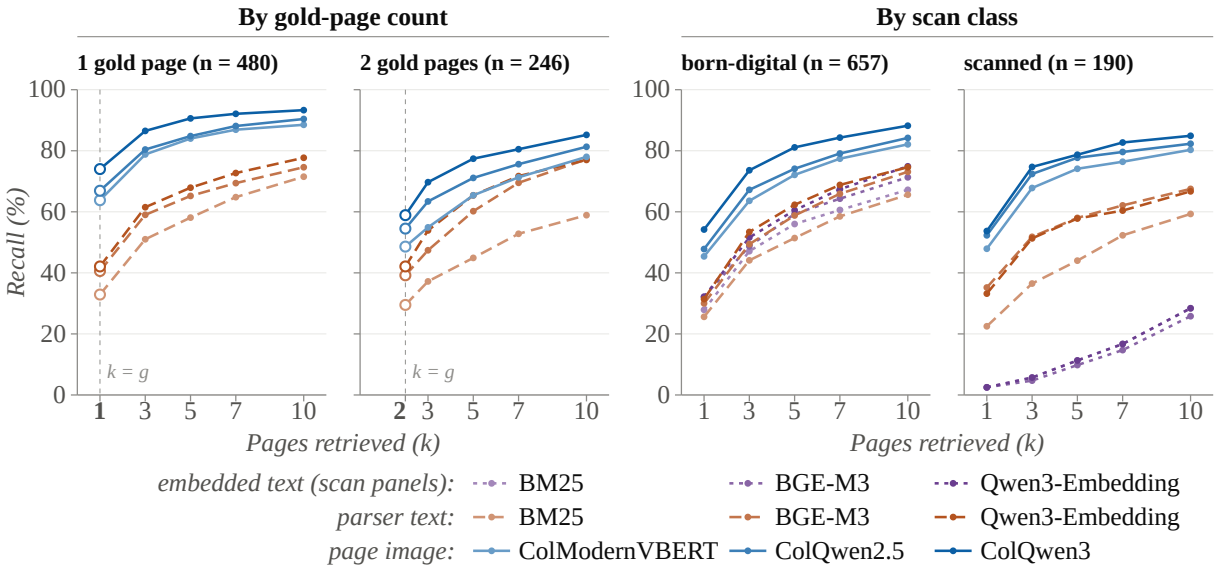


Figure B.1 Page-level recall on MMLongBench-Doc by retrieval depth, split by gold-page count and by scan status.

	Drop top		Drop bottom		Keep top		Keep bottom		
Source	−	+	−	+	−	+	−	+	n
Text	48.1	2.1	15.8	4.3	19.3	3.7	52.3	2.4	621
Layout	36.5	1.6	20.1	2.7	23.4	2.3	40.7	1.8	513
Table	32.2	2.6	15.9	3.9	17.2	4.3	33.0	1.7	233
Figure	44.2	2.8	22.7	4.4	26.3	5.6	48.2	2.4	251
All	43.2	2.0	19.6	3.6	22.2	3.5	46.5	1.8	1,093

Table B.6 LongDocURL verdict transitions after removing or retaining gold pages. Share (%) of multi-page questions whose TLV verdict moves from correct to incorrect (−) or incorrect to correct (+) against the complete gold set; n is the paired question count.

Coverage on LongDocURL. Table B.6 repeats the coverage experiment of Figure 3.5 on LongDocURL. Omission again breaks far more answers than it repairs. Unlike on MMLongBench-Doc, retrieval rank is informative here: removing the highest-ranked gold page breaks 43.2% of answers, against 19.6% for the lowest-ranked page. As §3.4.6 notes, LongDocURL's annotated pages need not all be required, which is consistent with the lowest-ranked gold page being the one a question can most often do without.

B.1.3 Reasoning

Further Reasoner Configurations. Table B.7 adds two comparisons to Table 3.9c. Qwen3-VL improves with scale at every representation, from 38.3% at 2B to 59.6% at 32B under **TLV**. The Thinking variant of Qwen3-VL-8B reaches 62.2% at **TLV**, 9.7 points above the Instruct model at the same weight footprint and above the 32B Instruct model.

Prompt Modes by Evidence Source. Table B.8 reports prompt modes by evidence source. `abstain` raises unanswerable refusal from 45.5% to 74.6% and answerable refusal from 6.2% to 27.7%, with figure and table questions refused most. Adding a sentence against declining supported questions (`abstain_balanced`) recovers little, and `grounded` alone lowers accuracy without improving

Reasoner	T	TL	TLV	V	GB
Scale (Qwen3-VL, bf16)					
2B	24.7	31.9	38.3	31.9	4.3
4B	32.3	38.6	50.4	42.4	8.9
8B	31.9	38.8	52.5	45.6	17.5
32B	35.5	43.9	59.6	56.7	66.7
Reasoning variant (Qwen3-VL-8B)					
Instruct	31.9	38.8	52.5	45.6	17.5
Thinking	34.8	42.9	62.2	52.1	17.5

Table B.7 Accuracy for further reasoner configurations. Accuracy (%) on answerable MMLongBench-Doc questions with gold pages; GB is the weight footprint.

	Answerable accuracy						Answerable refusal						Unans.
Prompt mode	Text	Layout	Table	Chart	Figure	All	Text	Layout	Table	Chart	Figure	All	refusal
`none`	**55.3**	50.8	48.4	45.5	44.8	52.5	7.3	0.8	6.7	7.3	7.6	6.2	45.5
`grounded`	51.2	42.4	40.1	33.7	42.4	45.7	7.0	0.0	7.2	5.1	7.2	6.0	37.3
`abstain`	45.7	41.5	39.2	28.7	34.5	41.4	**24.1**	**19.5**	**30.3**	**25.3**	**35.5**	**27.7**	74.6
`abstain_balanced`	47.4	41.5	37.3	30.9	36.2	42.6	19.2	18.6	25.5	18.5	31.4	23.2	68.9
`cot`	54.6	50.0	55.3	45.5	**45.2**	**53.1**	7.7	3.4	7.2	9.6	6.9	6.6	43.0
`extract_cot`	51.5	50.8	52.5	39.3	44.1	50.2	9.9	2.6	9.6	6.3	11.1	8.8	47.7
`abstain_cot`	54.6	**53.3**	**57.1**	**46.2**	39.6	53.0	22.0	15.9	16.3	22.5	28.0	19.9	**85.5**

Table B.8 Prompt-mode results at TLV. Rates (%) on MMLongBench-Doc: accuracy and refusal on answerable questions with gold pages, by evidence source, and refusal on unanswerable questions with BM25's top three pages, where refusing is correct.

refusal.

B.2 Statistical Uncertainty

Table B.9 gives a 95% confidence interval for the contrast behind each main finding. Each contrast pairs the two conditions within question, and the interval comes from 10,000 bootstrap resamples of documents, since questions on the same document are not independent. Because a contrast is computed over the questions both of its conditions completed, its difference can sit a few tenths of a point from the difference between two accuracies printed elsewhere, each of which is read on its own questions; the gap is largest for LongDocURL, where the paired difference is 3.4 points against 3.9 in Table 3.8.

Supported Contrasts. The intervals of the large effects exclude zero: adding page images to parser text, raising resolution under **V**, removing a gold page, the first three added pages for both reasoners, and the accuracy cost of `abstain`. The single-hop advantage over multi-hop questions is also clear at both ends of the scale range, at 21.9 points for 2B and 24.5 for 32B.

Contrasts Within Sampling Variation. Five intervals include zero, and we read these as ties. **TV** and **TLV** differ by 0.9 points, which supports calling them comparable. Resolution does not move **TLV** on table questions. The change in the hop gap from 2B to 32B is 2.5 points with an interval from -4.5 to $+9.6$, consistent with a gap that scale leaves in place. For Gemma-3-12B, neither the last distractor step nor the move from `abstain` to `abstain_cot` is separable from zero, so the recovery that `abstain_cot` brings is established for Qwen3-VL-8B only.

Contrast	Δ	95% CI	n
Modality ceiling			
TL → TLV	+**13.7**	[+10.2, +17.3]	847
V → TLV	+**7.0**	[+4.3, +9.7]	847
TV → TLV	+0.9	[−1.4, +3.3]	847
TL → TLV, LongDocURL	+**3.4**	[+1.6, +5.4]	2,325
Conversion fidelity, `low` → `high`			
V	+**12.4**	[+8.9, +16.0]	847
TLV	+**5.4**	[+3.1, +7.8]	847
V, table questions	+**17.1**	[+11.0, +22.8]	217
TLV, table questions	+0.0	[−4.5, +4.6]	217
Evidence coverage, one gold page removed			
lowest-ranked page	−**20.7**	[−26.6, −15.2]	358
highest-ranked page	−**23.3**	[−29.3, −17.5]	358

Contrast	Δ	95% CI	n
Distractor exposure			
Qwen3-VL-8B, +0 → +3	−**6.3**	[−9.8, −3.0]	847
Qwen3-VL-8B, +0 → +18	−**11.2**	[−14.8, −7.6]	847
Qwen3-VL-8B, +15 → +18	−**2.0**	[−3.8, −0.2]	847
Gemma-3-12B, +0 → +3	−**7.3**	[−10.3, −4.1]	847
Gemma-3-12B, +0 → +18	−**15.7**	[−18.9, −12.5]	847
Gemma-3-12B, +15 → +18	−1.5	[−4.1, +1.0]	847
Evidence integration, Qwen3-VL			
8B → 32B	+**7.4**	[+4.7, +10.2]	847
single- minus multi-hop, 2B	+**21.9**	[+14.7, +28.8]	847
single- minus multi-hop, 32B	+**24.5**	[+17.4, +31.1]	847
change in that gap, 2B → 32B	+2.5	[−4.5, +9.6]	847
Response calibration			
Qwen3-VL-8B, `none` → `abstain`	−**11.1**	[−13.7, −8.5]	847
Qwen3-VL-8B, `abstain` → `abstain_cot`	+**9.2**	[+5.9, +12.6]	847
Gemma-3-12B, `none` → `abstain`	−**10.7**	[−13.2, −8.1]	847
Gemma-3-12B, `abstain` → `abstain_cot`	+1.1	[−0.6, +2.8]	847

Table B.9 Bootstrap confidence intervals for the main contrasts. Δ is the accuracy difference in points over the questions both conditions completed, with a 95% interval from 10,000 resamples of documents; bold marks an interval that excludes zero. n is the number of questions in the contrast's pool.

B.3 Error Analysis

The error categories give a finer view of how the reasoner fails once representation and evidence selection are controlled. Unlike the main-text analyses, each category is reported as a share of the full question pool for its condition, so categories can be compared directly across prompt modes and evidence sources.

B.3.1 Error Categories

Error categories are assigned by an extension of the judge rubric (Figure 3.3) that adds one step after the verdict (Figure B.2). Each incorrect response receives exactly one category from Table B.10, the first that applies in the listed priority order. Correct and abstained responses receive none. Categories are assigned from the question, gold answer and response alone. The judge does not see the document pages, representation or retriever, so it cannot attribute an error to a pipeline stage it cannot observe; attribution to a locus comes from the controlled conditions, not from the judge. The category shares in this section, including their *incorrect* totals, come from this analysis pass, which is slightly stricter in practice than the rubric that scores accuracy (§3.4.7). They are therefore read against each other and not as 100 minus an accuracy reported elsewhere. In practice the taxonomy is nearly exhaustive: only one of the 1,853 incorrect responses across the four representations is assigned `other`.

```
⟨judge rubric of Figure 3.3, plus key error_category⟩

ERROR CATEGORY
Decide the verdict FIRST, using only the rules above, and do not
revisit it.  The error category explains a verdict you have already
reached; it never changes one.  In particular, a response that states
the gold answer in its own words, or surrounds it with explanation, is
still correct.

⟨the eleven categories of Table B.10, in priority order⟩
```

Figure B.2 Judge analysis rubric.

Category	Assigned when
`insufficient_evidence`	The response declines: the evidence does not support an answer.
`no_answer_reached`	It reasons but never commits, trailing off or stopping mid-calculation.
`answered_unanswerable`	The question is unanswerable and a substantive answer is given.
`equivocal`	The gold answer is present but not committed to.
`wrong_count`	A "how many" question is answered with a different count.
`wrong_value`	A non-count numeric answer differs from the gold value.
`wrong_entity`	A non-numeric answer names a different thing.
`incomplete_list`	The gold answer is a set and the response gives a subset.
`over_inclusive_list`	The response includes items the gold answer does not.
`unit_or_format_mismatch`	The same quantity in a non-matching unit, scale or granularity.
`other`	None of the above.

Table B.10 The error taxonomy in judge priority order.

B.3.2 Error Analysis Results

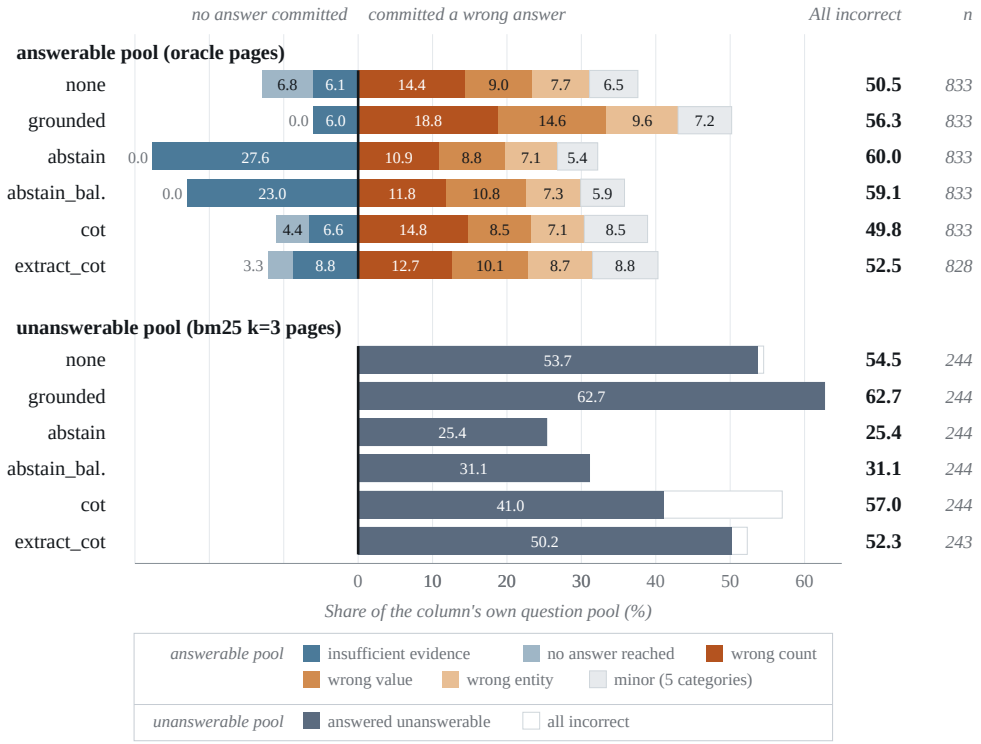


Figure B.3 Error profile by prompt mode on answerable and unanswerable questions.

Prompt Mode. The error distribution changes substantially with the reasoning instruction, showing that prompting alters the form of failure rather than uniformly removing it. Grounding largely eliminates cases where the model fails to reach an answer, but this is offset by committed errors such as incorrect counts, values, and entities. The abstention instruction shifts the distribution more strongly toward `insufficient_evidence`, consistent with the calibration results of §3.5.3: it prevents some unsupported answers but also introduces refusals on answerable questions. The `abstain_balanced` variant produces a very similar profile despite explicitly encouraging the model to answer when evidence

is sufficient, suggesting that this trade-off is not easily removed through wording alone. Chain-of-thought produces a different redistribution, changing the mix of substantive answer errors without the same increase in abstention, while `extract_cot` remains close to ordinary `cot` despite adding an explicit evidence-extraction step. Prompting therefore changes how the reasoner fails, but no mode removes the underlying reasoning errors across the board.

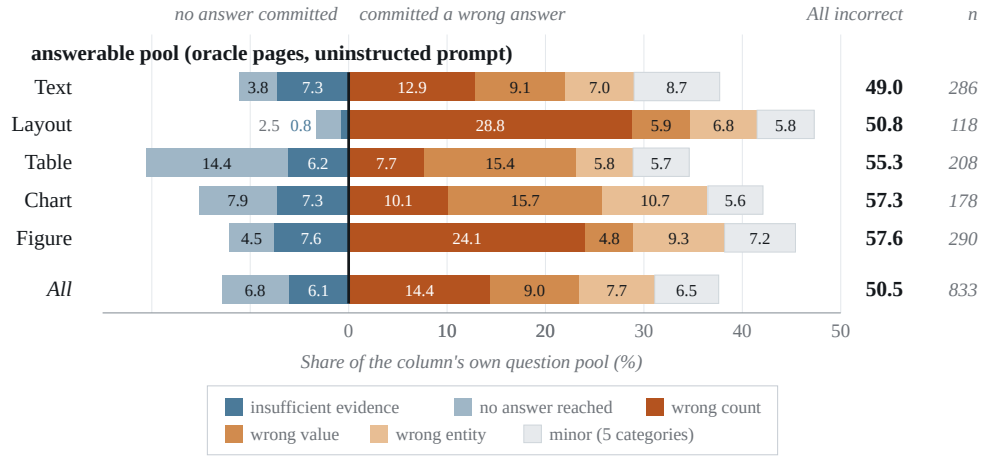


Figure B.4 Error profile by evidence source on the answerable pool with gold pages and no prompt instruction.

Evidence Source. Error type also varies substantially with the evidence a question requires. Under the uninstructed gold-page condition, 50.5% of answerable questions are incorrect overall under the analysis pass, but the dominant failure differs by source. Layout and figure questions are particularly prone to `wrong_count` errors, which account for 28.8% and 24.1% of their respective pools, compared with 14.4% overall. Tables and charts instead show the highest `wrong_value` rates, at 15.4% and 15.7%, suggesting that structured quantitative evidence more often fails during value extraction, comparison, or combination. Tables also stand out for `no_answer_reached`, which occurs on 14.4% of questions, more than twice the pooled rate of 6.8%. These differences indicate that providing the required multimodal evidence does not make downstream reasoning uniform: spatial and visual content more often fails during identification or aggregation, while quantitative structures produce a different mixture of extraction and comparison errors.

Appendix C

Additional Method Details

C.1 Implementation Details

This appendix provides the main implementation details for the evidence-management pipeline introduced in §4.1. We report the role-specific prompts and the configuration used for the MMLongBench-Doc experiments. Lower-level implementation details, including caching, batching, and output recovery, are omitted.

C.1.1 Role Prompts

The three roles share the same underlying vision-language model but receive different instructions and evidence. The `reader` extracts persistent evidence from each visited page and decides whether enough evidence has been collected to attempt an answer. The `synthesizer` reasons over the assembled evidence and produces the answer, while the `arbiter` determines whether the available evidence is sufficient or whether reading should continue. Figures C.1–C.3 show the principal instructions; repetitive formatting rules and error-recovery instructions are omitted for brevity.

C.1.2 Evidence Assembly

The `reader` maintains an append-only evidence record containing only information relevant to the question. Each record is associated with its source page, and pages considered likely to contain answer-bearing evidence are additionally marked for full presentation during synthesis. When the `synthesizer` is invoked, evidence is reordered by document position rather than retrieval rank. A marked page contributes the `reader`'s record, Unlimited-OCR text, and the rendered page image, while an unmarked page contributes only its retained record. Pages with an empty record are omitted. If no page is marked, the two highest-ranked pages already inspected are supplied in full as a fallback. This preserves broad evidence coverage while preventing every inspected page from entering the final multimodal context.

C.1.3 Configuration

Table C.1 details specific pipeline configurations. The `reader` traverses the fixed retrieval ranking until it proposes stopping or exhausts the page budget. A stopping proposal is ignored before k pages have been inspected. If the `arbiter` requests further evidence, at least m additional pages are read before another stopping proposal is accepted. At most C synthesis attempts are permitted. These controls bound evidence coverage and trajectory depth independently of the underlying model.

```
You are reading a long document one page at a time to answer a question.
Pages arrive in retrieval-ranked order, not document order.  You have
no memory of earlier turns beyond NOTES. A later turn will not see the
current page, so record any relevant evidence it contains.

Think step by step.  Then output these markers on separate lines:
REASONING: <your assessment of the page and notes>
RECORD: <evidence from this page, or exactly "none">
ANSWER PAGE: <true or false>
DECISION: <CONTINUE or STOP>

Record only what the CURRENT PAGE adds.  Preserve names, numbers, units,
table headers, and the details needed to interpret visual evidence.  For
a count or list, identify matching items on this page and where each
appears; end the record with COUNT ON THIS PAGE: <number> (<what was
counted>).  Do not copy evidence already in NOTES.

Mark ANSWER PAGE: true whenever the answer may depend on this page,
including when it supplies only part of a multi-page answer or you
are uncertain of its relevance.  Flagged pages will be shown to the
synthesizer in full.

Before the applicable reading floor, write DECISION: CONTINUE. After
that floor, write DECISION: STOP only when the accumulated evidence
appears sufficient to answer the question completely.  A later ranked
page may still contribute to a count or list.

QUESTION: {question}
NOTES: {reader records from earlier pages}
CURRENT PAGE: {document page number, parser text, page image}
```

Figure C.1 Reader prompt structure. The stopping instruction uses the initial floor k or the resumed floor m, as appropriate.

```
You are answering a question about a long document.  You cannot see the
whole document.  EVIDENCE is arranged one document page at a time, in
document order.  A block may contain a READER'S RECORD, PARSED TEXT,
and a PAGE IMAGE. Reader records may be incomplete or mistaken.  Use the
parsed text to check words, names, and numbers, and the image to check
visual and spatial details.

Use only the EVIDENCE. Think step by step, write that thinking after
REASONING:, and give a concise final answer after ANSWER:.

Check the question's premises before answering:  verify the particular
entities, dates, locations, figures, groups, and measurements it
assumes.  Do not substitute a nearby fact when a required detail
conflicts with the evidence.  Combine relevant facts across page blocks.
For a count or list, identify matching items page by page, exclude
duplicates and irrelevant items, and then aggregate them.

Answer when the evidence supports an answer.  Write "Not answerable"
only when a required fact is genuinely absent or contradicted.  Before
doing so, inspect the available parsed text and images and write:
MISSING: <the specific unsupported fact>.

REASONING: <premise checks and evidence integration>
ANSWER: <concise answer or exactly "Not answerable">

QUESTION: {question}
EVIDENCE: {page blocks containing records and, for flagged pages, parser
text and images}
```

Figure C.2 Synthesizer prompt structure. If the reader flags no page, the evidence includes the two highest-ranked inspected pages in full.

```
A reader gathered evidence from a long document, and a synthesizer
proposed an answer.  Decide whether the gathered evidence is sufficient
to answer the QUESTION. You may inspect the reader's records and the
flagged pages shown in full.  Your task is to decide whether to end or
read more pages; you cannot replace the proposed answer.

Think step by step.  Identify the facts the question requires, where the
evidence supplies them, and any specific fact still missing.  Inspect
the parsed text and images when a record may have omitted a relevant
detail.

Write DECISION: CONTINUE only if a required fact is missing and further
reading could plausibly find it.  Name that fact after MISSING:.  If the
evidence is sufficient but the proposed answer is mistaken, additional
reading is not a substitute for correcting the answer:  write DECISION:
END. When the proposed answer is "Not answerable" and pages remain,
check especially whether more reading could resolve the gap.

REASONING: <assessment of evidential sufficiency>
MISSING: <specific missing fact, or exactly "none">
DECISION: <CONTINUE or END>

QUESTION: {question}
RECORDS: {accumulated reader records}
FLAGGED PAGES: {parser text and medium-resolution images}
PROPOSED ANSWER: {synthesizer answer}
PAGES LEFT: {number of unread pages available}
```

Figure C.3 Arbiter prompt structure. A `CONTINUE` decision resumes reading at the next unread ranked page; the arbiter has no channel for issuing a replacement answer.

Setting	Value	Function
Retriever	ColQwen3.5	Ranks all pages once per question
Parser	Unlimited-OCR	Supplies the text channel
Shared backbone	Qwen3-VL-8B-Instruct	One resident model implements all three roles
Precision and decoding	4-bit NF4, FP16 compute, greedy	Runs on one NVIDIA V100 16 GB GPU
Reader pages	**TLV**, `high`	Parser text and images capped at 752,640 pixels per page
Synthesizer and arbiter pages	**TLV**, `med`	Flagged pages in full, with images capped at 501,760 pixels per page
B / p	30 / 1	Maximum pages read / pages per reader turn
k / m	6 / 4	Initial stopping floor / minimum further turns after reopening
Reader / synthesizer / arbiter limits	8192 / 4096 / 4096	Maximum generated tokens per role turn
Completion limit / C	1024 / 3	Tokens per completion pass / maximum completion passes per turn
Full-page fallback	Two pages	Highest-ranked inspected pages when none is flagged

Table C.1 Configuration of the simple adaptive-trajectory method.